



Enhancing Cyber Threat Forecasting with Comprehensive Architectural Comparison, Dark Web Signals and Explainable AI

[Comments]

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Declaration

I, Isobel Smith, declare that the work carried out and reported on by this document and the contents of this document, titled Enhancing Cyber Threat Forecasting with Comprehensive Architectural Comparison, Dark Web Signals and Explainable AI, and accompanying artefacts, except for where attributed to other sources by way of citations within the body of this document and listed under the References section of this document, is formed of my own original work, which has never before been produced for any other purpose, and has been created solely as part of delivery of an MSc in Data Science for or under the supervision of Dr Paul Yoo.

All information in this document and accompanying artefacts has been obtained and presented in accordance with the relevant ethical conduct and academic rules.

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Abstract

This project presents a novel approach to cyber threat forecasting by integrating dark web intelligence with explainable AI (XAI) techniques whilst comparing state-of-the-art architectures to establish which performs best in the operational context. This research contributes to the overall goal of shifting cybersecurity implementation from a reactive to a proactive posture to reduce the asymmetric advantage of threat actors. Building on Almahmoud's (Almahmoud, 2024) work on long-term forecasting, this research improves error rates via the integration of early dark web signals and the synthesis of an original architecture.

This first systematic comparison of deep learning architectures for cyber threat time-series reveals a trade-off between optimising for operational concerns and reducing error rates. The introduction of the spatio-temporal ensemble architecture used for this study achieves RAE 0.0258 and RRSE 0.058, which is an order of magnitude lower than competing models.

In addition, this study presents the first specialised cyber threat forecasting agentic LLM system, which boasts the strongest performance on operational readiness and adversarial robustness. Moreover, the addition of XAI strengthens model transparency for security decision-making and reveals insights about model behaviours that would benefit from further investigation.

The research program also introduces a robust, complex cybersecurity dataset, aggregating dark web data, web/social media data, industry specific metrics, and additional structured data sources, which builds a solid foundation for future research.

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List of Abbreviations

ACLED - Armed Conflict Location & Event Data Project

API - Application Programming Interface

APT - Advanced Persistent Threat

AUC-ROC - Area Under the Curve - Receiver Operating Characteristic

AutoML - Automated Machine Learning

B-MTGNN - Bayesian Multivariate Time-series Graph Neural Network

CAPTCHA - Completely Automated Public Turing test to tell Computers and Humans Apart

CVE - Common Vulnerabilities and Exposures

DDoS - Distributed Denial of Service

DiCE - Diverse Counterfactual Explanations

DORA - Digital Operational Resilience Act

FPR - False Positive Rate

GNN - Graph Neural Network

GPSCnv - General, Powerful, Scalable Graph Transformer

IDS/IPS - Intrusion Detection System/Intrusion Prevention System

KNN - K-Nearest Neighbors

LIME - Local Interpretable Model-agnostic Explanations

LLM - Large Language Model

LSTM - Long Short-Term Memory

mBERT - Multilingual BERT

MC - Monte Carlo

ML/DL - Machine Learning/Deep Learning

NLP - Natural Language Processing

NVD - National Vulnerability Database

ORI - Operational Readiness Index

PAT - Pertinent Alleviation Technology

RAE - Relative Absolute Error

RAG - Retrieval-Augmented Generation

RRSE - Root Relative Squared Error

RSS - Really Simple Syndication

SEC - Securities and Exchange Commission

SGD - Stochastic Gradient Descent

SHAP - SHapley Additive exPlanations

SLM - Small Language Models

SMOTE - Synthetic Minority Over-sampling Technique

TTC - Threat Trajectory Cycle

ViT - Vision Transformer

XAI - Explainable AI

1 Introduction

1.1 Problem Statement

As society grows increasingly reliant on digital infrastructure, which is in turn vulnerable to cyber sabotage, protecting this critical infrastructure becomes paramount to protecting the wellbeing of political institutions, public health, economies, livelihoods and ultimately personal safety. For instance, the 2021 Colonial Pipeline ransomware attack disrupted fuel supplies across the United States eastern seaboard (Wood, 2023). Yet, contemporary cybersecurity operations remain fundamentally reactive, with organisations responding to threats only after they appear rather than anticipating and mitigating them beforehand. This reactive paradigm creates an asymmetric advantage for threat actors, who can select targets, develop exploits, and coordinate attacks with relative impunity whilst defenders remain blind to emerging threats until they manifest in active incidents. The persistence of this reactive posture stems from three critical limitations in current threat forecasting approaches.

First, existing threat intelligence frameworks rely predominantly on historical attack data from public repositories. Whilst these sources provide valuable retrospective analysis, they lack the forward-looking signals necessary for genuine prediction. By the time a vulnerability is catalogued in these databases, threat actors have often already identified and begun exploiting it. Meanwhile, early warning signals, discussions of emerging exploits, vulnerability trading, and attack planning, proliferate across dark web forums and marketplaces, yet remain largely untapped by forecasting systems. This reliance on lagging indicators fundamentally constrains the lead time available for defensive preparation.

Second, recent advances in deep learning have produced increasingly sophisticated forecasting models, but these systems typically operate as black boxes, providing predictions without meaningful explanations of the underlying reasoning behind the outcomes. For security analysts tasked with operationalising these forecasts, allocating resources, prioritising patches, and implementing defensive measures, the absence of interpretable insights makes even accurate predictions difficult to action. Security teams require not only knowledge of what threats may emerge, but why the model believes they will occur, which signals inform predictions, and how confident the system is in its assessments. Without this transparency, organisations struggle to justify resource allocation, communicate risk to stakeholders, and integrate forecasting outputs into their security workflows.

Third, the cyber threat forecasting domain lacks consensus regarding optimal architectural approaches for temporal threat analysis. Whilst recent work (Almahmoud, 2024) demonstrates the potential of Bayesian Multi-Task Graph Neural Networks (B-MTGNNs) for long-term forecasting, the relative merits of alternative architectures, including Graph Transformers capable of modelling complex threat relationships, Vision Transformers designed to capture long-range temporal dependencies and agentic LLMs capable of autonomous decision-making, remain unexplored. This architectural uncertainty prevents the establishment of industry standards and limits the field's ability to leverage the most effective modelling approaches for different threat types and prediction horizons.

Predicting cyber threats is particularly difficult due to the sparsity of data, volatility of threat patterns, fragmentation of signals, and, at times, the anonymity of malicious actors. However, these research limitations collectively perpetuate the reactive cybersecurity paradigm, leaving organisations vulnerable to emerging threats and unable to mount proactive defences. Addressing these gaps requires a fundamental reconceptualisation of threat forecasting that integrates early warning signals from non-traditional sources, maintains transparency through explainable AI techniques, and systematically identifies optimal architectural approaches for temporal threat modelling.

1.2 Project Aims

The project aims to extend the foundational work (Almahmoud, 2024) by: evaluating and comparing the performance of BMTGNNs, Graph Transformers, Vision Transformers, a Spatio-Temporal Ensemble, and a specialised agentic LLM; integrating Bayesian uncertainty to Vision Transformers and Graph Transformers; introducing XAI techniques to cyber threat forecasting and integrating dark web data into long term cyber threat forecasting.

1.3 Research Objectives and Questions

This research addresses these limitations through three interconnected research questions, each targeting a specific gap in current threat forecasting capabilities.

Research Question 1: Can the integration of dark web signals improve the predictive accuracy of long-term cyber threat forecasting models?

This question examines whether incorporating intelligence sourced from dark web sources enhances forecasting performance compared to models using only public data repositories. The dark web hosts numerous forums, marketplaces, and communication channels where threat actors discuss vulnerabilities, trade exploits, and coordinate attacks, often weeks or months before these threats manifest. By extracting and integrating these early signals, forecasting models may achieve substantially extended lead times, providing organisations with critical additional preparation time.

Research Question 2: How do advanced deep learning architectures compare for cyber threat forecasting?

This question evaluates the relative performance of different model architectures in capturing the complex spatiotemporal patterns inherent in cyber threat evolution. Cyber threats do not evolve in isolation; they exhibit intricate dependencies across threat types, propagate through networks of vulnerable systems, and demonstrate temporal dynamics influenced by defender responses, tool development cycles, and attacker coordination. Different architectural paradigms offer distinct advantages for modelling these patterns.

Graph Transformers can explicitly represent relationships between threats, vulnerabilities, and affected systems whilst leveraging attention mechanisms to identify critical connections. Vision Transformers, originally designed for image analysis, can be adapted to treat temporal threat data

as sequences of feature maps, potentially capturing long-range dependencies that traditional recurrent architectures miss. Agentic LLMs for time-series forecasting can make autonomous, context-informed decisions, reducing the advantage of potentially slow response times for malicious actors. Bayesian MTGNNs, as demonstrated (Almahmoud et al., 2023), combine graph-based relational modelling with uncertainty quantification, enabling both prediction and confidence estimation. This research question asks: how do these architectures compare for different threat types and Pertinent Alleviation Technologies? Which architectures most effectively capture the early warning signals present in dark web data? Can ensemble or hybrid approaches that combine architectural strengths achieve superior performance to individual models?

Research Question 3: How can explainability techniques be effectively applied to complex temporal models for cyber threat forecasting?

This question explores methods for making sophisticated forecasting models transparent and interpretable to security professionals, addressing the critical gap between model predictions and actionable security decisions. Existing explainable AI (XAI) techniques have been developed primarily for static prediction tasks in domains such as image classification or credit scoring, and their applicability to temporal forecasting in cybersecurity contexts remains unclear.

Several sub-questions guide this investigation. Which explanation methods, including feature attribution techniques like SHAP (SHapley Additive exPlanations), attention visualisation approaches, counterfactual analysis, or causal inference methods, best capture the temporal dynamics central to threat forecasting? How can explanations address both the "what" (which specific threats are predicted) and the "why" (which signals and patterns support these predictions)? How can uncertainty information be effectively incorporated into explanations to support risk-informed decision-making?

These three research questions are fundamentally interconnected. Dark web signals may provide their greatest value when processed by architectures optimised for early signal detection and made actionable through effective explanation techniques. Similarly, architectural choices influence both the types of signals that can be effectively integrated and the explanation methods that can be applied. So, this research adopts a holistic approach that considers these interdependencies throughout the investigation.

1.4 Novel Contributions

This research makes several novel contributions to the cyber threat forecasting domain, advancing both theoretical understanding and practical capabilities.

1.4.1 Systematic integration of dark web intelligence with long-term threat forecasting.

Whilst previous research has explored dark web intelligence extraction and separate work has developed threat forecasting models, this project represents the first systematic integration of

these domains for long-term prediction. This contribution includes a dark web corpus with temporal labels linking discussion mentions to subsequent threat events, providing a foundational resource for future research.

1.4.2 Comprehensive architectural comparison for cyber threat time-series forecasting.

The research conducts the first systematic comparison of advanced deep learning architectures specifically for cyber threat forecasting, evaluating Graph Transformers, Vision Transformers, an agentic LLM, and Bayesian MTGNNs across multiple dimensions. This contribution establishes evidence-based guidance for architecture selection, supporting the development of industry standards.

1.4.3 Development of a spatio-temporal ensemble architecture

Through this research, it has been possible to develop a novel spatio-temporal ensemble model combining strengths from Graph Transformers and Vision Transformers, that presents the least errors of all the models studied, along with strong adversarial robustness and operational readiness. This is promising for cyber threat forecasting because, of the models studied, the spatio-temporal ensemble, on balance, proves the most suitable for operational deployment.

1.4.4 Bayesian uncertainty estimation applied to Graph and Vision Transformers

The application of Bayesian uncertainty estimation to Graph and Vision Transformers to quantify epistemic uncertainty addresses critical operational requirements. First, high uncertainty signals can guide where additional threat intelligence is needed, making resource allocation conversations more impactful. Also, security teams can prioritise response urgency based on confidence predictions, allowing for more target delivery of resources.

1.4.5 Proposal of a specialised agentic LLM framework for cyber threat forecasting

The use of a specialist agentic LLM framework for cyber threat forecasting presents another original contribution of this research. This is the first work to close the loop (Forecast, Explain, Recommend, Act) in one streamlined workflow, which thereby enables cooperation for security analysts with emerging regulatory compliance frameworks. By systematically exploring the operational constraints, potential accuracy trade-offs and interpretability of agentic LLMs in cyber threat forecasting, this research builds a foundation for both further academic study and justification of deployment in a business environment.

These contributions collectively address the critical gaps identified in the problem statement, providing both theoretical advances in understanding cyber threat dynamics and practical tools

for operationalising proactive cybersecurity. By integrating early warning signals, optimising architectural approaches, and ensuring transparency through explainability, this research enables the transition from reactive to proactive threat management that contemporary organisations urgently require.

1.5 Outline

This report explores the integration of dark web data and XAI into cyber threat forecasting and conducts an architectural comparison of state-of-the-art time-series forecasting models in this context. So, the first section gives a general introduction to cyber threat forecasting, dark web intelligence integration into this context, XAI for cyber security, and an overview of the architectures analysed.

The second section explains the methodological choices that characterise the project: the dataset and data collection, the preparation and pre-processing techniques, transfer learning and the model architectures treated, the integration of XAI, and the model evaluation process.

The third section describes the implementation process to ensure that the project and results are reproducible. The technical tools used, model specific design choices, challenges faced and how they were overcome are discussed.

The fourth section presents and analyses the results using different visualisations and performance metrics. This includes error analysis, operational readiness and adversarial robustness, directional accuracy for testing and validation plots, forecast and gap analysis discussion, and XAI results.

The fifth section offers an interpretation of these findings in a wider theoretical and practical context, as well as discussing the limitations of the work to situate the relevance of the contribution.

Finally, the sixth section outlines potential future work expansion and offers recommendations to future researchers.

2 Background and Literature Review

This section examines relevant literature across four interconnected domains to situate the original contribution of this research. This study explores integrating dark web intelligence with explainable AI (XAI) techniques for long-term cyber threat forecasting. While previous research has examined these domains independently, this attempts to synthesise them within a unified predictive framework that addresses both accuracy and interpretability.

2.1 Cyber Threat Forecasting: Evolution and Current State

The cybersecurity domain is transitioning from reactive, signature-based defence to proactive, predictive strategies (Mohamed, 2025). Traditional detection systems create a temporal lag, leaving organisations vulnerable to novel attacks (Buchta et al., 2024). Predictive approaches aim to identify threat trajectories before they materialise, enabling pre-emptive, proactive defence and more efficient resource allocation (Werner et al., 2017).

The Threat Trajectory Cycle (TTC) model by Almahmoud exemplifies this paradigm shift. It understands cyber threats as evolving through five phases: launch, growth, maturity, trough, and stability/decline, and forecasts these trajectories alongside relevant Pertinent Alleviation Technologies (Almahmoud, 2024). The study employs a Bayesian Multivariate Time-series Graph Neural Network (B-MTGNN), which integrates multi-source data (scientific literature, news, social media) and provides probabilistic forecasts with uncertainty quantification (Almahmoud, 2024). This represents a significant advance, achieving predictive accuracy for trends and PATs up to 36 months ahead, substantially outperforming traditional models.

Despite its contribution, the TTC model exhibits critical limitations that this project directly addresses. The model relies on public data sources, creating a blind spot to early warning signals from the dark web, where threat actors often discuss vulnerabilities and techniques months before public emergence (Almukaynizi et al., 2017). This sacrifices valuable lead time for proactive defence. Also, the B-MTGNN, while sophisticated, operates as a complex black box. It lacks transparency regarding which specific features or data sources drive its predictions, severely limiting its utility for security planning, stakeholder trust, and regulatory compliance.

More broadly, there are also recurring challenges across the threat forecasting landscape. Most approaches rely heavily on public data and overlook early-stage indicators from dark web forums and closed communities, creating a structural limitation in achievable lead times (Basheer & Alkhatib, 2021). Also, sophisticated models (e.g., LSTMs, complex neural networks) typically lack transparency and interpretability, constraining trust and evidence-based resource allocation (Rudin et al., 2021). Moreover, many models excel at either short-term tactical prediction or long-term forecasting, while few bridge temporal scales, so restrict the ability to provide coherent threat evolution patterns (Balasubramanian et al., 2025). Additionally, there is a lack of relational modelling. Sequential models inadequately capture the complex interconnections between threat actors, malware, vulnerabilities, and targets, missing crucial contextual intelligence (Fieblinger et al., 2024). Moreover, there is limited systematic research comparing the effectiveness of different deep learning architectures for capturing the various aspects of cyber threat evolution.

2.2 Dark Web Intelligence for Cybersecurity

The dark web is a critical, yet underutilised, **source of Cyber Threat** Intelligence. The ecosystem includes hacking forums, which are venues for knowledge exchange on vulnerabilities, attack techniques, and target discussion and marketplaces, which are platforms for trading illicit goods (malware, exploits, stolen data), where pricing and availability dynamics provide early warning signals.

DARKMENTION (Almukaynizi et al., 2017) operationalises dark web intelligence by correlating forum mentions with subsequent cyberattacks, providing actionable warnings for enterprises. This demonstrates the concrete value of dark web data. However, systems like DARKMENTION focus on near-term detection of imminent threats based on current dark web activity, not long-range forecasting of threat trajectories. The temporal limitation reduces the system's utility for strategic security planning. This highlights a critical integration gap. While dark web data is valuable for detection and classification, it is rarely systematically incorporated into formal, long-term forecasting frameworks. This project addresses this gap by integrating dark web signals into the proposed architectures to extend prediction horizons.

2.3 Explainable AI in Cybersecurity

The need for explainability in cybersecurity is acute due to high-stakes decisions, resource commitments, and regulatory implications. XAI techniques provide insights into model behaviour. Global explanations illuminate overall model behaviour and influential features while local explanations clarify the rationale behind individual predictions (Schwalbe & Finzel, 2023), essential for tactical validation and action. Global explanations can support strategic understanding of threat dynamics and inform decisions about data collection and model refinement while local explanations can support tactical decision-making, enabling security analysts to assess whether specific predictions align with domain expertise, identify potential model errors and understand the evidence supporting operational recommendations. Additionally, counterfactual explanations enable what-if analysis by showing how changes to inputs would alter predictions (Guidotti, 2022), supporting strategic planning by allowing security analysts to explore how different defensive measures, threat actor behaviour or environmental conditions might affect predicted threat trajectories.

Several established techniques show promise for threat forecasting. For instance, SHAP (SHapley Additive exPlanations) is a game-theoretic approach that provides mathematically principled feature importance scores for both global and local explanations (Lundberg & Lee, 2017). SHAP analysis can identify which specific data sources, temporal patterns or threat indicators most strongly influence predictions to offer insights into model behaviour. The consistency and additivity properties of SHAP values facilitate robust comparison of feature importance across different predictions.

Attention visualisations expose which parts of the input data a model, like a transformer focuses on (Vig, 2019), which is crucial for interpreting temporal forecasts so particularly valuable for time

series forecasting. Attention visualisations allow analysts to assess whether the model focus aligns with domain expertise and identify failure points where the model exhibits spurious correlation.

LIME (Local Interpretable Model-agnostic Explanations) creates simple, local surrogate models to approximate complex model behaviour for specific predictions (Ribeiro et al., 2016), offering intuitive explanations accessible to non-specialists, thereby supporting operation decision-making for security practitioners without deep learning expertise. Bayesian Causal Modelling employs a probabilistic framework for moving beyond correlation to understand the causal drivers of threat evolution, enabling more robust reasoning about changing conditions and reliable counterfactual uncertainty estimation (Nishikawa-Toomey et al., 2022).

Emerging regulations, such as the EU's DORA, Digital Operational Resilience Act, (Digital Operational Resilience Act (DORA) - European Insurance and Occupational Pensions Authority, 2025) and the US's SEC Disclosure Rules (Security and Exchange Commission, 2025), increasingly mandate transparency in risk assessment. Explainable forecasting models directly support compliance by providing transparent, auditable predictions with clearly articulated evidence.

2.4 Advanced Deep Learning Architectures

2.4.1 Bayesian Multivariate Time-series Graph Neural Networks (B-MTGNN)

The B-MTGNN architecture employed by Almahmoud represents a sophisticated fusion of multiple architectural innovations, combining the complementary strengths of graph neural networks (which excel at modelling relational structures and dependencies between interconnected entities) with temporal processing capabilities (which capture sequential patterns and time-based evolution) (Almahmoud et al., 2024). This architectural synthesis proves particularly well-suited for cyber threat forecasting, which fundamentally involves both relational and temporal dimensions. The Bayesian component provides probability distributions over predictions, rather than point estimates, quantifying prediction confidence. This is useful for resource allocation in operational contexts, as threats that are less uncertain may be assigned more attention.

2.4.2 Vision Transformers for Time-Series Analysis

Vision Transformers (ViTs), originally developed for image analysis, have recently been adapted for time-series forecasting with promising results (Du et al., 2024). ViTs use self-attention mechanisms to model relationships between elements in a sequence regardless of their temporal distance, contrasting with traditional recurrent architectures that process sequences incrementally and may struggle with very long-range dependencies (Han et al., 2023). This means ViT may be more effective at identifying correlations between temporally distant cyber threat indicators, which may be valuable for cyber threats that exhibit complex, non-adjacent temporal patterns.

ViT's effectiveness with temporal complexities could prove useful as cyber-attack patterns can exhibit non stationarity with long-range dependencies spanning multiple temporal scales. Also, ViT's ability to process entire sequences in parallel enables more efficient training. This computational efficiency gain allows for the exploration of longer historical contexts and more extensive hyperparameter searches. Finally, positional encoding capabilities allow for the

preservation of temporal order information and enabling flexible attention patterns. This could enhance ViTs ability to capture seasonal patterns and periodic threats.

2.4.3 Graph Transformers for Spatio-Temporal Modelling

Graph Transformers represent a cutting-edge architectural synthesis combining the strengths of standard Transformers (flexible attention mechanisms and long-range dependency modelling) with Graph Neural Networks (explicit relational structure modelling) to address both spatial and temporal dependencies in structured time-series data (Shehzad et al., 2025). This is promising for long-term cyber threat forecasting because the architecture's ability to efficiently model both local correlations (such as related attack vectors) and global patterns (such as coordinated campaigns across multiple attack categories) addresses the lack of relational modelling capabilities of atypical sequential or temporal architectures (like LSTMs or CNNs).

Additionally, graph transformers' unified attention might enable the identification of complex patterns, such as how threats propagate through network structures over time or how relationships between different attacks evolve. Also, specialised graph-structured positional encodings account for both graph structure and temporal position, which allows the model to use both types of structural information when computing attention weights. This may lead to better performance, as cyber threats exhibit both entity relations and temporal relationships between campaigns. Finally, sparse attention and hierarchical processing addresses scalability challenges, allowing for the tractable application of graph transformers to larger threat intelligence graphs.

2.4.4 Agentic LLMs for Time Series Forecasting

Agentic LLMs represent a shift from passive, reactive systems to autonomous, goal-oriented systems capable of operating independently to achieve specified goals through reasoning, planning, and taking action, executing complex tasks with minimal human supervision (Plaat et al., 2025). While traditional generative AI models respond to prompts, agentic LLM systems can take initiative, decompose problems, make decisions about tool usage, and adapt their strategies based on outcomes (Ke et al., 2025). Agentic LLMs can perceive their environment and understand objectives; reason about possible actions and outcomes; act by executing tasks and using tools; learn from results and adapt strategies and iterate until set goals are achieved (Li et al., 2024)

Applied to the context of cyber threat forecasting, agentic LLM systems treat forecasting as a sequential decision-making process, involving data preparation, model selection, validation, and ensemble analysis to create more flexible, interpretable, powerful forecasting systems. This is promising for cyber threat forecasting because autonomous agents can make strategic decisions about resource allocation and forecasting strategy faster than human analysts, potentially reducing the asymmetric advantage of malicious actors considerably.

Multi-agent architectures typically outperform traditional approaches. TimeSeriesScientist achieved a 10.4% error reduction over statistical baselines (Zhao et al., 2025). This is the first end-to-end agentic framework, using multimodal knowledge to automate the entire forecasting (Zhao et al., 2025) workflow. Comprised of four specialised agents, the framework transforms forecasting into an interpretable system that is extensible across tasks and domains (Zhao et al., 2025). The Agentic LLM in this paper is based on TimeSeriesScientist's framework.

Another notable multi-agent architecture is AgenticRAG, which retrieves relevant prompts from shared pools based on cosine similarity to allow models to adapt to non-stationary data through conditioning on past patterns using specialised sub-agents (Ravuru et al., 2024), and inspired the optional RAG component of the agentic system.

DCATS (Data-Centric Agent for Time Series) is also noteworthy for leveraging meta-data accompanying time series to strategically enrich training datasets (Yeh et al., 2025), which informed data pre-processing strategies for the sub-agents in the agentic system of this paper.

Table 1: Key agentic LLM systems and their contributions.

	Architecture	Key Innovation	Performance Gain
TimeSeriesScientist(Zhao et al., 2025)	4-agent system (Curator, Planner, Forecaster, Reporter)	Multimodal diagnostics with plot understanding	10.4% vs statistical, 38.2% vs LLM baselines
Agentic-RAG (Ravuru et al., 2024)	Hierarchical multi-agent with specialised sub-agents	Dynamic prompt retrieval from knowledge pools	MAE 13.01 vs 15.57 on an industry benchmark
DCATS (Yeh et al., 2025)	Single LLM agent with iterative refinement	Data-centric approach via metadata reasoning	6% error reduction across all models

2.4.5 Architectural Trade-offs and Selection Criteria

Applying deep learning to cyber threat forecasting requires careful consideration of multiple architectural trade-offs. Different architectures offer distinct advantages and limitations for cyber threat forecasting.

Table 2: Comparison of trade-offs across architectures

Architecture	Advantages	Limitations
B-MTGNN	Superior explicit relational modelling Principled uncertainty quantification Proven performance on cyber threat data Interpretable graph structure.	Computational complexity for very large graphs Requires careful tuning of graph learning components May overfit with limited training data.

Vision Transformer	Excellent long-range dependency capture Computational efficiency through parallelisation Strong performance on regular time-series.	Less explicit modelling of relational structure May require more training data Positional encoding less naturally suited for irregular or multi-scale temporal patterns.
Graph Transformer	Unified spatio-temporal modelling Flexible attention mechanisms Recent scalability improvements.	Relatively nascent architecture with less established best practices Higher computational requirements Complexity may hinder interpretability.
Agentic LLMs	More flexible model selection, as models are chosen dynamically based on data which may yield better results. Allows for systematic comparison across architectures.	Potential for hallucinations and errors in high stakes situations. Potential for lack of transparency and interpretability, if reasoning is improperly elicited.

The architectural exploration gap is clear. There is limited systematic comparison of these advanced architectures specifically for cyber threat forecasting, creating uncertainty about which is most effective.

2.5 Research Gaps and Strategic Contributions

This review identifies four interconnected research gaps that this project seeks to address:

- The failure to routinely incorporate dark web intelligence into long-term forecasting models, constraining lead times and accuracy.
- The black box nature of sophisticated forecasting models, limiting their operational trust and utility for decision-making.
- The lack of systematic comparisons of advanced deep learning architectures for cyber threat forecasting, hindering informed architectural selection.
- The paucity of models bridging short-term tactical predictions and long-term strategic forecasting to support strategic decision-making at multiple temporal scales.

This research makes a strategic contribution by synthesising these disparate elements into a unified framework. The specific objectives are to:

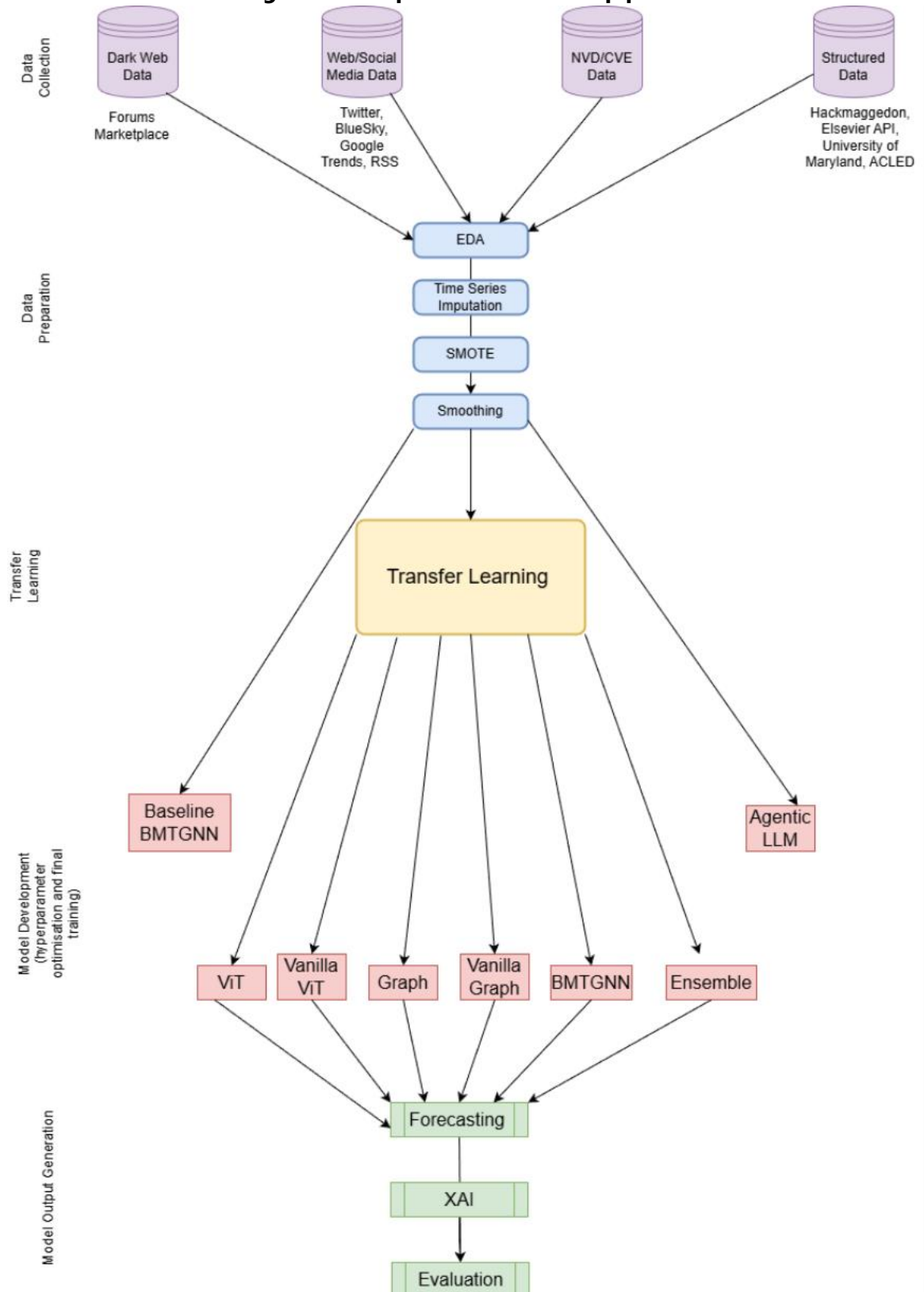
- Demonstrate the quantitative improvement in accuracy gained by integrating dark web intelligence.
- Apply XAI techniques to cyber threat time-series forecasting, ensuring explanations are both faithful and actionable.

- Compare B-MTGNN, Vision Transformer, and Graph Transformer architectures and an agentic LLM to establish their relative effectiveness for capturing different aspects of threat evolution.

3 Methodology and System Design

This section describes the data pipeline (as depicted in Figure 1, below) for this project. The system is designed with robustness, uncertainty analysis, interpretability and modularity as end goals. First, the data collection process and dataset are discussed. Then, data preparation techniques are detailed. Next, the process of model development, transfer learning, and training is explained for all the models evaluated. Special attention is afforded to the design specifics of model architectures. Then, the process of XAI integration is outlined. Finally, the model evaluation process is recounted.

Figure 1: A depiction of the data pipeline



3.1 Dataset and Data Collection

The dataset, originally developed by Almahmoud, has been transformed in both scale and scope (Almahmoud, 2024). The original dataset contained 142 temporal nodes (data points) with 143 features, covering July 2011 to December 2022 (Almahmoud, 2024). Through systematic enhancement, it has been expanded to 1,292 nodes and 2,126 features, now spanning until December 2024. This represents an 810% increase in temporal observations and a 1,385% increase in features, significantly enriching its analytical potential.

Table 3: Quantitative Summary of the Augmentation of the Dataset

Dimension	Original (Almahmoud 2023)	Augmented Dataset	Increase
Temporal Nodes	142	1,292 (smoothed)	+810%
Features (Columns)	143	2,126	+1,385%
Temporal Coverage	Jul 2011 - Dec 2022	Jul 2011 - Dec 2024	+24 months
Data Sources	5	11	+120%
Country Coverage	37	37 (maintained)	Maintained
Attack Taxonomies	2 (Hackmageddon, Elsevier)	6 (+ UMD, NVD, RSS, Dark Web)	+200%
Defence Taxonomies	1 (Elsevier PATs)	3 (+ RSS, Google Trends)	+200%

Table 4: Feature Breakdown by Source

Data Source	Fields Added	Temporal Span	Key Contribution
-------------	--------------	---------------	------------------

Hackmageddon Extension	1,036 (extended)	2023-2024	Incident continuity
Elsevier Extension	147 (extended)	2023-2024	Academic discourse
Python Holidays Extension	38 (extended)	2023-2024	Temporal context
RSS Feeds	161	2011-2024	Media perspective
Google Trends	86	2011-2024	Public awareness
Bluesky	37	2023-2024	Conflict tracking
University of Maryland	444	2014-2024	Technical taxonomy
ACLED	37	2017-2024	Political violence
NVD/CVE	40	2011-2024	Vulnerability linkage
Dark Web Counts	67	2020-2024	Underground economy
Dark Web Pricing	68	2020-2024	Threat commoditisation
Total New/Extended	2,161		

The dataset aggregates mention counts and attack counts. The augmentation integrates multiple new data streams to create a multi-faceted view of the cyber threat landscape.

To capture incident and technical data, the Hackmageddon (Passeri, 2024) incident count data was extended to 2024. Also, from the University of Maryland Cyber Event Database (2014-2024) (Harry, 2023), incident count data was added, providing a technical taxonomy of 12 attack methodologies (e.g., DDoS, data attacks, server exploitation), offering a complementary perspective to malware-focused classifications.

To capture media and public discourse data, a comprehensive collection of RSS¹ feeds (2011-2024) mention counts from security vendors and news outlets captures real-time

¹ Full list available in Appendix C

industry reporting and attack mentions. Also, to counter Twitter's API restrictions, Bluesky mention count data (2023-2024) (Bluesky, 2025) was integrated to monitor conflict-related discourse, allowing analysis of links between geopolitical tensions and cyber operations.

To capture public awareness data, Google Trends data (2011-2024) (Google, 2025) tracks worldwide search interest for 86 attack and defence-related topics, serving as an indicator of public concern and threat salience. Additionally, the academic research mention counts data from Elsevier (Toaza & Esztergár-Kiss, 2024) was extended until the end of 2024.

For vulnerability intelligence, National Vulnerability Database (NVD/CVE) mention count data (National Institute of Standards and Technology, 2025) (2011-2024) was incorporated, enabling the study of connections between disclosed software weaknesses and active attack trends. For conflict context dimensions, incident count data from the Armed Conflict Location & Event Data Project (ACLED) (2017-2024) (Raleigh et al., 2023) provides ground truth on political violence, enabling empirical analysis of correlations between conflict and cyber activity.

Finally, to better understand underground economy data, a major enhancement is the addition of Dark Web Marketplace Data (2020-2024). Sourced from 14 major platforms² using a scraping layer built on Tor manager, this includes mention frequencies and price data for 67 attack categories, from ransomware-as-a-service to exploit kits. This allows for the tracking of threat commoditisation, where price changes can indicate capability democratisation or demand shifts.

The attack categories correspond to the mention lists (code snippet 1) below.

Code Snippet 1: Mention List of Dark Web Data Categories

² Full list available in Appendix C

```
mention_list = [  
    # Initial Access  
    ('Phishing', 'Spear Phishing', 'Whaling', 'BEC', 'Email Spoofing'),  
    ('Social Engineering', 'Pretexting', 'Baiting', 'Vishing', 'Smishing'),  
    ('Email Bombing', 'Mailbomb', 'Spam Attack'),  
    ('Drive-by Download', 'Watering Hole Attack'),  
    ('Exploit Kit', 'Zero-day', '0-day'),  
    # Malware  
    ('Malware', 'Malicious Software'),  
    ('Ransomware', 'Crypto-ransomware', 'Locker Ransomware'),  
    ('Ransomware-as-a-Service', 'RaaS'),  
    ('Trojan', 'Banking Trojan', 'Dropper', 'Downloader'),  
    ('Backdoor', 'RAT', 'Web Shell'),  
    ('Info-Stealer', 'Spyware', 'Keylogger'),  
    ('Worm', 'Virus', 'Self-propagating Malware'),  
    ('Botnet', 'C2 Server', 'Command and Control'),  
    ('Rootkit', 'Bootkit'),  
    ('Adware', 'Malvertising', 'PUPs'),  
    ('Wiper Malware', 'Data Destruction Malware'),  
    ('Malware-as-a-Service', 'MaaS', 'CaaS'),  
    # Network & App Attacks  
    ('Denial-of-Service', 'DoS', 'DDoS'),  
    ('SQL Injection', 'SQLi'),  
    ('Cross-site Scripting', 'XSS'),  
    ('Remote Code Execution', 'RCE'),  
    ('Man-in-the-Middle', 'MITM', 'Session Hijacking'),  
    ('DNS Spoofing', 'DNS Hijacking', 'Pharming'),  
    ('Password Attack', 'Credential Stuffing', 'Brute Force'),  
    # Monetisation & Fraud  
    ('Carding', 'Card Fraud', 'Fullz', 'Card Dump', 'CVV'),  
    ('Account Takeover', 'ATO', 'Bank Log'),  
    ('Cryptojacking', 'Illegal Cryptomining'),  
    ('Data Breach', 'Data Leak', 'Data Exfiltration'),  
    ('Dark Web Market', 'Underground Market', 'Carding Forum'),  
    # Advanced Tactics  
    ('Lateral Movement', 'Privilege Escalation', 'Persistence'),  
    ('Advanced Persistent Threat', 'APT'),  
    ('Supply Chain Attack', 'Third-party Attack'),  
    ('Insider Threat', 'Malicious Insider'),  
    ('IoT Attack', 'OT Attack', 'ICS Attack'),  
    ('Adversarial Attack', 'AI Poisoning', 'Data Poisoning'),  
]
```

Impact and Analytical Value

This novel dataset enables a paradigm shift from single-source analysis to multi-perspective predictive intelligence. It allows for tracking across six key areas:

1. What attacks occurred? (Incident reports from Hackmageddon, UMD)
2. What is being studied? (Academic research from Elsevier)
3. What is being reported? (Media discourse from RSS feeds)
4. What is being searched? (Public awareness from Google Trends)
5. What is being commoditised? (Underground activity from the Dark Web)
6. What weaknesses are found? (Vulnerability disclosure from NVD)

The 13.5-year temporal coverage supports longitudinal analysis of trends across technological eras, economic cycles, and geopolitical events. This transformation creates

an intelligence platform that is predictive, proactive, and quantitatively robust, supporting operational defence, policy development, and scientific discovery in cybersecurity.

3.2 Data Preparation

To prepare the data for the models and mitigate data bias and inconsistencies, the following methods were implemented: exploratory data analysis to understand the data distribution, time series specific imputation for missing values, SMOTE for imbalanced classes and exponential and double exponential smoothing are applied to the raw dataset.

Exploratory Data Analysis

To explore the statistical and distributional properties of the dataset and then inform choices for building the model pipeline, various insights regarding data patterns and outliers were extracted at this stage.

Figure 2: Percentage of Outliers for Features with the Highest Number of Outliers

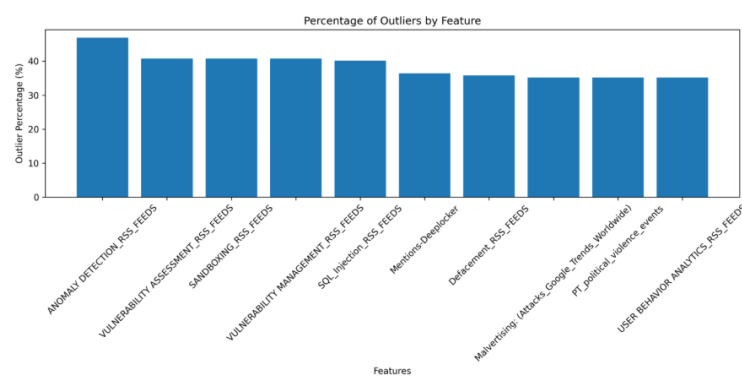


Figure 3: Feature Analysis by Combined Importance, Variance, Trend, Volatility and Outliers

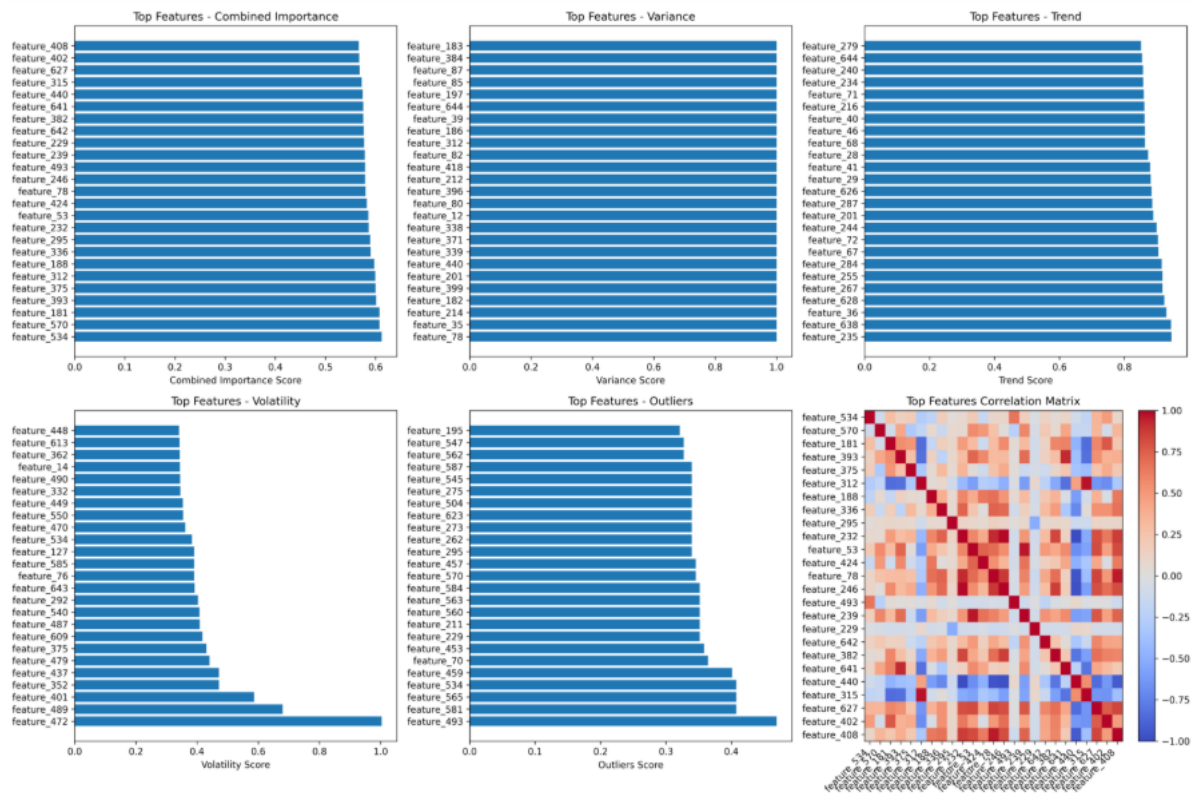


Figure 4: Feature Categories Distribution by Mutal Information Score

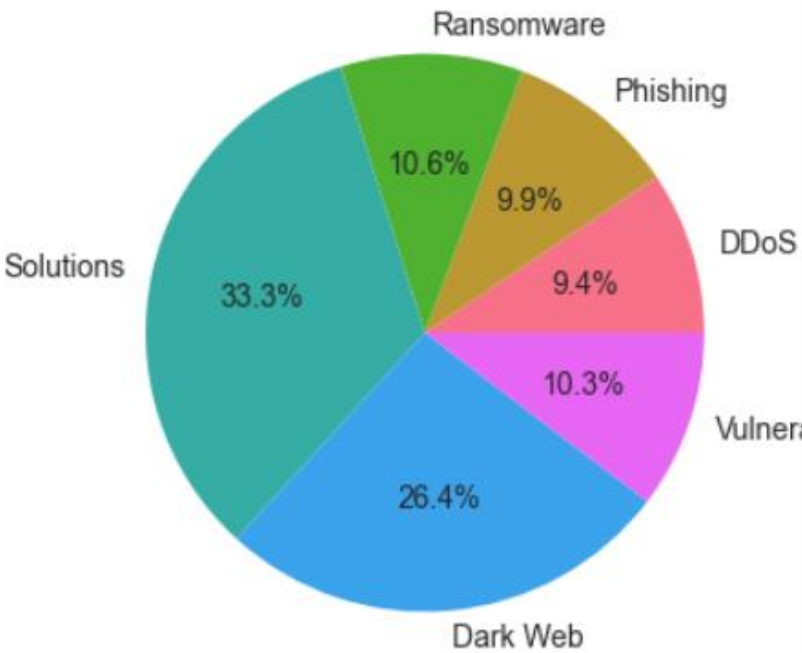


Figure 5: Distribution Plots of Six Key Features

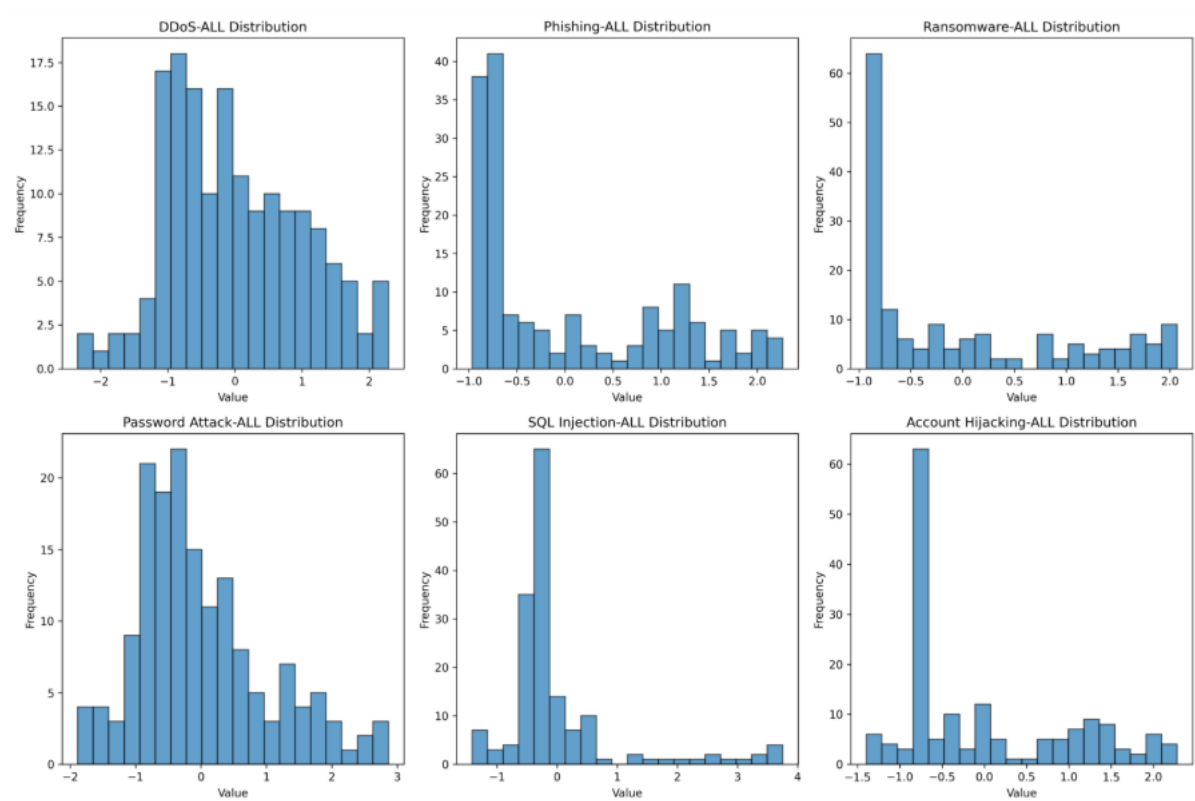
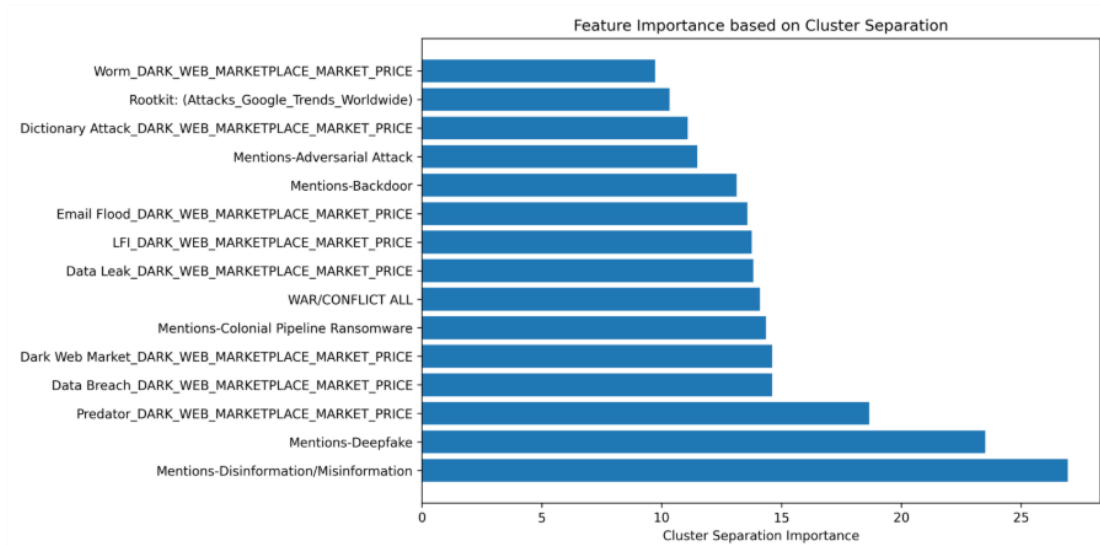


Figure 6: Feature Importance based on Cluster Separation

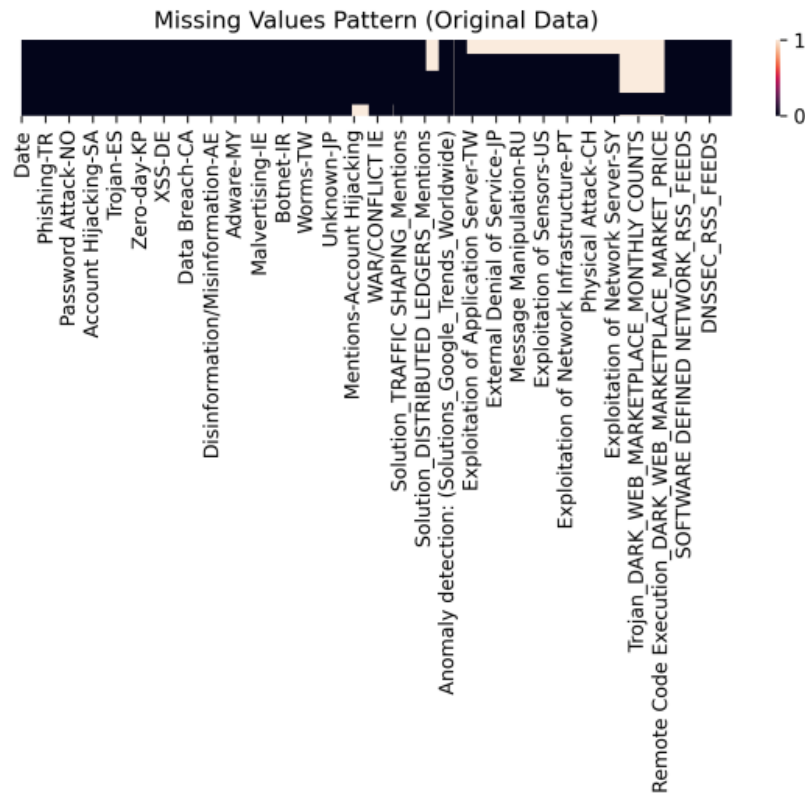


The high variance of some features and high levels of combined importance in Figure 3 suggests the data is information rich and there is a risk of the models overfitting. In Figure 4, the high mutual information of dark web data indicates that potential for high predictive power. Again, in Figure 6, the high feature importance for predominantly dark web features indicates a potential for strong influence on model predictions, suggesting that its inclusion would be beneficial to the model's predictive power.

IMPUTATION

The script applies a function for basic time series specific imputation: (1) forward fill, (2) backward fill, (3) linear interpolation, and (4) medians as a final fallback. Then, for values that were not imputed by the first function, KNearestNeighbours is applied. For market specific price data, forward fill is applied first, then backward fill, then the median of non-zero values. If there are no non-zero values, the overall median is applied. For country specific columns, consistency is enforced between country-specific columns and "ALL" aggregates. Temporal patterns are leveraged and time-based features created to make the logic more robust. Finally, the script provides comprehensive reporting on imputation quality, so that there are robust verification and validation protocols.

Figure 7: Missing Values Pattern Before Imputation



SMOTE

The SMOTE script supports a pipeline that detects imbalanced classes, which uses a threshold (that can be changed easily in the code) to determine which classes are imbalanced, by analysing the class ratio and imbalance ratio and returns list of threats needing SMOTE. This automatically selects relevant numerical features. Then, SMOTE is applied to the most imbalanced threats and the performance compared before and after applying SMOTE (using a Random Forest Classifier to evaluate performance). To maintain data integrity while adding balanced samples, a new dataset is created with synthetic samples of minority classes. Country-wise threat analysis, temporal analysis, and performance comparison across threats is then analysed to facilitate a decision as to whether the SMOTE augmented dataset is used.

SMOOTHING

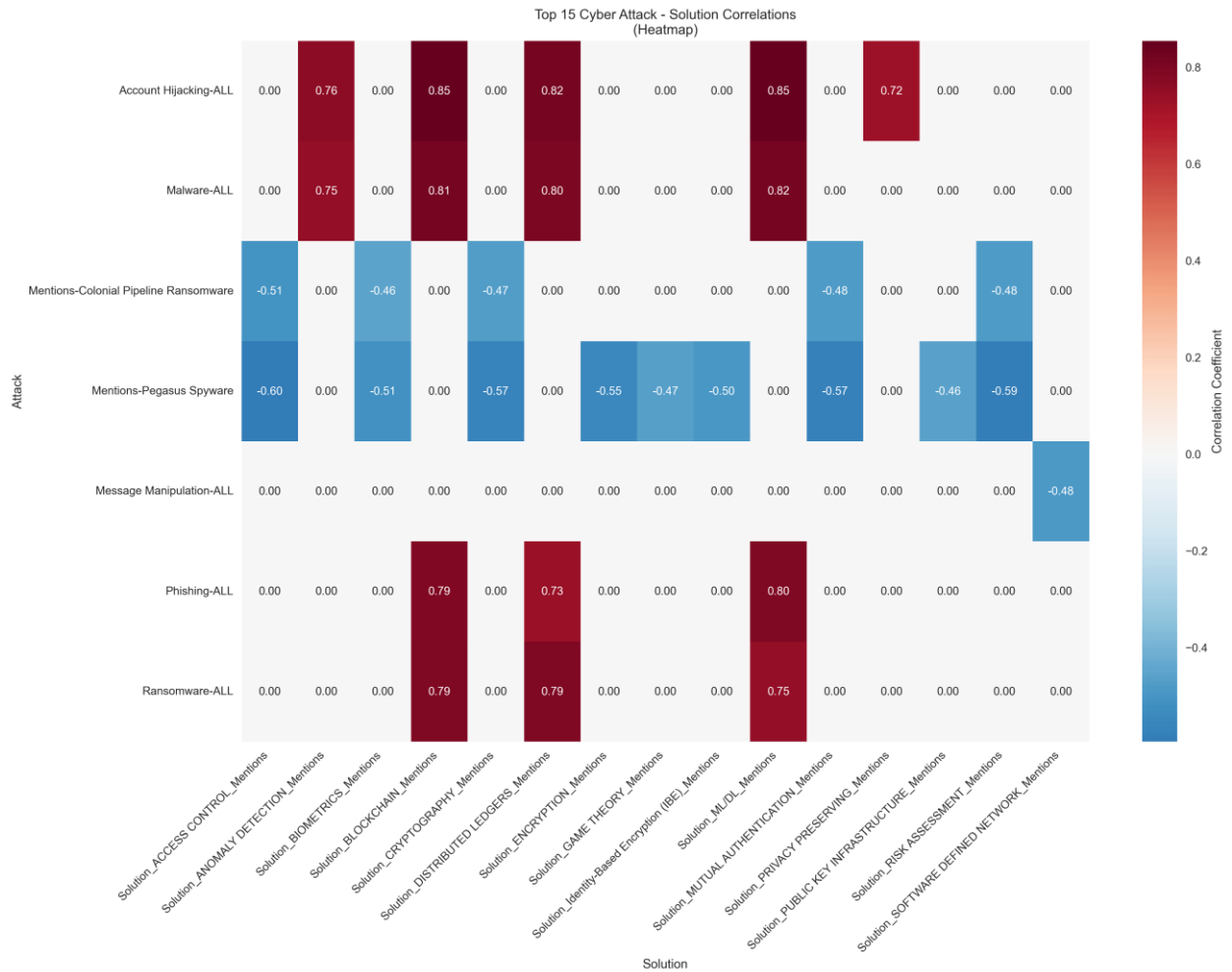
As exponential smoothing and double exponential smoothing are applied to the dataset used for Almahmoud's (Almahmoud, 2024) work, smoothing was applied to the dataset before ingestion, to ensure equal comparison to the baseline. This technique addresses several challenges inherent in cyber threat trend data. First, it facilitates noise reduction by filtering spurious fluctuations in threat observation frequencies. Second, trend preservation is enabled, maintaining underlying directional patterns whilst suppressing high-frequency variation. Finally, smoothing provides interpolated values for incomplete temporal records.

Graph Representation

The adjacency structure encoding relationships between attack vectors and pertinent technologies captures domain knowledge regarding which defensive technologies are

theoretically applicable to specific attack types, enabling the model to learn generalised patterns of threat-mitigation relationships. This graph has been updated to capture the new attack types offered by the University of Maryland Cyber Event Database, using both a python script for identifying correlations between attack vectors and technologies, and domain knowledge.

Figure 8: Attack to Solution Heatmap



3.3 Model Architectures

3.3.1 Transfer Learning

Due to the relatively limited scope of the period of the target data, each model architecture (excluding the agentic LLM and baseline B-MTGNN) has been pretrained on data from related domains, from the following sources:

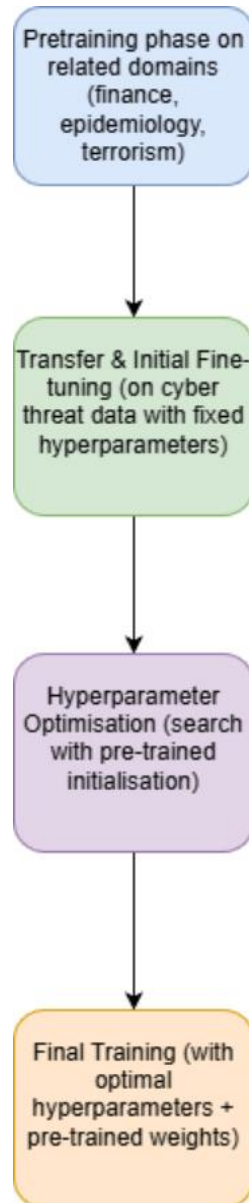
- Office for National Statistics (Office for National Statistics (ONS), 2025)
- Yahoo Finance stock data (*API Documentation — Yahoofinance Documentation*, 2025)
- John Hopkins global covid data (Dong et al., 2022)
- University of Maryland Global Terrorism data (University of Maryland, 2025)
- Federal Reserve Economic Data (U.S. Bureau of Labor Statistics., 2025)

These three domains (finance, epidemiology, and terrorism) were selected because they exhibit complimentary patterns which support transfer learning. Financial contagion data (i.e, black swan events, volatility of predictions) mirrors cyber threat propagation due to the inherent opaqueness of the domain, malware spreads in a similar way to epidemic viruses and global terrorism events are often directly linked to cyber threat attacks.

Transfer learning was chosen as a strategy because of the potential to

- increase computational efficiency by providing better initialisation, so hyperparameter search converges faster (fewer epochs needed per trial) (**Murel & Kavlakoglu, 2025**).
- allow for better hyperparameter search, as starting from pre-trained weights makes the search space smoother. (Vo et al., 2023)
- reduce the chance of getting stuck in poor local minima, leading to more stable training, as pre-trained models are less sensitive to hyperparameter choices, making optimisation more robust (Murel & Kavlakoglu, 2025).

Figure 9: Transfer Learning Pipeline



Each model was first pretrained on six csv files containing extensive data from the related domains. The weights learned were transferred and the model was fine-tuned to the target data. Then, hyperparameter optimisation took place and final training using the optimal hyperparameters and pre-trained weights. This strategy proved beneficial in terms of model performance. For instance, ViT showed a 9.1% improvement in validation loss as a direct result.

For all architectures discussed, random search was employed for hyperparameter optimisation, given the computational expense of training deep transformer models and the high-dimensional hyperparameter space, a random search approach balances exploration efficiency with resource constraints. The hyperparameter optimisation procedure includes validation-based selection, where hyperparameters are chosen based on performance against held-out validation data, Root Relative Squared Error (RRSE) and Relative Absolute Error (RAE) as joint optimisation objectives and performance evaluated across all nodes, with aggregate statistics informing hyperparameter selection. Training was deterministic to ensure a reproducible strategy.

The models employed a rigorous evaluation protocol, as in the original paper. In the validation phase, performance was assessed on temporally sequenced validation data, ensuring the model

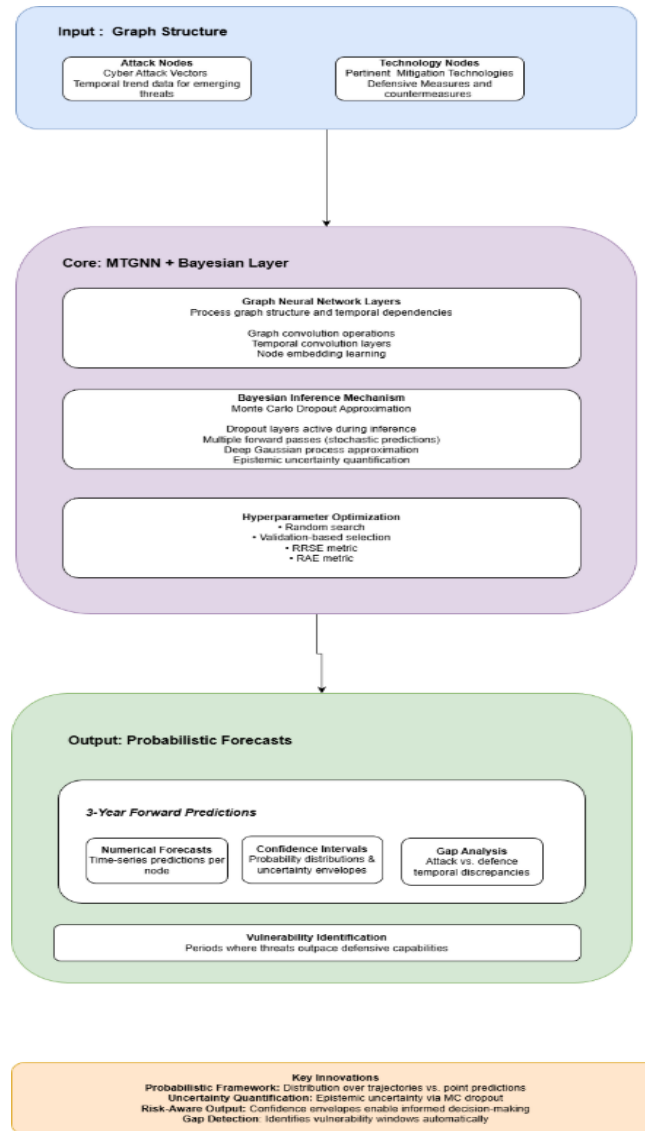
generalised to unseen time periods. In the testing phase, final performance was evaluated on held-out test data, with comprehensive visualisations comparing predicted trajectories against actual observations for each node. RSE and RAE provided complementary perspectives on prediction quality, with RSE penalising large deviations and RAE offering scale-invariant assessment. The operational model was then trained on the complete historical dataset using optimal hyperparameters identified during validation. This approach maximised the temporal coverage of training data, enhancing the model's capacity to capture long-term patterns in threat evolution.

The trained models produced three-year forward prediction across the entire graph. The output was composed of numerical forecasts (time-series predictions for each node), gap analysis (identification of projected temporal discrepancies between attack emergence and availability of pertinent mitigation technologies), where gaps represented periods of heightened vulnerability where threats outpace defensive capabilities, and visualisation (plotting of historical trends, forecasted trajectories, and confidence intervals). In these visualisations, gap periods were highlighted through distinct visuals, facilitating rapid identification of critical vulnerabilities.

3.3.2 **Bayesian MTGNN**

BMTGNN (Bayesian Multivariate Time-Series Graph Neural Network) provides a probabilistic framework for long-term cyber threat forecasting. Developed by Almahmoud (Almahmoud, 2024), the model extends the MTGNN architecture (Wu et al., 2020) through the incorporation of Bayesian inference mechanisms, enabling quantification of epistemic uncertainty in cyber threat predictions.

Figure 10: B-MTGNN System Architecture Diagram



B-MTGNN is the foundational architecture for this research, serving as the baseline model (when run on the same dataset as the original paper, updated to the end of 2024) and an architecture for legitimate comparison (when run on the full dataset). The B-MTGNN was selected as the baseline because, as the best performing model in the original paper, improving its performance was not only the logical point of extension, but also essential for an authentic contrast to state-of-the-art models. The baseline B-MTGNN, trained on a limited subset of features, operates on a graph-based representation of the cyber threat landscape, comprising of 124 nodes organised into two distinct categories: 26 attack nodes, which represent rapidly emerging and increasing cyber-attack vectors; and 98 pertinent technology nodes, which represent potential mitigation technologies and defensive measures.

The B-MTGNN with the full dataset operates 645 nodes, representing the attack nodes, pertinent technology nodes, dark web feature nodes, and other database enhancement nodes.

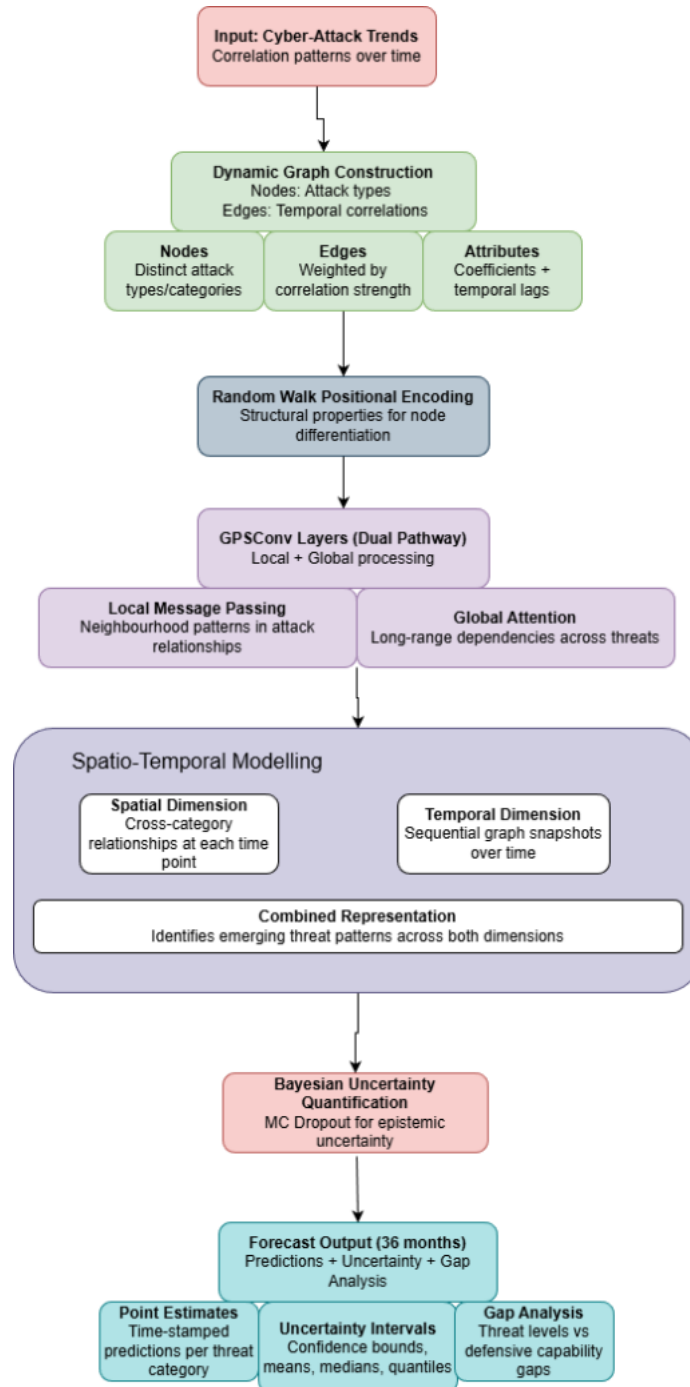
Each node encodes temporal trend data, transforming the forecasting challenge into a multivariate time-series problem embedded within a graph structure. This representation captures both the temporal evolution of individual threats and the structural dependencies between attacks and their corresponding defensive technologies.

The Bayesian treatment of uncertainty through Monte Carlo dropout approximation is advantageous to cyber threat forecasting. For instance, rather than producing deterministic point predictions, the model generates probability distributions over future threat trajectories. This is achieved by maintaining dropout layers active during inference, with multiple forward passes yielding a distribution of predictions that characterises model uncertainty. So, the output comprises not merely a predicted trend trajectory, but a confidence interval around predictions, enabling risk-aware decision-making in cybersecurity planning.

3.3.3 Graph Transformer

By combining the representational power of GPS-enabled graph transformers (Rampášek et al., 2023) with Bayesian uncertainty quantification, the graph transformer approach delivers actionable intelligence that supports proactive cybersecurity strategies. This implementation is based on a `pytorch_geometric` tutorial (PyG Team, 2025). The vanilla version has the same architecture without MC dropout.

Figure 11: Graph Transformer System Architecture Diagram



The architecture uses GPSConv (General, Powerful, Scalable Graph Transformer) layers, which combines local message passing, a traditional graph neural network mechanism for capturing neighbourhood-specific patterns in entity relationships, and global attention, transformer-style attention mechanisms, which could enable the model to identify long-range dependencies across the entire threat landscape. This dual-pathway design addresses a fundamental limitation in conventional graph neural networks, specifically the inability to efficiently model both local correlations (e.g., related attack vectors within a threat family) and global patterns (e.g., coordinated campaigns affecting multiple attack types simultaneously).

The graph structure is dynamically constructed from correlation patterns between cyber-attack trends, where nodes represent distinct cyber-attack types or threat categories, edges represent weighted connections based on temporal correlation strength between attack trends and edge

attributes encode correlation coefficients and temporal lag relationships. The maximum number of edges per node is pre-configured to prevent hub domination. This adaptive correlation-based approach might enable the model to learn both direct threat relationships and indirect dependencies arising from shared infrastructure or threat actor behaviour.

To enhance graph topology capabilities, random walk positional encoding is implemented, which encodes structural properties through random walk statistics on the graph. This enables the attention mechanism to distinguish between structurally similar but positionally distinct nodes, with the goal of improving the model's capacity to differentiate between attack types based on their position within the broader threat ecosystem.

For each node i , it performs random walks of length x :

$$PE_i = [p^1, p^2, \dots, p_x]$$

where p_i is the landing probability at step i , calculated through the normalised adjacency matrix

$$\tilde{A} = D^{-1}A$$

giving:

$$pe(v) = [\tilde{A}v, \tilde{A}^2v, \dots, \tilde{A}^xv]$$

(Huang et al., 2024)

Additionally, the architecture explicitly captures both spatial and temporal dimensions. The integration of spatial modelling, where the graph structure and GPSConv layers may capture relationships across different threat categories at each time point, and temporal modelling, where sequential processing of graph snapshots may capture the evolution of threats over time, allows for spatio-temporal representations which could enable the model to identify emerging threat patterns that manifest across both dimensions. This capability could be particularly valuable for detecting coordinated attack campaigns and understanding threat evolution in the context of the broader cyber landscape.

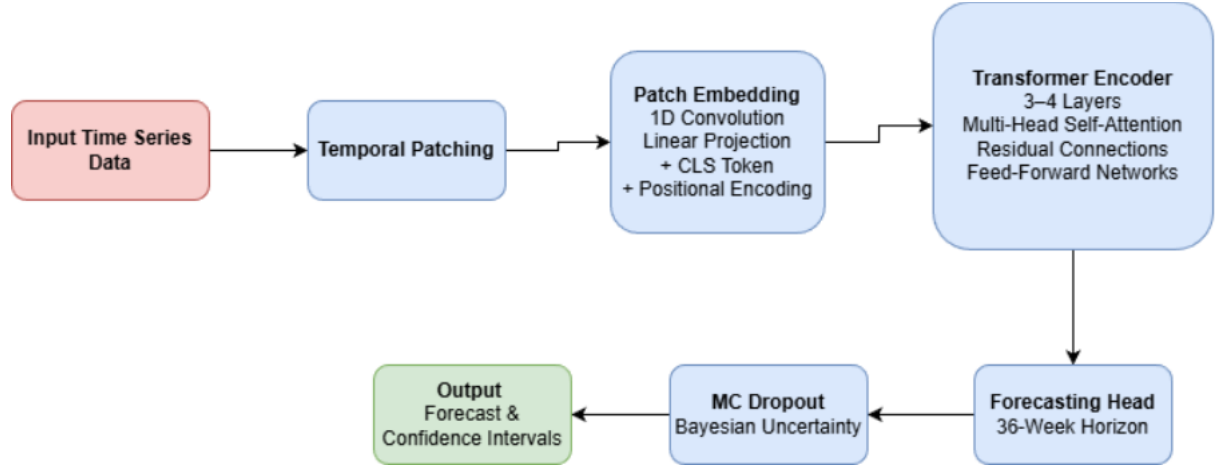
The graph transformer architecture also incorporates Bayesian approximation through Monte Carlo (MC) dropout to quantify epistemic uncertainty, providing confidence bounds around point forecasts to enable risk-aware decision-making. This has practical benefits for cybersecurity, as it facilitates dynamic resource allocation and allows professionals to prioritise based on decision confidence.

This graph transformer approach addresses several gaps in cyber threat forecasting literature. It is the first application of GPS-enabled graph transformers for cyber threat time-series forecasting. The integration of uncertainty quantification through Bayesian approximation tailored to cybersecurity operational requirements for graph transformers is also novel. These contributions help to enable a shift from reactive incident response to anticipatory defence posture, extend defender lead time, reduce threat actor advantages, support compliance with frameworks like DORA through transparent, quantified risk assessments and facilitate resource optimisation through informed and strategic allocation.

3.3.4 Vision Transformer (ViT)

The Vision Transformer (ViT) architecture, originally designed for computer vision tasks, is adapted for multivariate time series forecasting of cyber threats through treating temporal sequences like spatial patches in images (Dosovitskiy et al., 2020). This implementation, based on (Google, 2021), leverages self-attention mechanisms to capture long-range temporal dependencies whilst incorporating Bayesian uncertainty quantification through Monte Carlo dropout. The vanilla version has the same architecture without MC dropout.

Figure 12: Vision Transformer System Architecture Diagram



The temporal data is segmented into fixed-size patches rather than processing individual time steps. For a sequence length of 12 time steps with a patch size of 4, the input is divided into 3 patches. Each patch undergoes linear projection through a 1D convolutional layer (kernel size = patch size, stride = patch size) to transform multi-dimensional input into the embedding dimension space. Also, a learnable CLS token is added to the patch sequence, and positional encodings are added to preserve temporal ordering information. This approach enables the model to capture local temporal coherence within patches whilst facilitating global relationship modelling across patches.

Then, transformer encoder, which comprises stacked blocks, each containing multi-head self-attention layers computes attention in parallel.

Each transformer encoder layer applies scaled dot-product attention:

$$MultiHead(Q, K, V) = Concat(head1, \dots, headx)W_0$$

Where Q, K, V are query, key, and value matrices derived from linear projections. (GeeksforGeeks, 2025)

The multi-head architecture could be valuable for cyber threat forecasting, as different heads can specialise in capturing distinct temporal patterns: short-term tactical sequences, medium-term operational campaigns, long-term strategic trends, and cross-feature correlations between attack types and mitigation technologies.

Each complete encoder consists of L stacked transformer blocks made up of multi-head self-attention with residual connections; layer normalisation for training stability; position-wise feed-forward networks providing non-linear transformations and residual connections for preserving gradient flow. The final projection layer (the forecasting head) maps encoded representations to

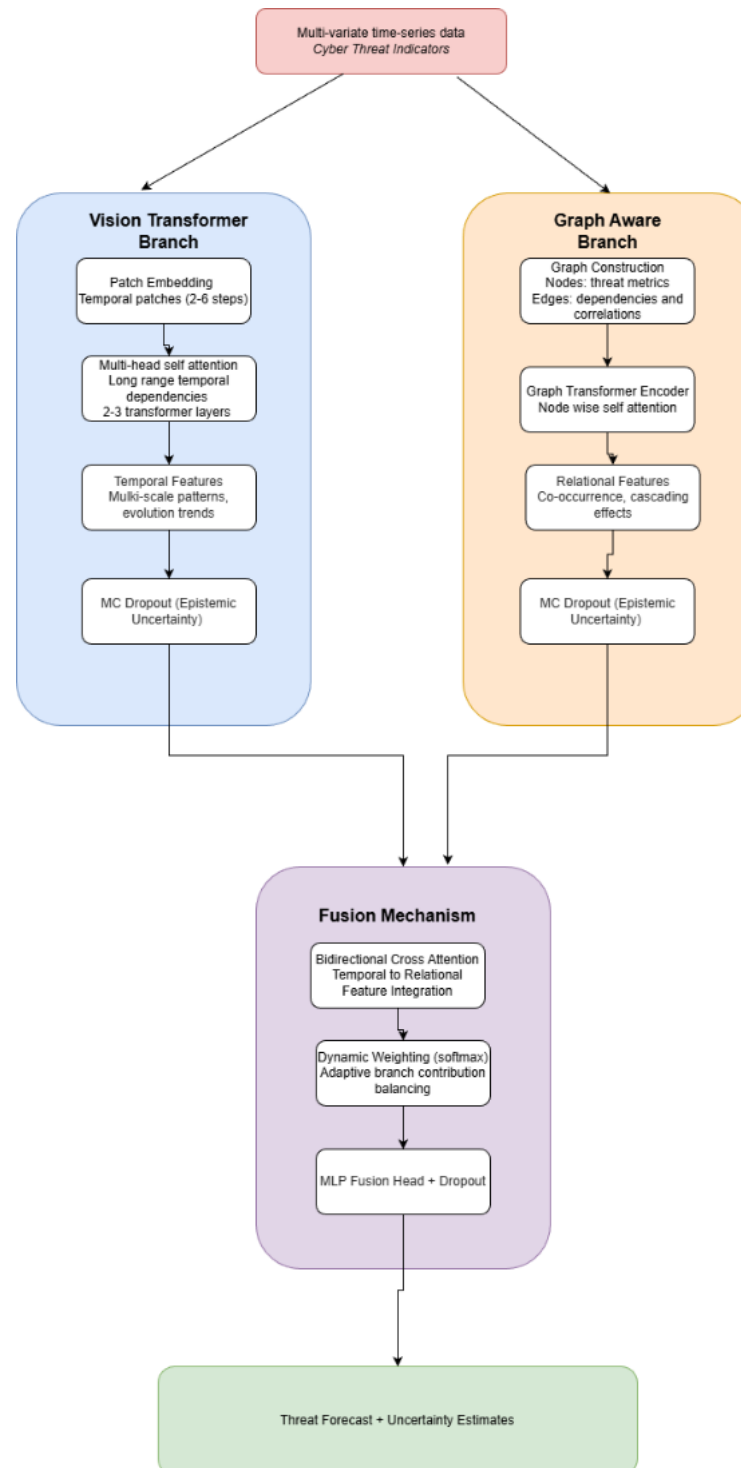
future threat predictions over a 36-week (three-year) horizon. The forecasting head is made up of a two-layer feed-forward network with ReLU activation and dropout regularisation.

Epistemic uncertainty is quantified through Monte Carlo dropout, implementing an approximate Bayesian inference framework. During inference, forward passes are performed with active dropout, generating an ensemble of predictions. The mean provides the point forecast, whilst the standard deviation yields calibrated confidence intervals. Higher uncertainty signals novel attack patterns, regime shifts in threat actor behaviour, or insufficient intelligence for threat vectors.

3.3.5 Ensemble Variant

This novel ensemble combines Vision Transformer and Graph Transformer architectures for long-term cyber threat forecasting. The hybrid model addresses a fundamental challenge in cybersecurity time-series analysis: capturing both temporal evolution patterns and relational dependencies amongst threat vectors simultaneously. By leveraging the strengths of each architecture, this ensemble provides strong predictive capability whilst maintaining computational efficiency and interpretability, critical requirements for operational threat intelligence systems.

Figure 13: Ensemble Variant System Architecture Diagram



The ensemble architecture employs a parallel dual-branch design, where each branch specialises in extracting distinct feature representations from the input threat data. This dual-modelling approach is particularly suited to cyber threat forecasting, where threats exhibit both temporal evolution (e.g., campaign lifecycles, seasonal variations) and complex interdependencies (e.g., threat actor relationships, attack chain correlations).

The Vision Transformer branch treats the multivariate time-series as a visual signal, partitioning the temporal sequence into non-overlapping patches. Each patch undergoes linear embedding before being processed through multi-head self-attention layers. This patch-based approach enables the model to capture long-range temporal dependencies and multi-scale patterns

inherent in cyber threat evolution. The architecture employs configurable parameters including patch size, embedding dimensions, and transformer depth, allowing adaptation to varying forecast horizons.

The Graph Aware branch models the relational structure amongst threat indicators, treating the multivariate time-series as a dynamic graph where nodes represent individual threat metrics and edges encode their dependencies. This uses a simplified graph encoder based on transformer layers, which applies self-attention across the node dimension to capture correlation patterns.

The implementation employs a specialised fusion strategy. Cross-attention integration allows the model to learn which temporal patterns are most relevant for specific relational structures and vice versa. This bi-directional information flow enriches the feature representation beyond what either branch could achieve independently. Through dynamic weighting, a learnable parameter vector is applied to softmax-normalised weights to balance contributions from each branch. This adaptive mechanism allows the model to emphasise different branches based on the characteristics of the input data, providing robustness across varying threat scenarios. Finally, an enhanced fusion head lets the combined features pass through a multi-layer perceptron with progressive dimensionality reduction. This non-linear transformation enables the model to learn complex interactions between temporal and relational features, whilst dropout regularisation prevents overfitting.

Both branches employ Monte Carlo dropout during inference, where dropout masks are sampled multiple times to generate prediction distributions. This technique provides computationally efficient Bayesian approximation, enabling confidence interval estimation for each forecast.

This ensemble architecture is well suited to integrating dark web intelligence and extending forecast lead time because the architecture leverages both temporal patterns and relational structures within the threat landscape, whilst incorporating Bayesian uncertainty quantification to support proactive cybersecurity decision-making. The Vision Transformer branch has the potential to capture temporal patterns in both historical CVE data and early indicators from dark web forums, identifying lead-lag relationships that extend prediction horizons. The Graph-Aware branch can model correlations between traditional threat metrics and emerging dark web signals, enabling the system to learn which early indicators are most predictive of specific threat types. Moreover, the dynamic fusion mechanism may allow the model to adaptively weight Dark Web Intelligence features based on their relevance to the current threat landscape, supporting the transition from reactive to proactive cybersecurity postures.

This combination of features captures architectural advantages from both the ViT and Graph Transformer to deliver accuracy improvements whilst remaining operationally viable.

3.3.6 **Agentic LLM**

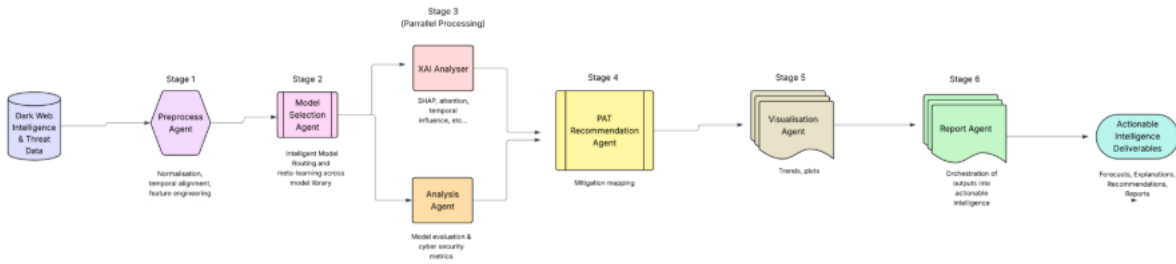
The agentic LLM system is based on TimeSeriesScientist, specialised to the domain of cyber threat forecasting with specific augmentations (e.g. XAI and PAT sub-agents and optional RAG³ capabilities to ensure reliable outputs) that prove useful in the target domain. The model is run through Ollama, using a local instance of deepseek r1:8b to avoid rate limiting. Deepseek r1:8b

³ the RAG knowledge base includes NVD/CVE metadata going back to 1999 and the MITRE ATT&CK framework.

was selected for reasoning and inference capabilities as well as outstanding performance compared to benchmarks (Ollama, 2025).

By using a modular agentic architecture, the system employs the collaboration of LLM powered agents to deliver end-to-end threat analysis, mitigation recommendations, and XAI plots. The system is comprised of seven distinct agents, with the ModelSelectionAgent at the core, each responsible for key aspects of the forecasting pipeline.

Figure 14: Agentic LLM System Architecture Diagram



The LLM is prompted to provide highly detailed and verbose reasoning for each choice at each stage to further enhance the interpretability of the system.

First, the PreprocessAgent is the data ingestion and normalisation layer, delivering feature engineering to derive relevant features, rolling statistics to characterise threat evolution, and quality assurance through anomaly detection, flagging data quality issues (i.e, missing values, outliers) and completeness verification to ensure sufficient data density across strata.

Then, the ModelSelectionAgent implements an LLM-informed dynamic selection of forecasting architectures based on data characteristics (passed forward by PreprocessAgent), enabling adaptive model selection through meta-learning. The LLM reasoning engine contextually deliberates over data characteristics and model suitability, generating justified recommendations and detailed reasoning. The model selection framework maintains a comprehensive model repository and coordinates parallel evaluation when multiple architectures are deemed appropriate. The model repository contains sixteen distinct forecasting models including, classical statistical methods, machine learning regressors, deep learning architectures, and Graph and Vision-Based models, which allows both systematic architectural comparisons based on the data qualities and operational robustness. The modular design means that new models can be integrated, along with new selection heuristics.

Code Snippet 2: Selection Rationale for ModelSelectionAgent

```
# Score based on model characteristics and data characteristics
if model == 'ARIMA' and is_stationary:
    score += 10
elif model == 'ARIMA' and has_trend:
    score += 8
elif model == 'SARIMA' and has_seasonality:
    score += 9
elif model == 'Prophet' and (has_trend or has_seasonality):
    score += 8
elif model in ['LSTM', 'VisionTransformer', 'SimpleVisionTransformer']:
    score += 7 # High versatility for complex patterns
elif model in ['RandomForest', 'GradientBoosting']:
    score += 6 # Good for non-linear relationships
elif model == 'LinearRegression' and has_trend:
    score += 5
elif model == 'ExponentialSmoothing' and has_seasonality:
    score += 7
elif model in ['BMTGNN', 'GraphTransformer', 'SimpleGraphTransformer', 'SpatioTemporalEnsemble', 'VisionTransformerForTimeSeries']:
    score += 8 # Advanced models for complex patterns
else:
    score += 3 # Base score

return score
```

After model selection and forecast generation, the ModelSelectionAgent communicates results to the downstream agents AnalysisAgent, XAIAnalyser, VisualiationAgent and PATRecommendationAgent.

- AnalysisAgent receives raw forecasts (with uncertainty quantification) for statistical validation, anomaly detection, and trend analysis to enhance prediction reliability, contextualise forecasts in historical threat landscapes, and support model comparison.
- XAIAnalyser provides model agnostic post-hoc interpretability to ensure transparency and actionability of model predictions, such as feature importance analysis, attention visualisations, counterfactuals, confidence assessments, faithfulness checks and stability analysis.
- VisualisationAgent receives forecast outputs to transform analytical outputs into visualisations, facilitating robust reporting.
- PATRecommendationAgent translates probabilistic threat forecasts into defensive recommendations with LLM-powered semantic matching for novel or evolving threat patterns.

Finally, the ReportAgent is responsible for orchestration and synthesis of analytical outputs into actionable intelligence reports, using the LLM for narrative generation and technical writing.

3.4 XAI Integration

To enhance interpretability, each model undergoes a sophisticated suite of XAI analysis techniques. This serves to improve the transparency and trustworthiness of operational outputs. In tandem with forecasting, comprehensive XAI evaluation is undergone by the models to validate the fidelity, consistency, and actionability of the outputs.

Code Snippet 3: XAI analysis methods listed with descriptions

```
'analysis_types': {  
    'prediction_analysis': 'Prediction vs actual statistics and error analysis',  
    'prediction_uncertainty_analysis': 'MC dropout uncertainty estimation',  
    'permutation_feature_importance': 'Feature importance via permutation',  
    'attention_visualization': 'Transformer attention heatmaps',  
    'temporal_attention_analysis': 'ViT branch temporal patterns',  
    'shap_analysis': 'SHAP global and local feature importance',  
    'lime_analysis': 'LIME local explanations',  
    'causal_analysis': 'Bayesian network causal relationships',  
    'counterfactual_analysis': 'Minimal changes for target outcomes',  
    'dice_counterfactual_analysis': 'DiCE counterfactual explanations',  
    'dynamic_weight_interpretation': 'Temporal vs relational branch dominance',  
    'branch_specific_explanations': 'Modality-specific explanation decomposition',  
    'explanation_faithfulness_metrics': 'Quantitative validation of explanation reliability',  
    'adversarial_robustness_explanations': 'Identification of adversarial vulnerabilities'  
}
```

The XAI framework considers both global and local explanations, as well as causal and uncertainty-based methods to offer detailed insight into the models' processes.

3.5 Model Evaluation

The evaluation process for all models involves a rigorous combination of measuring against both established and custom metrics. Established metrics include accuracy metrics, like RRSE and RAE, and detection quality metrics, like AUC-ROC and False Positive Rate.

Code Snippet 4: Error metrics and detection metrics listed

```
# Group metrics by category  
error_metrics = {k: v for k, v in self.metrics.items()  
                 if 'MAE' in k or 'RMSE' in k or 'MAPE' in k or 'SMAPE' in k}  
rel_error_metrics = {k: v for k, v in self.metrics.items()  
                    if 'RRSE' in k or 'RAE' in k}  
detection_metrics = {k: v for k, v in self.metrics.items()  
                    if 'AUC' in k or 'FPR' in k or 'Coverage' in k or 'Precision' in k or 'Operational_Readiness' in k or 'Adversarial_Robustness' in k}
```

Custom cybersecurity specific metrics include measuring an operational readiness index to assess the model's readiness for deployment, adversarial robustness score to show resilience to malicious perturbations, directional accuracy to determine how often the model is correct about the trend of attacks, quantile loss showing uncertainty quantification at different probability levels, and Jensen-Shannon divergence for detecting data drift. The combination of these metrics gives a robust view of model performance against a gamut of metrics relevant to cybersecurity and operational deployment challenges.

Code Snippet 5: Formula for calculating operational readiness index

```
def calculate_operational_readiness_index(self, precision, recall, fpr, resource_efficiency):  
    """Calculate composite Operational Readiness Index"""  
    # Normalise metrics to 0-1 scale  
    precision_norm = precision  
    recall_norm = recall  
    fpr_norm = 1 - fpr # Lower FPR is better  
    resource_norm = min(resource_efficiency, 1.0)  
  
    # Weighted combination  
    ori = (0.3 * precision_norm + 0.3 * recall_norm +  
          0.2 * fpr_norm + 0.2 * resource_norm)  
  
    return ori
```

4 Implementation

This section describes the technical details of the research. First, the technologies and Python libraries used are detailed. Then, model specific design choices are explained. Finally, challenges faced and how they were overcome are discussed.

4.1 Tools

To enforce reproducibility, the GitHub repo and appendix contain a DockerImage, a compose.yaml, a requirements.txt, and an environment.yml. Throughout the project, matplotlib, seaborn, plotly were used for visualisations and numpy and pandas for data manipulation.

In the data collection stage, requests for HTTP requests; holidays for public holiday data collection; BeautifulSoup4 for HTML parsing; selenium for web browser automation; webdriver-manager for automatic Selenium webdriver management through Firefox and elsapy for the Elsevier API client. Additionally, for dark web data collection, subprocess was used to construct a firewall for system hardening; stem and Tor Manager for anonymous connection to the dark web; and urllib.parse to validate .onion web addresses.

At the data preparation stage, sklearn was used for IterativeImputer and KNNImputer for time series aware imputation and imblearn was used for SMOTE resampling.

All the models are built in pytorch. The graph transformer models are built with torch_geometric also (specifically GINEConv, GPSConv, global_add_pool). In all models, the random package is used to set a random seed for reproducibility, and the seed is set as deterministic.

Code Snippet 6: Function for setting random seed

```
# Set random seeds for reproducibility
def set_random_seed(seed):
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)
    torch.backends.cudnn.deterministic = True
    torch.backends.cudnn.benchmark = False
```

Pickle files were used to store model_state_dicts. For ViT, ensemble, and graph transformers, ReLU was used to introduce non-linearity. For the B-MTGNN the engineering choices reflect the original paper for equal comparison.

The Agentic LLM uses statsmodels to import ARIMA, ExponentialSmoothing and SARIMAX; RandomForestRegressor and LinearRegression are imported from sklearn; the prophet model is imported from the prophet library; LLM interface libraries are imported from langchain (like ChatOpenAI and ChatAnthropic) but the current system is hosted on Ollama. FAISS and HuggingFaceVectorEmbeddings are used for building a knowledge base for RAGSystem.

For evaluation, roc_curve is imported from sklearn. Jensen-Shannon divergence and Kolmogorov-Smirnov tests are calculated with scipy.

Finally, the XAI scripts were built with networkx for network analysis, shap, DiCE and lime for core XAI plotting and pgmpy for Bayesian Network analysis.

4.2 Model Specific Design Choices

For all models, a wide hyperparameter search space is employed to capture a comprehensive search and account for a diversity of scenarios.

4.2.1 ViT

ViT uses AdamW optimiser with decoupled weight decay because this separates weight decay from gradient updates, preventing regularisation from interfering with adaptive learning rates. It handles sparse gradients effectively (crucial for attention mechanisms) so weight decay acts as true L2 regularisation, improving generalisation. The learning rate scheduler is ReduceLROnPlateau because this allows the model to fine-tune at optimal pace without predetermined milestones and the patience (up to 10 epochs) prevents premature convergence. Regularisation takes a multi-level approach (dropout, MC dropout, weight decay, and gradient clipping) to prevent co-adaptation of attention heads and ensure training stability with variable sequence lengths. Early stopping is configured to prevent stopping due to floating-point precision artifacts and dynamic patience. This avoids both premature stopping and excessive training.

Three-phase adaptive sampling is employed for dynamic adaptation to new best configurations without re-running experiments. The first phase is broad exploration, the second focused refinement, and the third is fine-tuning. This allows for the preservation of universal patterns for optimal transfer learning.

Code Snippet 7: Adaptive Sampling Implementation

```
# Adaptive sampling strategy
best_configs_history = []

for iteration in range(args.iterations):
    print(f"\n=== Iteration {iteration + 1}/{args.iterations} ===")

    # Adaptive sampling based on iteration phase
    if iteration < 20:
        # Phase 1: Broad exploration
        hp = sample_hyperparameters(search_space, validation_rules, data)
    elif iteration < 40:
        # Phase 2: Focused refinement - sample near best-performing configs
        if best_configs_history:
            # Use best config as base and perturb
            base_hp = random.choice(best_configs_history[-5:]) # Use recent best configs
            hp = perturb_hyperparameters(base_hp, search_space, validation_rules, data)
        else:
            hp = sample_hyperparameters(search_space, validation_rules, data)
    else:
        # Phase 3: Fine-tuning - narrow ranges around optimal architecture
        if best_hp is not None:
            hp = fine_tune_hyperparameters(best_hp, search_space, validation_rules, data)
        else:
            hp = sample_hyperparameters(search_space, validation_rules, data)

    print(f"Hyperparameters: {hp}")
```

Table 5: Vision Transformer hyperparameter space

Hyperparameter	Search Range	Default
Architecture Parameters		
sequence length	[6, 48]	12
patch size	[2, 6]	4
Embedded dimensions	[64, 256]	128

Number of heads	[4, 16]	8
hidden dimensions	[128, 512]	256
Number of layers	[2, 6]	3
Regularisation Parameters		
dropout	[0.0, 0.3]	0.1
MC dropout	[0.0, 0.4]	0.2
weight decay	[1e-5, 1e-1]	1e-2
Gradient clipping value	[0.5, 2.0]	1.0
Label smoothing	[0.0, 0.2]	0.1
Optimisation Parameters		
learning rate	[1e-5, 1e-2]	1e-3
batch size	[4, 64]	8
scheduler patience	[3, 10]	5
scheduler factor	[0.3, 0.7]	0.5
Early stopping patience	[5, 20]	10
Positional Encoding		
Positional encoding type	['learned', 'sinusoidal']	'learned'
Uncertainty Quantification		
MC inference runs	[5, 50]	20
Activation Function		
Activation function	['relu', 'gelu', 'silu']	'gelu'
Fixed/Derived Parameters		
Forecast horizon	From data	3
Number of nodes	From data	Varies

4.2.2 Graph Transformers

The graph transformers use AdamW for decoupled weight decay regularisation, for its stable convergence and better weight decay handling, which is crucial when fine-tuning pre-trained representations. They also support SGD with momentum as a fallback. Cosine is the default learning rate scheduler because it balances exploration (periodic learning rate increases) with exploitation (annealing), critical for transfer learning where the model needs to both preserve pre-

trained features and adapt to new data. Multi-level regularisation (dropout, attention dropout, weight decay, gradient clipping) addresses different failure modes where dropout prevents feature co-adaptation, weight decay prevents parameter norm explosion, and gradient clipping ensures numerical stability.

Early stopping involves a two-tier patience system where trivial improvements are ignored to avoid stopping on noise; then the globally best model is saved, rather than just recent improvements. Also, extended patience gives the model extra epochs after initial patience expires; and loss-based stopping monitors training loss to detect divergence. Extended patience allows recovery from temporary validation degradation and prevents premature stopping during the adaptation phase, where transferred features are being fine-tuned, which avoids underfitting when model needs time to forget pre-trained task-specific features.

Differential learning rates are implemented through a three-tier strategy, where transferred layers have 10x reduced learning rate to preserve pre-trained features, and new layers have the full learning rate for rapid adaptation with separate parameter groups in the optimiser. This allows for feature preservation, by preventing catastrophic forgetting of pre-trained representations; selective fine-tuning as new layers learn task-specific patterns while transferred layers make minor adjustments.

Code Snippet 8: Function for differential learning

```
def create_optimizer_with_differential_lr(model, config, transferred_layers):
    """Create optimiser with different learning rates for transferred vs new layers"""
    base_lr = config['learning_rate']

    # Parameters for transferred layers (lower learning rate)
    transferred_params = []
    # Parameters for new layers (higher learning rate)
    new_params = []

    for name, param in model.named_parameters():
        if any(transferred_layer in name for transferred_layer in transferred_layers):
            transferred_params.append(param)
        else:
            new_params.append(param)

    print(f"Transferred parameters: {len(transferred_params)}, New parameters: {len(new_params)}")

    # Use lower learning rate for transferred layers, higher for new layers
    optimizer = optim.AdamW([
        {'params': transferred_params, 'lr': base_lr * 0.1}, # Lower LR for fine-tuning
        {'params': new_params, 'lr': base_lr} # Normal LR for new layers
    ], weight_decay=config['weight_decay'])

    # Store initial LR for warmup
    for param_group in optimizer.param_groups:
        param_group['initial_lr'] = param_group['lr']

    return optimizer
```

The robust training pipeline automatically falls back to default hyperparameters when failures occur, implements detailed logging of each optimisation iteration and handles inf/nan losses. The flexible architecture supports 3 GNN types (GINE, GCN, GAT) and multiple activation functions (ReLU, GELU, SiLU).

Table 6: Graph Transformer Hyperparameter Space

Category	Parameter	Search Range	Default
Architecture	Channels	[32, 64, 128]	64
	Number layers	[2, 3, 4, 5]	3
	Number of heads	[2, 4, 8]	4
	Head hidden dimensions	[[64,32], [128,64], [256,128]]	[128,64]
	Postional encoding dimensions	[4, 8, 16]	8
	Positional encoding walk length	[10, 20, 30]	20
Attention	Attention type	['multihead', 'performer']	'multihead'

	Attention dropout	[0.0, 0.1, 0.2, 0.3]	0.2
	Performer number of features	[64, 128, 256]	128
	Performer redraw intervals	[500, 1000, 2000]	1000
Regularisation	dropout	[0.1, 0.2, 0.3, 0.4, 0.5]	0.3
	Weight decay	[1e-5, 5e-5, 1e-4, 5e-4, 1e-3]	1e-4
	Gradient clip	[0.5, 1.0, 2.0]	1.0
	Label smoothing	[0.0, 0.05, 0.1]	0.1
Optimisation	Learning rate	[1e-4, 5e-4, 1e-3, 5e-3]	5e-4
	Optimiser type	['adamw', 'sgd']	'adamw'
	Learning rate scheduler	['cosine', 'plateau', 'onecycle']	'cosine'
	Warmup epochs	[5, 10, 20]	10
	Batch size	[4, 8, 16, 32]	8
	Transferred learning rate multiples	[0.01, 0.1, 0.5]	0.1
Graph Construction	Correlation threshold	[0.1, 0.2, 0.3, 0.4, 0.5]	0.3
	Max edges per node	[5, 10, 15, 20]	15
Task-Specific	Sequence length	[12, 18, 24]	18
	forecast horizon	[12, 24, 36]	36
GNN Type	Local gnn type	['GINE', 'GCN', 'GAT']	'GINE'
	Global model type	['Transformer', 'None']	'Transformer'
Normalisation	Normalisation type	['batch', 'layer', 'instance']	'batch'

Activation	Head activation	['relu', 'gelu', 'silu']	'relu'
Early Stopping	Patience	[15, 20, 30]	20
	Minimum delta	[1e-5, 1e-4, 1e-3]	1e-4
Uncertainty	MC samples	[10, 20, 50]	20
Training	Use amp	[True, False]	False
	epochs	Fixed at runtime	100

4.2.3 Ensemble

The ensemble chooses between adam and adamw for an optimiser at the hyperparameter stage, with adamw as the default because the decoupled weight decay prevents over-regularisation of adaptive learning rates, crucial for the ensemble architecture where different branches may require different regularisation strengths. The learning weights are in a conservative, lower range to prevent catastrophic forgetting and ensure stable gradient flow through cross-attention fusion. Dropout is applied at the transformer encoder layers, forecast head intermediate layers and graph-aware encoder to prevent co-adaptation between ensemble branches while maintaining representation capacity for temporal patterns. L2 regularisation penalises large weights in AdamW's decoupled form to prevent overfitting the training period while allowing the model to capture genuine temporal dynamics.

The learning rate scheduler (ReduceLROnPlateau) is adaptive to convergence and considers multiple branches simultaneously (plateau detection allows slower branches to catch up), enabling better convergence.

Code Snippet 9: Learning Rate Scheduler Implementation

```
# Configure optimiser based on hyperparameters
if hp['optimizer_type'] == 'adamw':
    optimizer = optim.AdamW(model.parameters(), lr=hp['learning_rate'],
                             weight_decay=hp['weight_decay'])
else:
    optimizer = optim.Adam(model.parameters(), lr=hp['learning_rate'],
                             weight_decay=hp['weight_decay'])

scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer,
                                                  patience=hp['scheduler_patience'],
                                                  factor=hp['scheduler_factor'])
```

Adaptive gradient clipping adjusts the clip threshold based on current gradient magnitude because different branches produce gradients of different magnitudes, so adaptive clipping prevents one branch from dominating.

Code Snippet 10: Adaptive Gradient Clipping Implementation

```
# Monitor gradients
total_norm = 0
for p in model.parameters():
    if p.grad is not None:
        param_norm = p.grad.data.norm(2)
        total_norm += param_norm.item() ** 2
total_norm = total_norm ** 0.5

# Adaptive gradient clipping
clip_value = max(0.1, min(1.0, total_norm))
torch.nn.utils.clip_grad_norm_(model.parameters(), clip_value)
```

Early stopping with a patience of 10 epochs (using RRSE) allows the model to recover from temporary plateaus. Early stopping is also applied per hyperparameter iteration to prevent wasting compute on poor hyperparameter choices while allowing good configurations to fully converge. A dynamic weighting mechanism is applied to allow for learnable branch weighting, so the model can adaptively balance temporal (ViT branch) and relational (graph branch) features based on data characteristics.

Code Snippet 11: Dynamic Weighting Implementation

```
# Dynamic weighting
weights = torch.softmax(self.attention_weights, dim=0)
weighted_vit = weights[0] * vit_cls
weighted_graph = weights[1] * graph_global

# Feature fusion with cross-attention
fused_features = torch.cat([weighted_vit, weighted_graph], dim=1)
```

Table 7: Ensemble Model Hyperparameter Space

Category	Hyperparameter	Search Space	Default
Temporal Configuration			
Input Window	Sequence length	[12, 24, 36]	12
Output Horizon	Forecast horizon	[36]	36 (fixed)

Vision Transformer Branch			
Patch Size	ViT patch size	[2, 3, 4, 6]	4
Embedding Dim	ViTembed dimensions	[64, 96, 128, 160]	128
Attention Heads	ViT number of heads	[4, 8]	8
Hidden Dim	ViT hidden dimensions	[128, 192, 256, 320]	256
Encoder Depth	ViT number of layers	[2, 3, 4]	3
Graph-Aware Branch			
Hidden Dim	Graph hidden dimensions	[64, 96, 128, 160]	128
Encoder Depth	Graph number of layers	[1, 2, 3]	3
Fusion Mechanism			
Fusion Dimensions	Fusion dimensions	[128, 192, 256, 320]	256
Regularisation			
Standard Dropout	dropout	[0.1, 0.2, 0.3]	0.2
MC Dropout	MC dropout	[0.1, 0.2, 0.3, 0.4]	0.2
Weight Decay	Weight decay	[0.001, 0.01, 0.05]	0.01
Ensemble Decay	Ensemble weight decay	[0.0, 0.001, 0.01]	0.01
Optimisation			
Learning Rate	Learning rate	[0.0001, 0.0003, 0.0005, 0.001, 0.002]	0.001
Optimizer Type	Optimiser type	['adamw', 'adam']	adamw
Batch Size	Batch size	[4, 8, 16]	8
Gradient Clip	Gradient clipping	[0.5, 1.0, 2.0]	1.0
Learning Rate Scheduler			
Scheduler Patience	Scheduler patience	[5, 10, 15]	10
Scheduler Factor	Scheduler factor	[0.3, 0.5, 0.7]	0.5
Ensemble Configuration			
MC Samples	Number of MC samples	[10, 15, 20]	15

4.2.4 Agentic LLM

The agentic LLM imposes temperature 0.1 to maximise reproducibility, to reduce the likelihood of agent hallucination, and to introduce an element of determinism. The system supports several providers.

Code Snippet 12: Function for Initialiasing the LLM Provider

```
def _initialize_llm(self):
    """Initialize LLM based on provider"""
    if self.config.llm_provider == "ollama":
        print(f"LLM initialized successfully with model: {self.config.llm_model}")
        return Ollama(
            model=self.config.llm_model,
            base_url=self.config.base_url or "http://localhost:11434",
            temperature=0.1
        )
    else:
        raise ValueError(f"Unsupported LLM provider: {self.config.llm_provider}")

    """elif self.config.llm_provider == "openai":
        if self.config.api_key:
            openai.api_key = self.config.api_key
            return ChatOpenAI(
                model_name=self.config.llm_model,
                temperature=0.1,
                max_tokens=4000
            )
        elif self.config.llm_provider == "anthropic":
            return ChatAnthropic(
                model=self.config.llm_model,
                temperature=0.1,
                max_tokens=4000
            )
    """
```

To ensure modularity and flexibility, a config class controls the configuration for the LLM, data, validation and testing split, agents, XAI, and RAG, as well as the models introduced to the system and the preprocessing outlier detection field.

Code Snippet 13: Forecast Configuration Class

```
class ForecastConfig:
    """Configuration for the forecasting system"""
    # LLM Configuration
    llm_provider: str = "ollama"
    llm_model: str = "deepseek-r1:8b"
    api_key: Optional[str] = None
    base_url: Optional[str] = "http://localhost:11434"

    # Data Configuration
    data_path: str = "data/sm_data_g.csv"
    graph_path: str = "data/graph.csv"
    forecast_months: int = 36
    target_columns: List[str] = None

    # Model Configuration
    validation_split: float = 0.8
    seasonal_period: int = 12

    # Output Configuration
    output_dir: str = "forecast_results"
    plot_style: str = "default"

    # Agent Configuration
    enable_agents: bool = True
    max_retries: int = 3

    # XAI Configuration
    enable_xai: bool = True
    xai_methods: List[str] = None

    # RAG Configuration
    enable_rag: bool = False
    rag_db_path: str = "cyber_threat_db"

    # Report Configuration
    enable_report: bool = True
    report_path: str = "forecast_report.pdf"

    # New Models
    enable_lstm: bool = True
    enable_prophet: bool = True
    enable_validation: bool = True
    validation_config: Dict[str, Any] = field(default_factory=lambda: {
        'n_candidates': 3,
        'k_models': 3,
        'cv_folds': 5,
        'optimization_method': 'grid_search'
```

4.3 Challenges

At the data collection stage, the Twitter API, used in the original paper, was no longer accessible for academic use, so the dataset has been augmented with BlueSky insights instead. Collecting data from all relevant sources identified proved unfeasible, for example, because of paywalls with APIs or due to web scraping not being feasible for forums because of the complexity of the web page structures, so the scope of data collection was reduced to match the time frame. Again, dark web scraping for the full period proved intractable due to the structural complexity of the dark web, so features were collected from 2020 – 2024 only. Finally, some scripts took so

long to run they were intractable in the time period so parallel processing with tqdm was implemented.

In data preparation, a bias towards more common threats was noticed (which makes it harder for the models to predict black swan events), so SMOTE for imbalanced threats was applied. There were also missing values present, which were mitigated with time-series-aware imputation.

For the baseline BMTGNN, the run time for hyperparameter optimisation was intractably high (over 36 hours), the data set was too complex so the script could not run to completion on a local machine, so feature selection was executed to lower training times and computational cost.

For all the models, poor model accuracy was observed, so transfer learning applied to improve model performance. For ViT and graph transformers, the impact of Bayesian MC dropout could not be evaluated, so vanilla models (without Bayesian dropout) were initiated for comparison. Also, training instability was also observed, so the ensemble model was added and data drift was monitored through Jensen-Shannon divergence. For the ensemble, ViT and graph transformers, overfitting and slow training times were spotted, so early stopping was implemented. For the graph transformers, training on a local machine was intractable due to a high compute burden, so Google GPUs were used for faster training and hyperparameter optimisation.

For the agentic LLM, finding optimal prompt and context engineering strategy proved difficult, so zero shot prompting, few shot prompting, role assignment and Chain of Thought prompting were tested and compared, with role assignment mixed with Chain of Thought selected as the most effective strategy. There was also difficulty with rate limiting, so a locally hosted model on Ollama was used as the primary instantiation, with model agnostic support for other providers.

Model integration also presented challenges: managing numerous forecasting architectures in one program made modularity challenging. So, graceful degradation was used, so rather than failing completely when optional dependencies are unavailable, the system sets feature flags, and the agent only recommends models available in the environment. Moreover, as the system needs to support multiple LLM providers without creating provider-specific code branches throughout the agents implementations, and local models (like Ollama) require different connection patterns than cloud APIs, implementing LLM provider abstraction also posed a considerable problem. So, provider-specific details are encapsulated within LLManager so agents remain completely provider-agnostic. LLM errors logged but do not crash calling agents, instead they return an empty string on failure, allowing agents to implement fallback strategies (creating extensibility without active dependency requirements).

Agent communication state management was another issue, as the seven agents must: share intermediate results without tight coupling; maintain consistent state across pipeline stages; handle failures in individual agents without system collapse; support optional agent execution. To address this, the system implements optional execution flags to allow configuration-driven agent activation and clean separation between agent availability and agent execution, as well as structured results where each agent returns typed dictionaries with standardised keys which enables type-safe result passing without tight interface coupling. Finally, a single comprehensive

results dictionary is returned at pipeline completion which enables atomic success/failure for the entire system.

Parsing LLM outputs also presented a considerable challenge as LLM responses are generally unreliable for structured output generation: the JSON output may be malformed, misstructured (e.g. in a markdown file), or contain unforeseen variations (such as conversational preamble or schema inconsistencies). So, the system operates regex-based extraction, silent exception handling (so any parsing failure immediately falls through to deterministic fallback), and never operates with incomplete configuration objects.

Finally, for the XAI integration, there were complexity trade-offs with handling novel methods, so the scope of the XAI implementation was reduced to match the timeframe and project needs. Accessing attention weights required considerable script and model modifications so that the weights were explicitly available. Finally, some methods had too burdensome a cognitive load to be useful in a cybersecurity environment, so these outputs were simplified to improve actionability.

5 Results and Analysis

This section analyses the empirical results of this research project. First, the error analysis of the models is presented and discussed. Second, the operational readiness and adversarial robustness of these models is reviewed to encapsulate potential deployment concerns. Third, the testing and validation plots are compared with the model of the original paper. Fourth, the forecast plots and gap analysis are presented and analysed. Finally, the XAI plots are evaluated.

5.1 Error Analysis

Table 8: Relative Absolute Error (RAE) and Root Relative Squared Error (RRSE) across all models.

	Average RAE	Average RRSE
Original Paper (Almahmoud, 2024)	0.77	0.83
Baseline BMTGNN	0.637	0.819
BMTGNN	0.9944	0.9991
Graph Transformer	0.504	0.522
Vanilla Graph Transformer	0.498	0.556
Vision Transformer	0.3887	0.7771
Vanilla Vision Transformer	0.3831	0.7659
Ensemble Architecture	0.0258	0.058
Agentic LLM	0.2985973078145388	0.34673338675158105

The incorporation of additional data yielded notable improvements in both the RAE and RRSE when compared with the metrics reported in the original study. This enhancement is evidenced by the marked reduction observed between the original paper's results and the baseline figures established in the current investigation. The baseline BMTGNN model demonstrated RAE and RRSE values of 0.637 and 0.819 respectively, representing a considerable improvement over the original paper's (Almahmoud, 2024) values of 0.77 and 0.83.

Introducing dark web data revealed interesting patterns across the various architectures examined. Whilst most models exhibited significant reductions in both error metrics, the BMTGNN model presented an anomalous behaviour, showing a deterioration in performance with RAE and RRSE values increasing to 0.9944 and 0.9991 respectively. This counterintuitive result warrants further investigation to determine whether the BMTGNN architecture possesses inherent

limitations in processing dark web data, or whether other confounding factors may have influenced these outcomes.

The transformer-based architectures demonstrated considerably more promising results. The Graph Transformer achieved RAE and RRSE values of 0.504 and 0.522 respectively, whilst its vanilla counterpart recorded marginally superior RAE performance at 0.498, though with a slightly elevated RRSE of 0.556. These results suggest that the graph-based transformer architectures are particularly well-suited to capturing complex relationships inherent in the augmented dataset.

The Vision Transformer variants also exhibited strong performance characteristics, with the standard Vision Transformer achieving an RAE of 0.3887 and RRSE of 0.7771. The Vanilla Vision Transformer demonstrated a marginal improvement in RAE (0.3831) and a more notable enhancement in RRSE (0.7659), suggesting that the simplified architecture may offer advantages in terms of generalisation capability.

Most notably, the Ensemble Architecture delivered the strongest performance, achieving the lowest error metrics across all models tested, with an RAE of merely 0.0258 and an RRSE of 0.058. This represents a substantial improvement compared to the baseline and demonstrates the benefits of using multiple architectural approaches in tandem. The ensemble's superior performance suggests that the synthesis of model strengths may enable more robust pattern recognition and enhanced predictive accuracy when processing dark web data.

The Agentic LLM approach yielded strong results, with an RAE of approximately 0.299 and an RRSE of 0.347. These figures also represent a considerable improvement over the original paper and the baseline BMTGNN.

These findings indicate that the incorporation of dark web data, with the deployment of advanced neural architectures, particularly transformer-based models and ensemble methods, can substantially diminish the incidence and magnitude of predictive errors. The exception presented by the BMTGNN model's deterioration highlights the importance of careful architectural selection when working with novel data sources, whilst the ensemble architecture's exceptional performance highlights the considerable value of multi-model integration strategies in achieving optimal predictive accuracy.

5.2 Operational Readiness Index (ORI) and Adversarial Robustness

Table 9 shows the adversarial robustness (showing the model's resilience to adversarial examples, where malicious actors attempt to manipulate the model with adversarial inputs, a high score indicates that they are likely to be unsuccessful) and the Operational Readiness Index, a custom metric measuring multiple operational concerns into a single readiness score, where the weighted combination analyses detection quality (precision, recall), operational burden (FPR), and practical constraints (resource efficiency).

Table 9: ORI and Adversarial Robustness Scoring Across All Models

	Operational Readiness Index	Adversarial Robustness
Baseline BMTGNN	0.486	0.107
BMTGNN	0.5903	0.9259
Graph Transformer	0.522	0.926
Vanilla Graph Transformer	0.556	0.926
Vision Transformer	0.510	0.926
Vanilla Vision Transformer	0.408	0.926
Spatiotemporal ensemble	0.7453	0.9261
Agentic LLM	0.8678571423635204	0.871908431572

The ORI and Adversarial Robustness metrics reveal different performance patterns across the architectures examined, offering insights into their practical deployment viability.

Regarding adversarial robustness, a separation emerges between the baseline BMTGNN and all subsequent models. The baseline demonstrates poor resilience at 0.107, rendering it highly vulnerable to adversarial manipulation. However, the introduction of dark web data appears to have dramatically enhanced the BMTGNN's defensive capabilities, elevating its robustness to

0.9259. All models incorporating dark web data, except for the baseline, achieved consistently high adversarial robustness scores clustering tightly around 0.926, with the Agentic LLM performing marginally better at 0.872. This uniformly strong performance suggests that exposure to dark web data, which likely contains naturally adversarial patterns, has effectively protected these models against malicious manipulation attempts.

For ORI, the Agentic LLM architecture demonstrates superior operational readiness with a score of 0.868, indicating an optimal balance. The spatiotemporal ensemble follows with 0.745, suggesting strong overall operational characteristics despite its exceptional predictive accuracy. The BMTGNN variants show moderate improvement from baseline (0.486 to 0.590), whilst the transformer architectures display considerable variability, ranging from 0.408 for the Vanilla Vision Transformer to 0.556 for the Vanilla Graph Transformer.

The ensemble architecture, whilst achieving the lowest error rates, does not perform best on ORI, suggesting potential operational overheads or resource constraints. Yet, the Agentic LLM strikes a strong balance, combining strong adversarial robustness with the highest operational readiness, making it particularly suitable for real-world deployment scenarios where practical constraints and operational burden are key considerations.

5.3 Testing and Validation

Figure 15: Testing plot for DDoS Vision Transformer

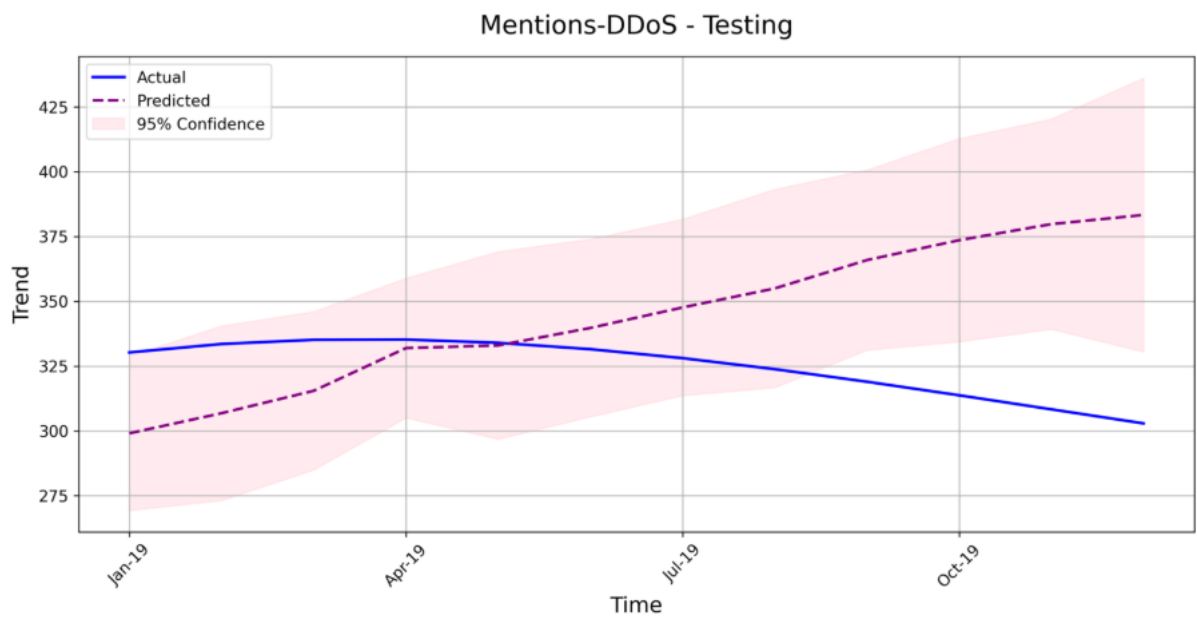


Figure 16: DDoS Testing (Almahmoud, 2024)

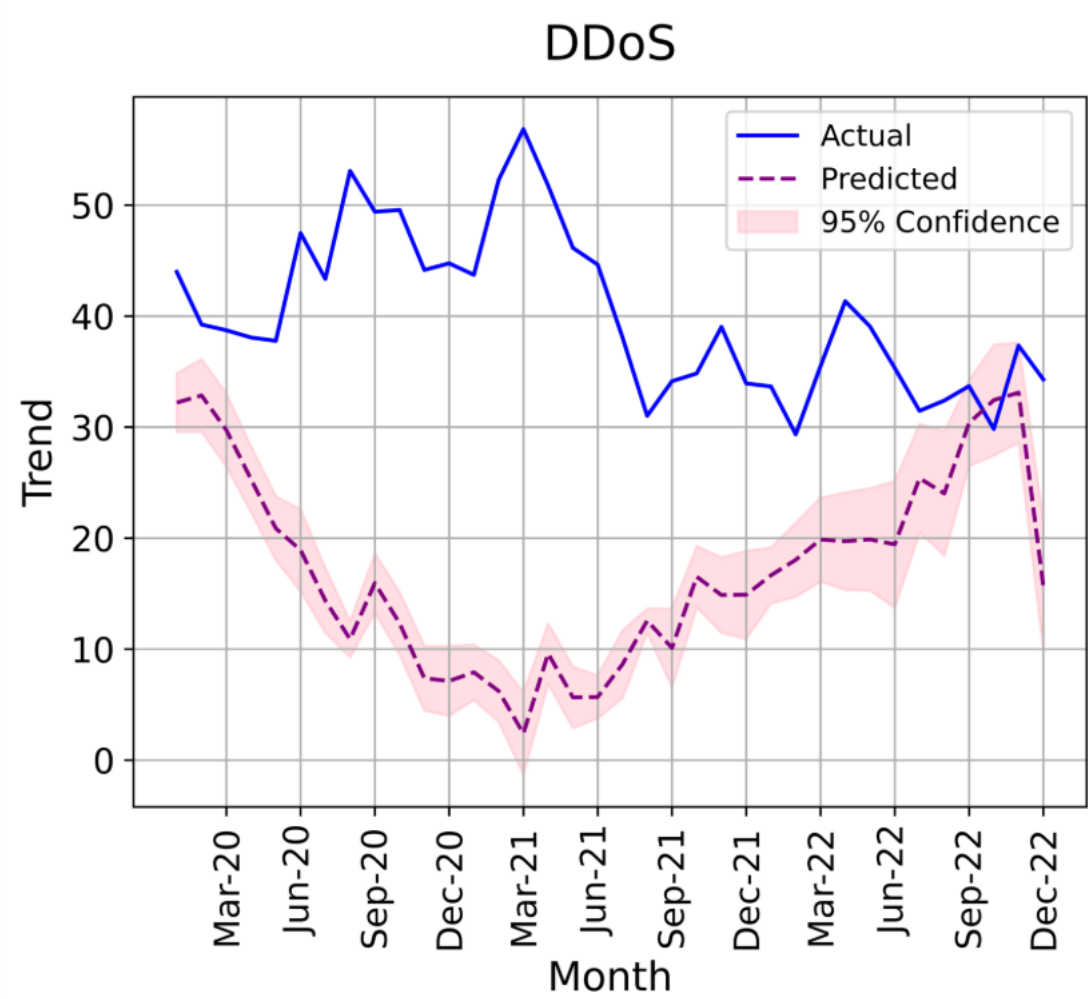


Figure 17: Validation plot for Botnet-ALL Vision Transformer

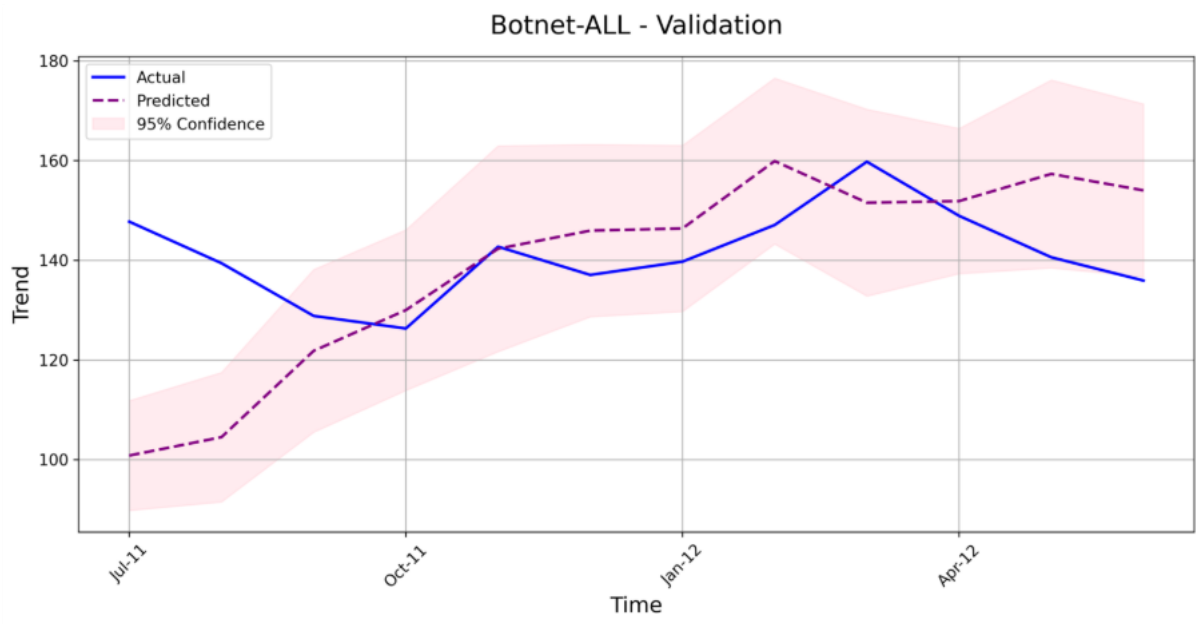
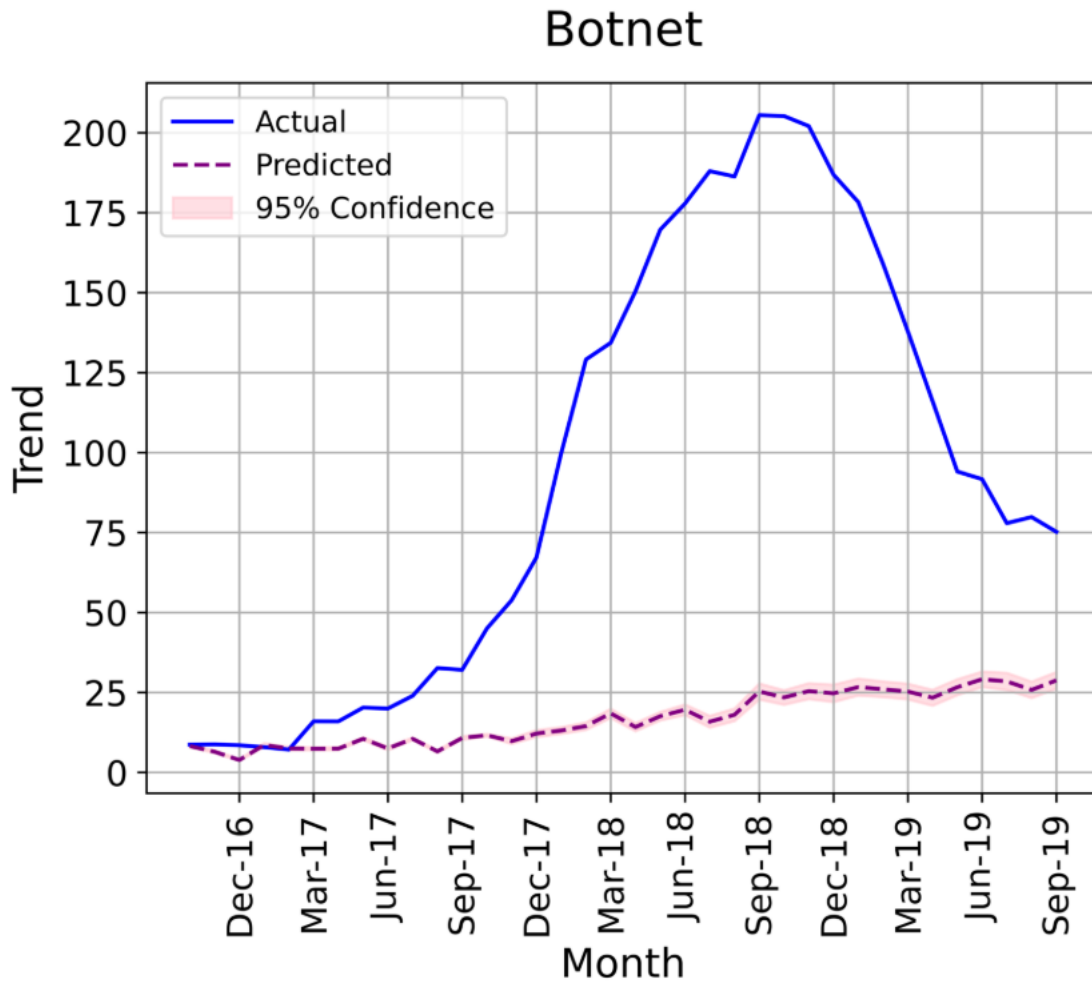


Figure 18: Botnet Validation Plot (Almahmoud, 2024)



In both the testing (Figures 15 and 16) and validation (Figures 17 and 18) plots, the comparison architectures exhibit higher directional accuracy than the B-MTGNN in the baseline paper. Considered in tandem with the improvements in RAE and RRSE, this suggests that these architectures are better suited to cyber threat forecasting and that the addition of dark web data improves the accuracy of trend identification.

5.4 Forecast and Gap Analysis

Figure 19 shows the ensemble model's forecast for Account Hijacking and Table 10 shows the corresponding gap analysis.

Figure 19: Account Hijacking (Ensemble Forecast)

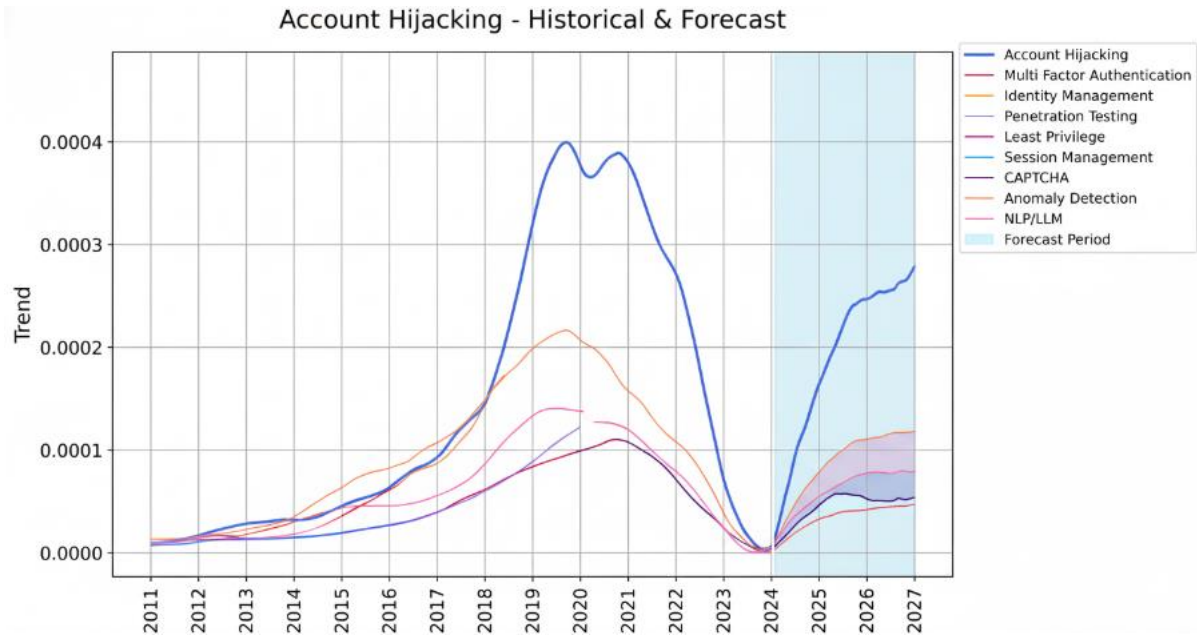


Table 10: Account Hijacking Gap Analysis (Ensemble)

Solution	2025	2026	2027
Encryption	-0.000513746	-0.000952591	-0.000860666
Cryptography	-0.000287764	-0.000494786	-0.000441309
ML/DL	-0.000134007	-0.000405689	-0.000424779
IDS/IPS	-6.58E-05	-0.000146417	-0.000134812
Access Control	-2.91E-05	-6.79E-06	3.1343930459115654e-05
Anomaly Detection	6.755646480390473e-05	0.00014104374107167436	0.00015842484814735752
NLP/LLM	0.00012588202074918325	0.00025450602424825775	0.00028619423786343157
CAPTCHA	0.00013477743302322173	0.0002744950663403264	0.0003075197391808615
Identity Management	0.0001358632807371881	0.00027766835131615153	0.00031144989965999537
Least Privilege	0.0001340699619163388	0.0002791157169307705	0.0003129240531004977
Multi Factor Authentication	0.0001350280660972203	0.00027888269670957777	0.0003123727758425806
Penetration Testing	0.00013642125426827079	0.00027952829987517214	0.0003114894917170356
Session Management	0.0001379961152612926	0.000280480980171897	0.0003134833143465521

Figure 20: Exploitation of End Hosts (Ensemble Forecast)

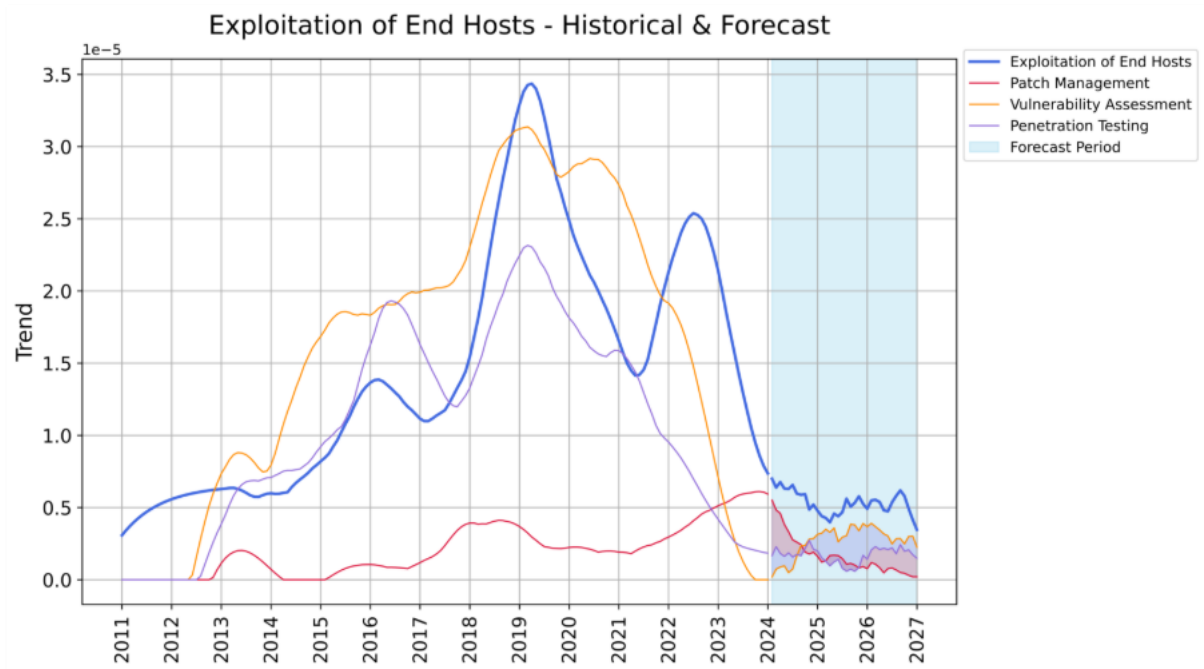


Table 11: Exploitation of End Hosts Gap Analysis (Ensemble)

Solution	2025	2026	2027
<i>ML/DL</i>	-0.000266368	-0.000681446	-0.000733164
<i>IDS/IPS</i>	-0.000198137	-0.000422174	-0.000443198
<i>Anomaly Detection</i>	-6.48E-05	-0.000134713	-0.00014996
<i>Vulnerability Assessment</i>	4.400618323074923e-06	1.4535991113007185e-06	2.1260311579377835e-06
<i>Penetration Testing</i>	4.060713356087339e-06	3.77105888086741e-06	3.1043614683312626e-06
<i>Patch Management</i>	2.9016384909634025e-06	3.6251839977315585e-06	4.516979726550592e-06

The forecast for cyber threats and the corresponding PATs over 3 years are shown in Figures 19 and 20. The ensemble model is chosen for this analysis because it exhibits the lowest error rate. Here, the PATs shown are those for which the trend was forecasted to be below the trend of the relevant threat. The gap area is represented using the same colours as the corresponding PAT curve.

Tables 9 and 10 show the gap trend for these attacks. The low results across the board (i.e., small gap) indicate that the model is confident in the predictions, which makes the outputs more actionable and potentially reliable.

5.5 XAI Results

SHAP (SHapley Additive exPlanations)

Figure 21: Vision Transformer SHAP Beeswarm Summary Plot



SHAP uses game-theoretic Shapley values to assign each feature an importance score representing its contribution to predictions. This provides both global (overall feature importance) and local (individual prediction) explanations, which is critical for understanding which threat indicators (e.g., dark web forum activity, exploit pricing, vulnerability metrics) most influence forecasts. Therefore, users can identify which signals warrant closest monitoring and validate the model's focus on genuine threat indicators rather than spurious correlations. Figure 21 indicates that the model is highly influenced by public attention (e.g. Google Trends), low-level technical chatter (e.g. RSS feeds), and defensive infrastructure signals (e.g. honeypots), pointing to a coherent, directionally consistent model that captures the transition from information awareness to actual cyber activity.

LIME (Local Interpretable Model-agnostic Explanations)

Figure 22: ViT LIME Feature Importance Analysis

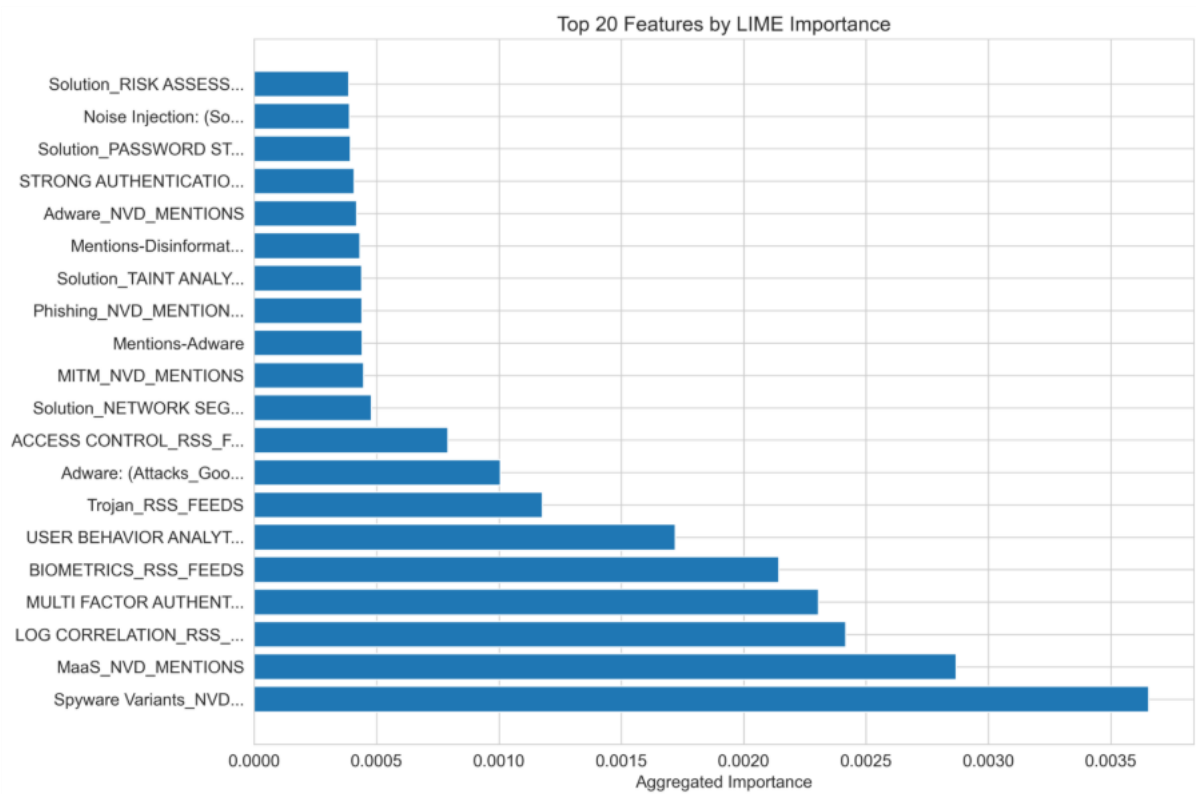
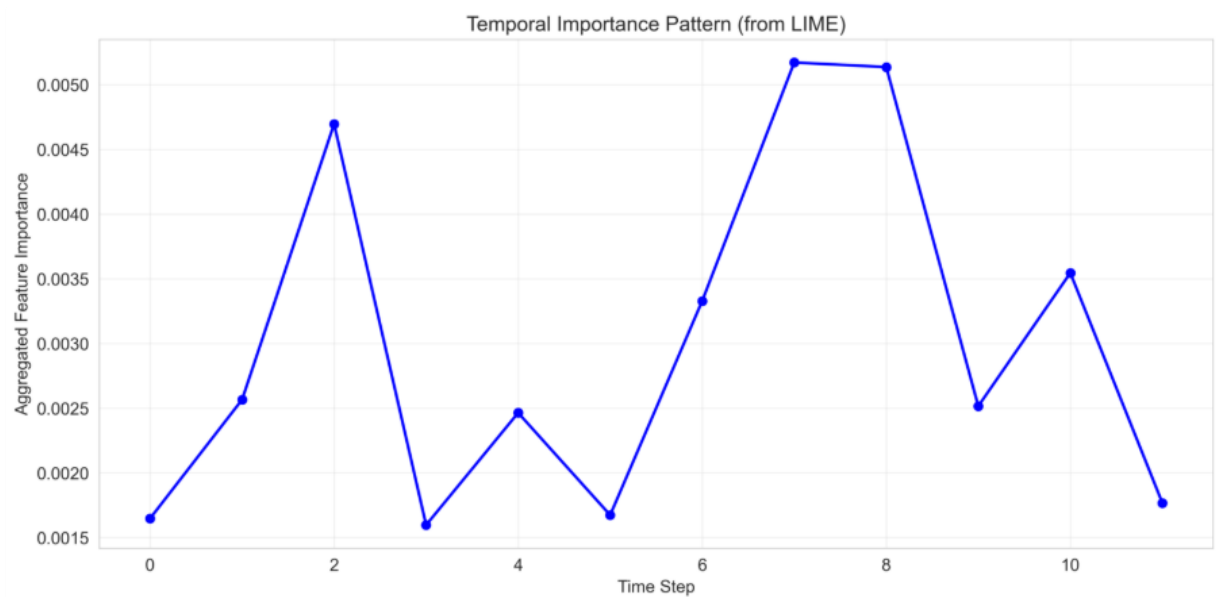


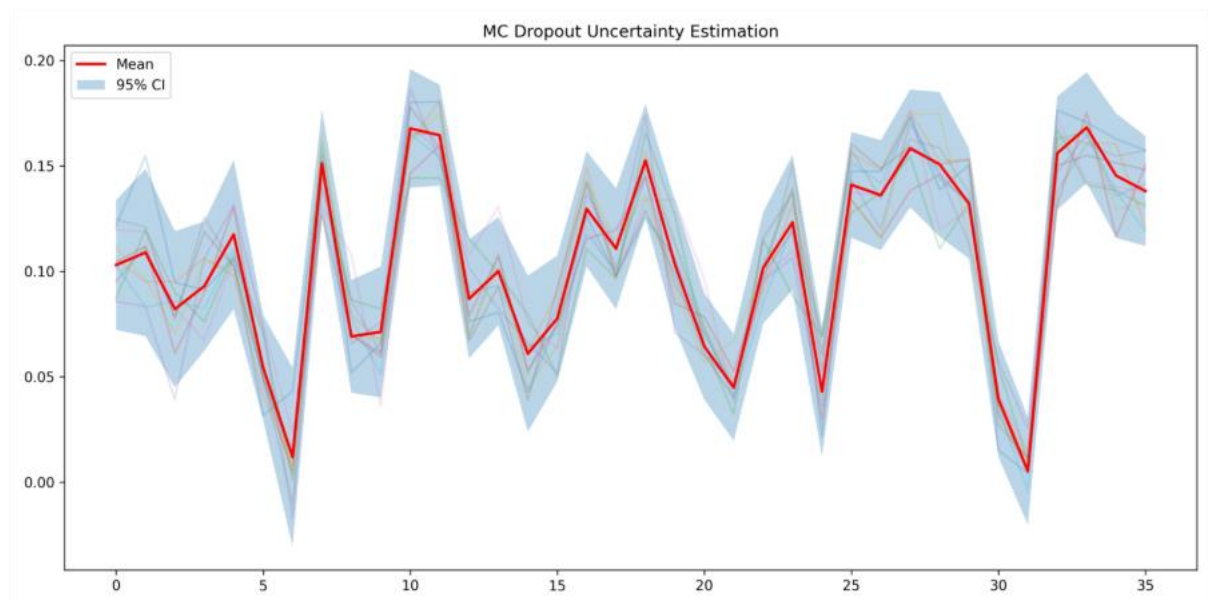
Figure 23: ViT LIME Temporal Importance Pattern over Time Steps



LIME creates simplified linear models that locally approximate the complex model's behaviour around specific predictions by perturbing inputs and observing prediction changes. This enables security teams to understand why a specific threat was flagged at a particular time, providing actionable insights and supporting incident response prioritisation. In Figure 22, the highest ranked features reflect concrete, technical security signals, indicating predictions are locally driven by concrete malware variants and insights from operational security documents. In Figure 23, we see that importance is non-monotonic over time, with clear peaks around mid-to-late time steps so early and very late steps contribute less. So, ViT relies most heavily on mid-window temporal context, not just the most recent observation, suggesting the model is learning temporal build-up patterns rather than reacting instantaneously. This is ultimately a positive sign for model fidelity as the model appears to be learning from the full context window.

Monte Carlo Dropout for Uncertainty Quantification

Figure 24: Ensemble Uncertainty Estimation Plot



As in the paper this project builds on (Almahmoud, 2024), this estimates prediction uncertainty by performing multiple forward passes with dropout enabled, generating a distribution of predictions. The variance indicates model confidence. This communicates forecast reliability to security teams through risk scores, where high uncertainty flags novel threat patterns requiring human analyst review, while high confidence predictions enable automated response workflows. This is critical for distinguishing actionable intelligence from ambiguous signals. In Figure 24, the

width of confidence intervals indicates high uncertainty, consistently over time steps, suggesting that the model is providing stable confidence intervals.

DiCE (Diverse Counterfactual Explanations)

Figure 25: Graph Transformer DiCE Counterfactual plot

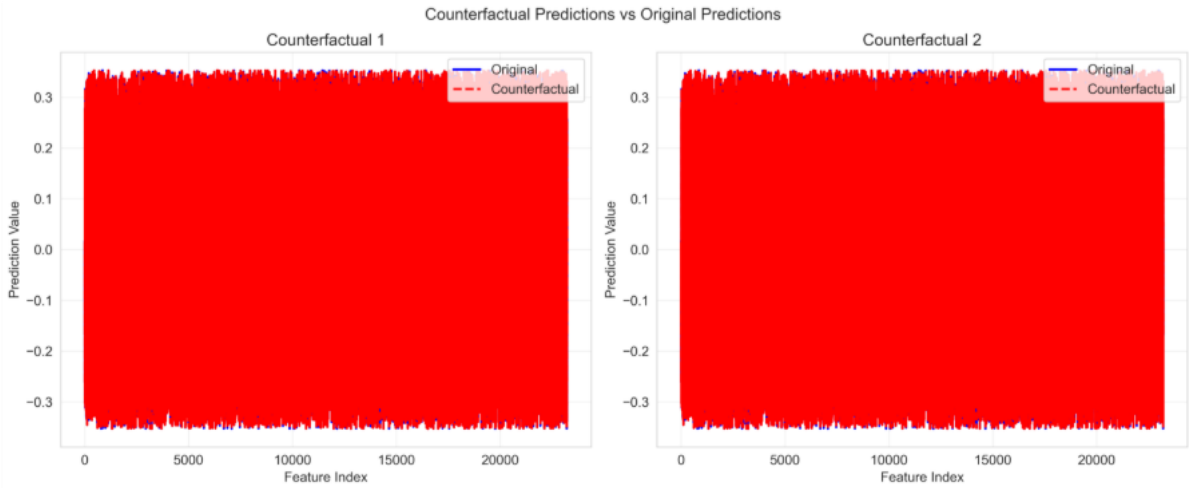
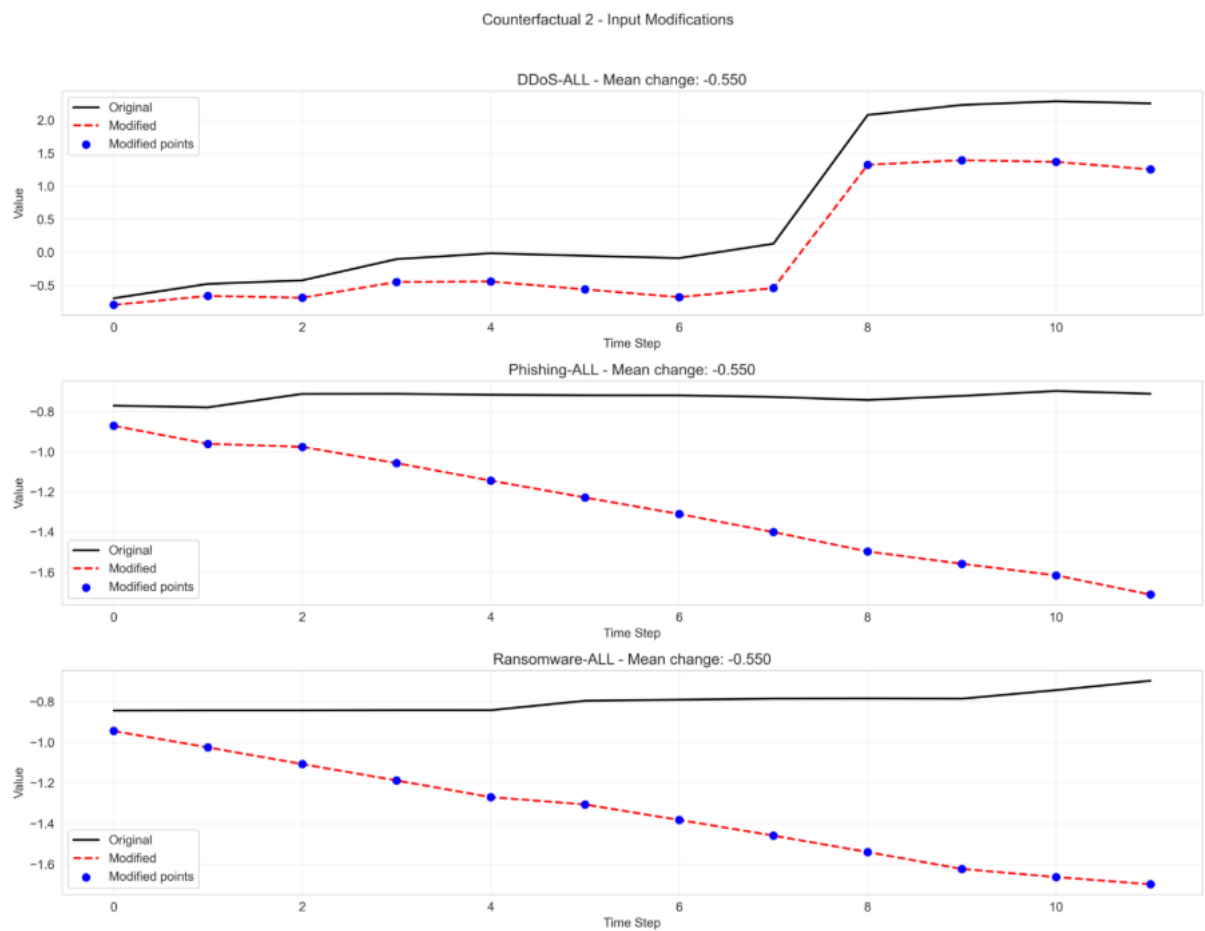


Figure 26: Vision Transformer Counterfactual Perturbation Analysis



Counterfactuals generate minimal modifications to inputs that would change predictions to a desired outcome. This identifies intervention points, supporting proactive defence by revealing which threat indicators, if disrupted, would most effectively prevent attacks. DiCE extends counterfactuals by generating multiple diverse scenarios achieving the same outcome change, providing security teams with various intervention options rather than a single path. This offers actionable alternatives for threat mitigation, giving defenders flexibility in choosing interventions based on available resources and operational constraints. In Figure 25, the counterfactuals are sparse, only features near indices 0 and ~20,000 show variation (the white noise pattern), indicating these are the decision boundaries the model relies on most heavily, meaning predictions can be changed by modifying only a small, targeted subset of features. This suggests that the model focuses on specific feature regions while being relatively insensitive to most features in the middle range. In Figure 26, the large gaps between the modified and original trend lines indicate that the model is sensitive to these perturbations, and so is behaving as expected and consistently relying on features for predictions.

Causal Analysis with Bayesian Networks

Figure 27: Ensemble Model Causal Influence Analysis

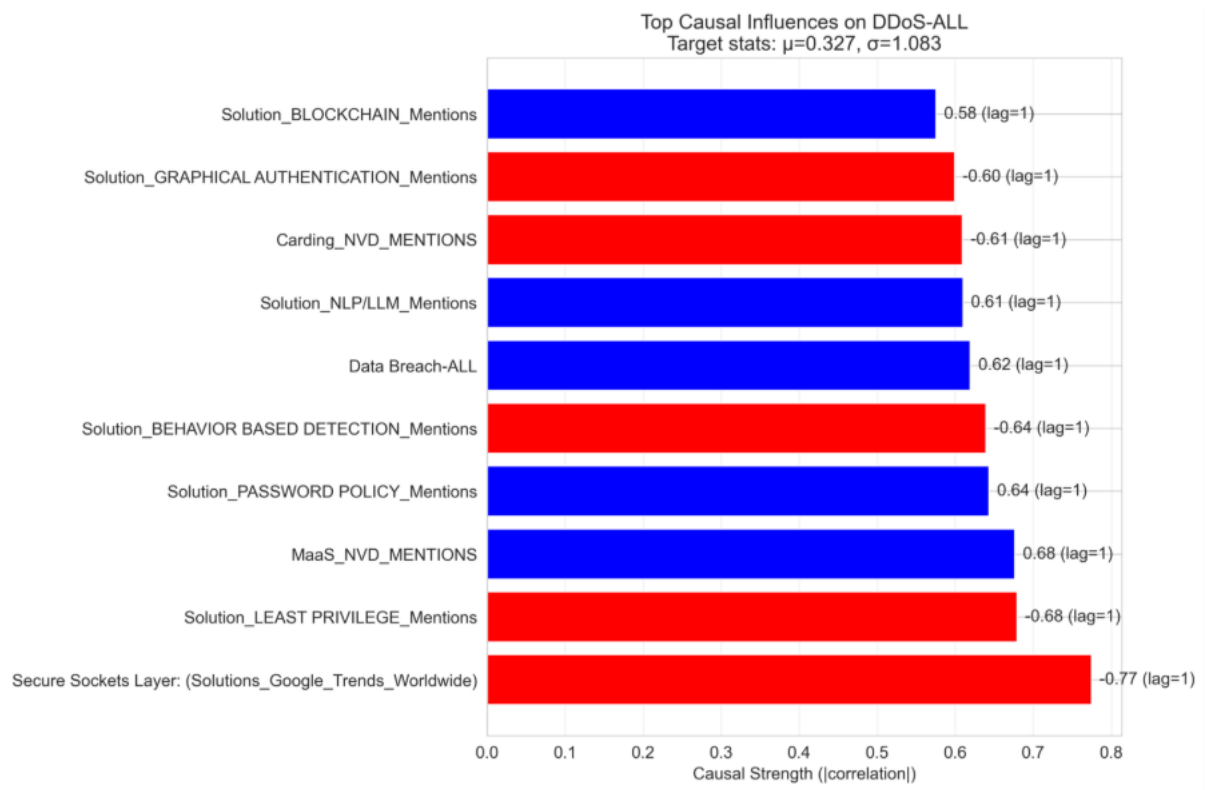


Figure 28: Graph Transformer Network Correlation Analysis

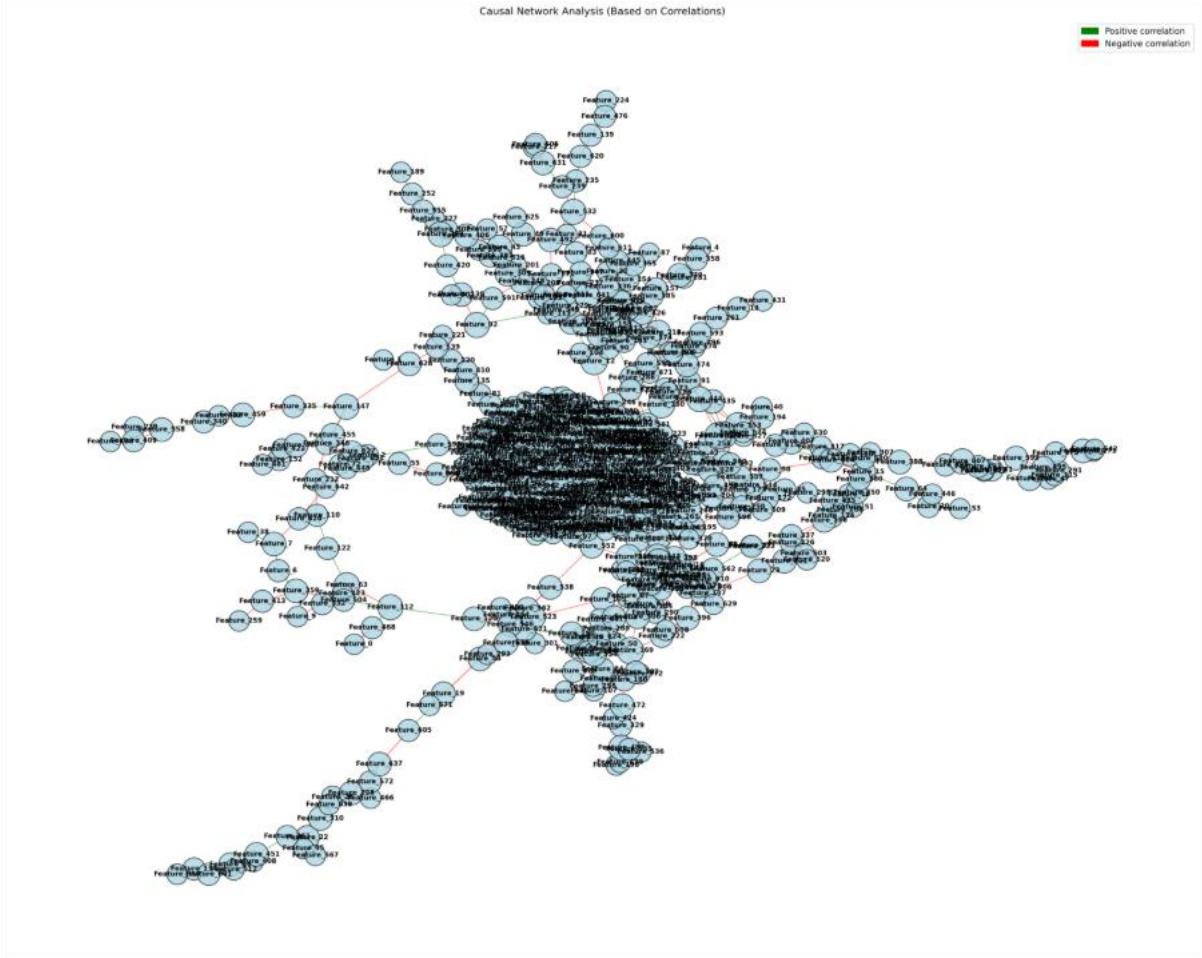
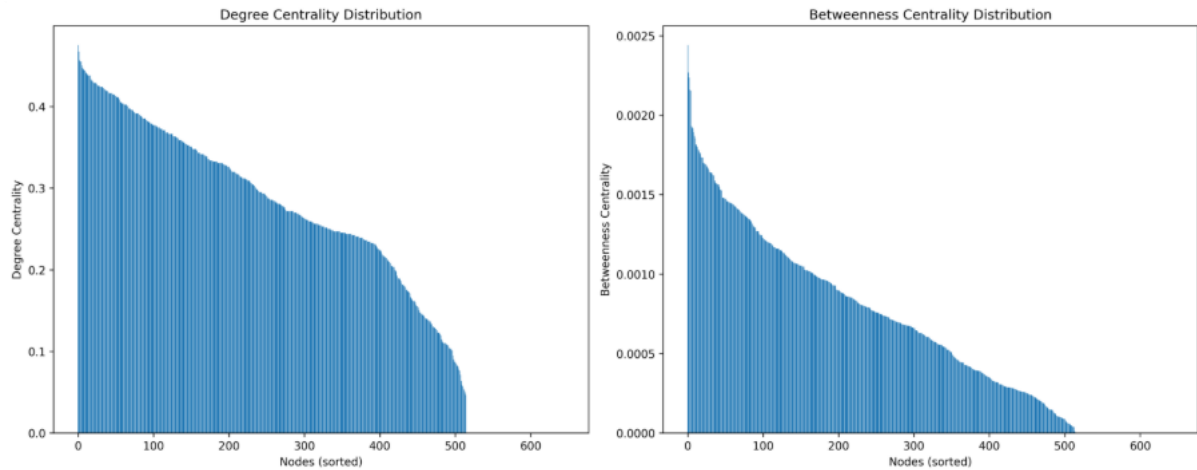


Figure 29: ViT Network Centrality Distributions

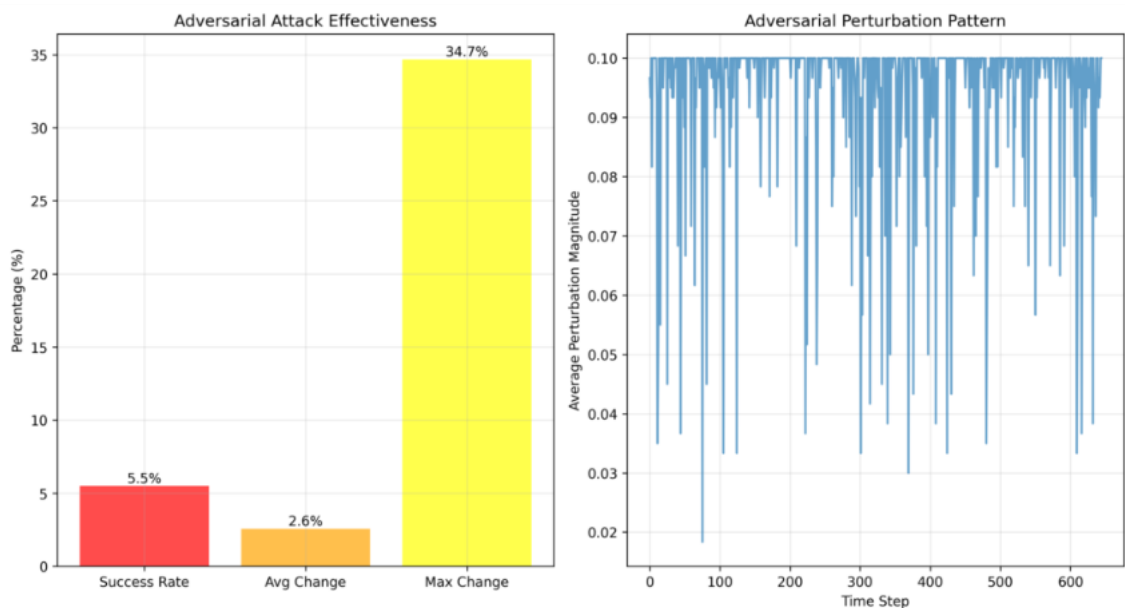


Causal Analysis constructs probabilistic graphical models representing cause-effect relationships between variables, distinguishing genuine causal drivers from mere correlations, using conditional independence testing and intervention calculus.

This may separate true threat precursors from coincidental correlations which prevents over-reliance on spurious indicators. For example, distinguishes whether dark web chatter causes attack surges or merely correlates due to a common upstream factor (e.g., major vulnerability disclosure). In Figure 27, the negative correlations are unexpected, but the strength indicates that the model relies heavily on this subset of features. In Figure 28, the density of connections towards the middle is expected with a graph structure, suggesting the model is behaving as intended (creating graph connections between nodes), so indicates consistent behaviour. In Figure 29, this structure (e.g. hub dominance and heterogenous power-law-like distribution) indicates a self-organising network and suggests the model has specialised certain components as information aggregators or bottlenecks. This is expected with graph transformers, so indicates strong fidelity.

Adversarial Robustness Explanations

Figure 30: Ensemble Model Adversarial Robustness by Attack Effectiveness and Perturbation Pattern



This identifies features most vulnerable to adversarial manipulation, where small perturbations could flip predictions, revealing which inputs attackers could exploit to game the forecasting system. This exposes how threat actors could manipulate dark web signals to evade detection (e.g., slightly altering forum language to avoid triggering high-risk classifications) which enables defensive hardening. In Figure 30, the resistance to high magnitude perturbations across time steps and a low success rate of attacks demonstrate consistently high adversarial robustness. This means that the ensemble model is more difficult for bad actors to manipulate, which is key in operational contexts where AI security is paramount.

Attention Weights Visualisation

Figure 31: Vision Transformer Attention Weights Visualisation

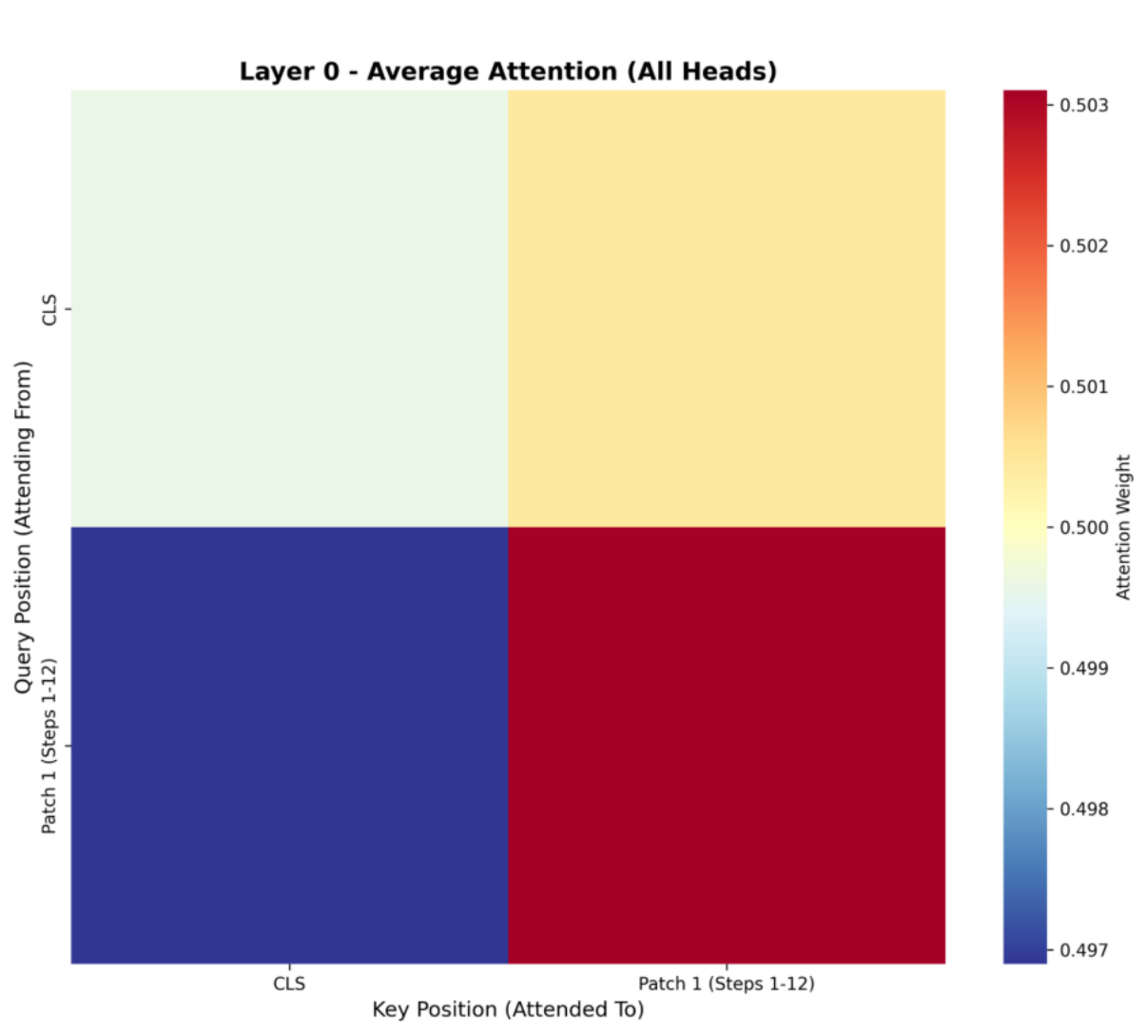
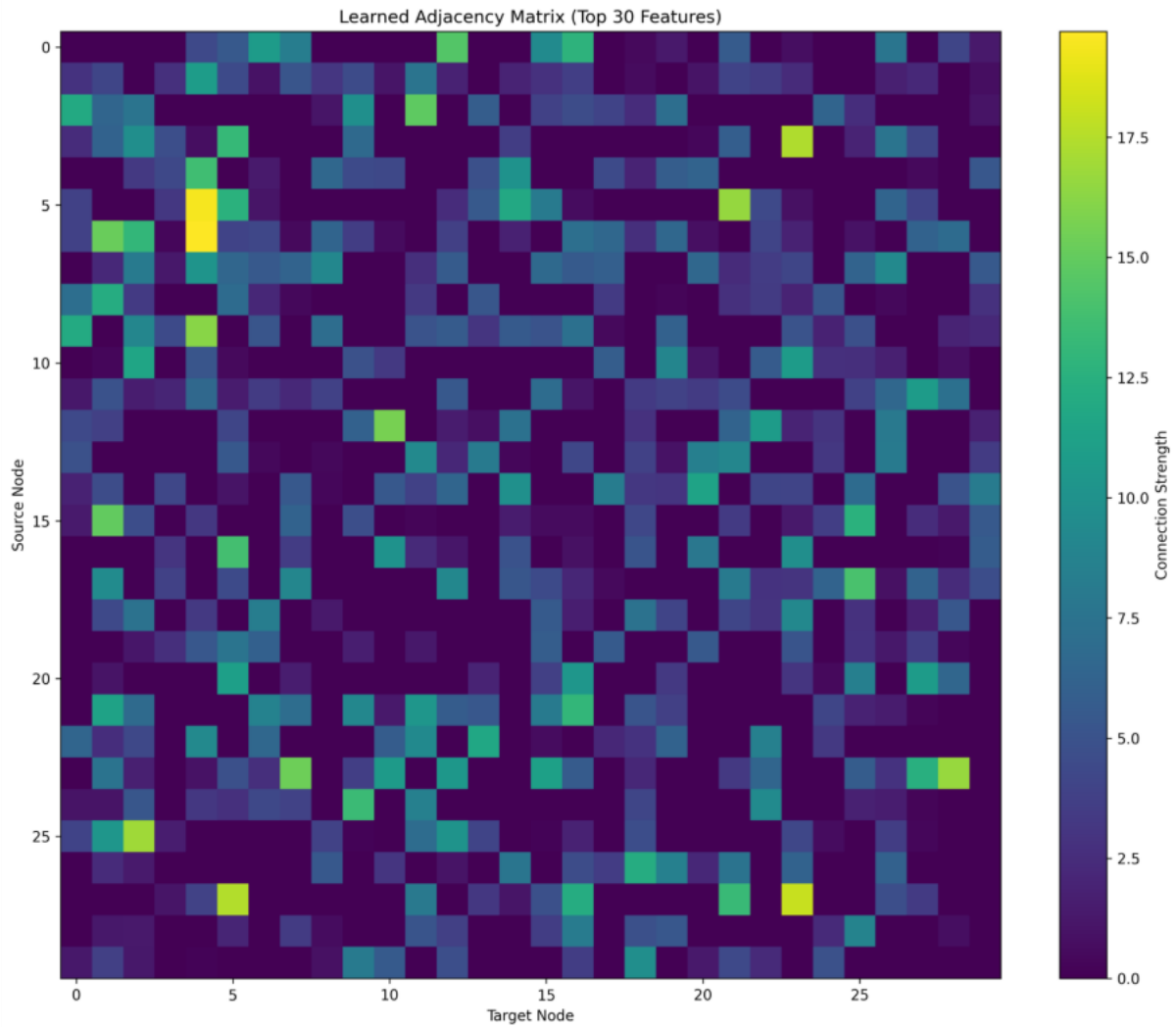


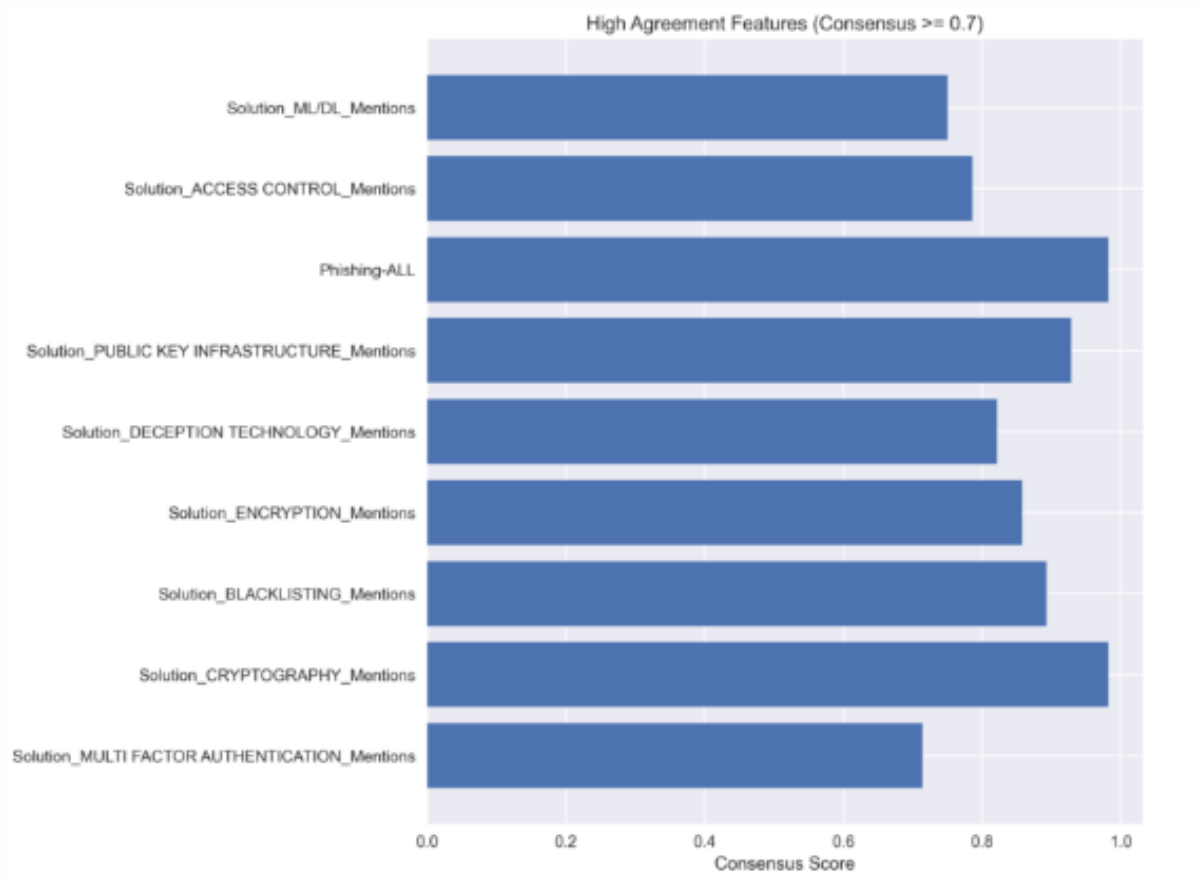
Figure 32: BMTGNN Learned Adjacency Matrix Plot



Visualising attention weights extracts and visualises attention mechanisms learned weights, revealing which input tokens/patches/nodes the model focuses on when making predictions, demonstrating how forecasts are driven. Figure 31 shows clear task separation as different attention weights are best suited to different tasks. The symmetrical nature indicates that the model is learning basic spatial relationships at this stage, therefore behaving as expected and indicating strong fidelity. In Figure 32, the predominantly low connectivity indicates sparse connectivity where most feature pairs do not have strong relationships, suggesting that the model avoided spurious connections, which also suggests strong fidelity.

Multi-Model Explanation Consensus

Figure 33: Agentic LLM Multi-modal consensus plot



Consensus compares explanations across different architectures, where high agreement indicates robust, reliable explanations and disagreement highlights areas needing investigation. This validates threat intelligence by cross-referencing architectures. The high consensus in Figure 33 indicates that certain features have high predictive power, regardless of architecture. This suggests consistency amongst the models, which is a positive sign for reliability.

6 Critical Evaluation and Discussion

This section analyses the findings of this research in a wider academic context. First, the empirical results are considered with respect to dark web intelligence, the ensemble architecture, the agentic LLM, and adversarial robustness. Next, the implication of this interpretation is discussed. Finally, the limitations of this research are assessed.

6.1 Interpretation of Findings

The empirical results suggest a nuanced portrait of architectural performance that challenges expectations whilst also validating several hypotheses regarding dark web intelligence integration and advanced forecasting architectures.

6.1.1 Dark Web Intelligence

The integration of dark web data produced divergent effects across architectures, suggesting that raw data enrichment alone is insufficient, as the architectural capacity for data processing also plays a role. The baseline B-MTGNN demonstrated improvement over the original paper (RAE reduction from 0.77 to 0.637) (Almahmoud, 2024), which may be due to dataset augmentation and extended temporal coverage.

However, the introduction of dark web signals degraded B-MTGNN performance to potentially uninformative predictions (RAE 0.9944, RRSE 0.9991). This may be because the B-MTGNN's graph learning mechanism, optimised for 143 features and 124 nodes, may lack the representational capacity to process 2,126 features and 645 nodes without substantial architectural modifications. The increase in dimensionality could have overwhelmed the graph learning component, forcing the model into a solution space where predictions collapse towards mean values. Conversely, it could be because dark web data, whilst containing early warning signals, introduces substantial noise from forum discussions, pricing volatility, and marketplace churn.

In contrast, the transformer architectures performed better with the integration of dark web data. For instance, the Graph Transformer achieved RAE 0.504 and RRSE 0.522, representing a 35% improvement over the baseline B-MTGNN, whilst the Vision Transformer demonstrated even stronger performance with RAE 0.3887. These results suggest that transformer architectures, with their flexible attention mechanisms and capacity for modelling long-range dependencies, possess inherent advantages for processing heterogeneous, multi-scale threat intelligence. The self-attention mechanism enables selective focus on relevant dark web signals whilst filtering noise, a capability absent in the B-MTGNN's fixed graph structure.

6.1.2 The Ensemble Architecture

The spatiotemporal ensemble's performance (RAE 0.0258, RRSE 0.058) is significant, achieving error rates an order of magnitude lower than competing architectures. This 95% error reduction compared to the Vision Transformer alone (RAE 0.3887) suggests genuine architectural complementarity, where the Vision Transformer captures temporal evolution patterns whilst the

Graph-Aware branch models relational dependencies, with the cross-attention fusion mechanism enabling synergistic integration.

The ensemble's success validates the theoretical premise that cyber threats exhibit both temporal and relational characteristics that single-paradigm models struggle to capture simultaneously. The ensemble architecture captures both dimensions, enabling superior predictive accuracy. However, the gap between predictive accuracy (RAE 0.0258) and operational readiness (ORI 0.745) reveals a critical tension. The ensemble's computational overhead introduces resource constraints that diminish its operational viability relative to simpler architectures. This illustrates that predictive accuracy alone cannot determine model value, operational constraints are also key in real-world deployment scenarios.

6.1.3 The Agentic LLM

The agentic LLM framework achieved RAE 0.299 and RRSE 0.347, revealing strong performance. The true advantage emerges in operational metrics: the highest operational readiness index (0.868) combined with strong adversarial robustness (0.872) positions this architecture as the most deployment-ready solution despite worse raw accuracy than the ensemble. This illuminates a trade-off in forecasting systems, the pursuit of accuracy gains through architectural complexity can sacrifice operational viability. The agentic framework's modular design, interpretable decision-making process, and graceful degradation capabilities address practical deployment concerns that fixed-architecture models ignore. Security professionals need systems that integrate into existing workflows, provide actionable intelligence, and fail gracefully under adverse conditions, which the agentic LLM delivers while the ensemble cannot.

By treating model selection, feature engineering, and validation as autonomous decision-making processes, the agentic framework demonstrates that meta-learning approaches can achieve competitive accuracy whilst maintaining transparency and adaptability. This represents a conceptual departure from traditional deep learning, where model architecture is fixed at design time, towards systems that dynamically adapt their internal structure to data characteristics.

6.1.4 Adversarial Robustness

The dramatic improvement in adversarial robustness across all dark web-enhanced models (baseline 0.107 to >0.926) indicates that exposure to dark web data, which inherently contains adversarial patterns, like manipulated forum posts, strategic pricing signals, deceptive marketplace listings, effectively protects models against adversarial attacks. The dark web, as an adversarial training ground, naturally incorporates the perturbations and manipulations that adversarial robustness techniques artificially introduce. So, rather than treating adversarial robustness as a separate concern requiring dedicated techniques (adversarial training, certified defences), incorporating naturally adversarial data sources may provide inherent robustness. The 0.926 robustness score indicates these models would resist common evasion attempts, a critical requirement given the incentives for threat actors to manipulate forecasting systems.

6.2 Implications

This research may advance theoretical understanding of graph-based temporal forecasting.

6.2.1 Graph Neural Networks and High-Dimensional Data

The BMTGNN's performance degradation with dark web data exposes a fundamental limitation in graph neural network architectures: fixed graph structures struggle to accommodate radical dimensionality expansion without proportional capacity increases. This research suggests that graph capacity scales do not linearly increase with feature dimensionality. The density from the increase in features potentially overwhelms the B-MTGNN's capacity to maintain distinct node representations.

6.2.2 Transformer Attention Mechanisms

The performance of transformer architectures with dark web data suggests attention mechanisms' capacity for heterogeneous signal processing is worth further investigation. Unlike GNNs, which impose structural assumptions through adjacency matrices, transformers learn data-driven relationships through attention weights. The Vision Transformer's attention visualisations reveal task-specific specialisation, where different attention heads focus on different signal types, processing each type independently before integration.

This suggests that attention mechanisms implement ensemble learning across signal types. Each attention head might act as a domain-specific expert, processing signal characteristics, with the multi-head structure enabling parallel processing of diverse patterns, suggesting they can successfully decompose complex problems into specialised sub-problems.

6.2.3 Bayesian Uncertainty Quantification

The Monte Carlo dropout implementation across transformer architectures provides empirical validation for Bayesian deep learning theory whilst also exposing its limitations in operational contexts. The ensemble's uncertainty quantification, visualised through confidence intervals, correctly identifies high-uncertainty regions corresponding to novel threat types and data-sparse temporal windows. This suggests dropout-based Bayesian approximation captures epistemic uncertainty well, but not aleatoric uncertainty, which is also key for informed decision-making.

6.2.4 Agentic Systems and Meta-Learning for Forecasting

The agentic LLM framework's performance suggests that meta-level decision-making, choosing appropriate models, features, and validation strategies, contributes comparably to model optimisation. The agentic system achieved RAE 0.299 by dynamically selecting from several forecasting models, adapting to data characteristics that would have been fixed architectural choices in traditional approaches.

This finding aligns with meta-learning theory (Hospedales et al., 2020) but extends it to temporal forecasting domains. Meta-learning typically focuses on few-shot classification or rapid task

adaptation; this research demonstrates its applicability to long-horizon forecasting through model selection as sequential decision-making. This suggests that effective forecasting systems may require model-level parameter optimisation (traditional deep learning) and meta-level architecture optimisation (model selection, feature engineering).

6.3 Limitations

This research exhibits several limitations that qualify findings and suggest directions for future investigation.

6.3.1 Data Limitations

Temporal Coverage Heterogeneity

The augmented dataset exhibits uneven temporal coverage across sources: Hackmageddon and Elsevier span 2011-2024, whilst dark web data covers 2020-2024. This heterogeneity introduces structural bias: the models' exposure to dark web signals concentrates in the most recent period, potentially causing them to overweight recent patterns whilst underweighting long-term trends.

Sample Selection Bias in Dark Web Data

The dark web dataset derives from 14 marketplaces selected for accessibility and stability during the collection window (2020-2024). This introduces the potential for survivor bias: marketplaces that remained operational throughout this period represent a non-random sample, potentially excluding volatile markets with higher-risk offerings or law enforcement pressure. Consequently, the dark web signals may underrepresent the most dangerous threat actors who operate in more protected venues. Additionally, the reliance on mention counts and pricing data captures marketplace activity misses qualitative intelligence, vendor reputation, product descriptions, buyer reviews, that human threat analysts routinely exploit.

Geographic and Linguistic Constraints

The dataset exhibits anglo-centric bias as data is from English-language sources. This limitation proves particularly acute for dark web intelligence, where significant threat actor communities operate in many languages. The 37-country coverage mitigates this somewhat, but language barriers persist in the data (forum posts, vendor communications). This limitation likely causes the models to underestimate threats originating from non-Anglophone threat ecosystems, particularly state-sponsored actors who predominantly communicate in native languages.

6.3.2 Methodological Limitations

Hyperparameter Optimisation

The hyperparameter search employed random search rather than more sophisticated approaches (Bayesian optimisation, neural architecture search) due to computational constraints. This limitation suggests reported performance metrics represent lower bounds as optimally tuned

architectures would likely achieve superior results. However, computational realities constrain operational systems similarly as random search reflects practical deployment conditions.

6.3.3 Architectural Limitations

Ensemble Complexity

The ensemble architecture's exceptional accuracy comes at operational costs inadequately captured by the ORI metric. Beyond computational overhead the ensemble exhibits deployment complexity, as the ensemble coordinates two separate model instances, managing their weight files, and ensuring consistent preprocessing. This increases deployment failure modes compared to single-architecture models.

Agentic LLM

The agentic LLM framework's reliance on large language models introduces failure modes absent in fixed-architecture systems. There is a hallucination risk. LLMs occasionally generate plausible but factually incorrect outputs. In the forecasting context, this could be selecting inappropriate models, generating invalid feature engineering code, or producing misleading explanations. The extensive prompt engineering (zero-shot, few-shot, chain-of-thought) mitigates but doesn't eliminate this risk.

Additionally, there are reproducibility concerns, as even with temperature=0, LLMs exhibit stochastic behaviour. Identical inputs may produce different model selections across runs, violating reproducibility requirements for operational systems. Also, the system's performance critically depends on the underlying LLM's reasoning capabilities. The current implementation uses DeepSeek R1:8B; but improvements in base LLMs might enhance agentic system performance yet also introduce dependency on external API providers and associated costs.

These limitations do not invalidate the agentic approach but require careful operationalisation: comprehensive testing across multiple runs to understand stochastic variation, fallback mechanisms for LLM failures, and a detailed cost-benefit analysis of LLM provider selection.

Graph Structure Learning

The Graph Transformer constructs dynamic graphs based on correlation patterns using fixed thresholds. Yet, small threshold changes can dramatically alter graph structure. If the threshold is too high, the graph becomes disconnected (isolated nodes) and if it is too low, it becomes fully connected (every node connects to every other), destroying structural information. The random search explored threshold ranges [0.1-0.5], but optimal values likely vary across threat types and temporal windows.

Also, correlation-based graphs capture statistical associations but miss causal relationships. Two attack types may correlate not because one causes the other, but because both respond to a common upstream factor.

6.3.4 **Validation Limitations**

Single-Organisation Perspective

Different organisations face different threat landscapes and the models' performance on this general-purpose dataset may not generalise to organisation-specific threat profiles. This limitation proves particularly acute for rare threat types. Black swan events, zero-day exploits, novel attack vectors, unprecedented threat actor behaviours, all lack historical precedent. Models trained on historical data will inevitably fail to forecast genuinely unprecedented threats. The SMOTE augmentation mitigates this for statistically rare threats with some historical precedent but cannot address truly novel threats.

Temporal Validation

The three-year forecast horizon, whilst operationally relevant, limits validation possibilities. Forecasts extend to December 2027, beyond available ground truth. Consequently, validation relies on held-out test data from the training period rather than true prospective validation against future events.

This limitation means reported metrics may overestimate operational performance. Models optimised on historical data may exploit patterns that don't persist into the future. True validation requires generating forecasts, waiting for events to unfold, then comparing predictions against reality.

6.3.5 **Scope Limitations**

Attribution and Actor-Specific Forecasting

The research forecasts aggregate threat trends (DDoS attacks, ransomware) without differentiating threat actor types (nation-states vs cybercriminals, APT groups vs hacktivists). Different actor classes exhibit different motivations, capabilities, and behaviours. Actor-specific forecasting would provide superior operational intelligence. The current models conflate actor types, potentially obscuring actor-specific trends. Incorporating actor attribution would require additional data sources and modified architectures. This represents a substantial scope expansion that would significantly enhance operational value.

Omitted Threat Vectors

The dataset, whilst comprehensive, omits several emerging threat vectors, like AI-enabled attacks and quantum threats, which reflects data availability constraints (i.e, emerging threats, a lack of historical data), limiting the models' ability to forecast cutting-edge threat evolution.

7 Future Work and Recommendations

There is a wide gamut of directions that future work could take.

7.1 Dataset expansion

Future research might pursue uniform temporal coverage through historical dark web data reconstruction. Whilst active scraping cannot recover closed marketplaces from 2011-2019, alternative sources may be accessed: academic datasets, law enforcement disclosures, and archived forum snapshots. Integrating these sources would enable truly longitudinal analysis, differentiating genuine architectural improvements from recency bias. Also, integrating data from other social media platforms, such as Instagram or Reddit, might further improve the predictive power of the models.

Due to the bias of not sourcing data from multiple sectors, future research could validate these architectures across multiple datasets representing diverse organisational contexts. Collaborative benchmarking initiatives could establish standardised cyber threat forecasting datasets, enabling cross-study comparisons and validating generalisation across threat landscapes.

To mitigate validation concerns, researchers should consider establishing longitudinal validation studies tracking forecasting performance as models encounter genuine future data.

To address language-based data bias concerns, future research should integrate natural language processing techniques (BERT embeddings, topic modelling) to extract semantic intelligence from forum posts and marketplace listings, moving beyond quantitative metrics to qualitative threat characterisation. It would also be advisable to incorporate multilingual NLP models and region-specific dark web sources to achieve geographic representativeness.

To improve the scope of attacks covered, future research should proactively incorporate emerging threat categories even with limited historical data, potentially using techniques like zero-shot learning or expert elicitation to generate synthetic training data.

7.2 Architectural extensions

Future research with greater computational resources could identify performance improvements, but the operational applicability of such computationally expensive optimisation remains questionable. Also, to further improve adversarial robustness, future research should develop realistic threat models incorporating domain-specific attack strategies and evaluate robustness against these contextually appropriate threats.

In relation to the agentic LLM, future work might explore new techniques like Mixture of Experts, to assess how they affect operational performance and accuracy. Also, using an SLM (Small Language Models), which may be easier to deploy in operational contexts because of the lightweight architecture, would be a worthwhile extension. Additionally, employing federated

learning techniques with organisations may gain corporate trust (meaning potential access to richer datasets) and yield accuracy improvements.

Moreover, developing model-specific, cyber security sensitive XAI techniques may lead to improved actionability in an operational context.

Finally, in connection to graph-based architectures, researchers should explore adaptive thresholding mechanisms that dynamically adjust based on graph connectivity metrics (average degree, clustering coefficient) or employ causal discovery algorithms to construct graphs representing genuine threat dependencies rather than statistical correlations.

7.3 Evaluation enhancements

Future work in evaluation could implement user studies with cybersecurity professionals to assess the usability of the system. Also, a longitudinal performance assessment may prove advantageous to understand the long-term dynamics of the system. Finally, implementing more ablation studies would lead to a better understanding of the inner workings of the system.

8 Conclusion

As civilisation increasingly depends upon digital infrastructure for sectors like utilities, healthcare and government services, the protection of such critical infrastructures becomes more imperative. Cyber threats are no longer just technical nuisances or economic inconveniences, but are existential risks to critical infrastructure, democratic institutions, public health systems, and personal safety, as cybersecurity failures are known to cascade from technical domains into physical consequences, economic disruption, and human suffering. The capacity to anticipate these risks offers concrete protection at an individual level, an organisational level, and a societal level. When prediction fails, imperfect forecasts can still contribute to harm reduction by enabling graduated response preparations, leading to more impenetrable defences.

For these reasons, this work extended Almahmoud's (Almahmoud, 2024) work on long-term forecasting, by improving error rates via the integration of early dark web signals and the synthesis of an original architecture. The objective was to contribute to the overall goal of shifting cybersecurity implementation from a reactive to a proactive posture, reducing the asymmetric advantage of threat actors. The systematic comparison of deep learning architectures for cyber threat time-series revealed that the introduction of a pioneering spatio-temporal ensemble architecture boasted excellent, best in class performance, RAE 0.0258 and RRSE 0.058, which is an order of magnitude lower than the competing models. The integration of dark web signals degraded the performance of the BMTGNN when compared to the baseline, indicating that graph capacity scales do not linearly increase with feature dimensionality. Yet, for all other models, the introduction of dark web signals facilitated a meteoric rise in performance. The study also presented the first specialised cyber threat forecasting agentic LLM system, which achieved the strongest performance on operational readiness and adversarial robustness. Moreover, the addition of XAI strengthened model transparency for security decision-making and revealed insights about model behaviours that warrant further investigation. XAI integration reduces the immense cognitive burden faced by security analysts of identifying genuine threats amongst overwhelming noise whilst under time pressure, allowing for more reliable, consistent threat identification and so, in turn, safer digital systems. Finally, the release of the augmented dataset as a research resource (subject to appropriate access controls) democratises access to sophisticated threat intelligence in service of future research, building on this foundation.

This research represents a foundation for continued advancement towards proactive, resilient, and humane cybersecurity. The technical achievements, improved forecasting accuracy, architectural innovations, explainability frameworks, are necessary but insufficient conditions for safer digital societies. These capabilities must be accompanied by institutional reforms, policy developments, and ethical commitments. Ultimately, this work's value lies in potentially preventing harm and therefore suffering by extending the horizon of anticipatory intelligence. So, this research expands the realm of preventable harm, contributing to digital environments where individuals can work, communicate, create, and participate in civic life with increased confidence that the systems supporting these activities will remain operational.

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9 Appendix A

The source code and data used is available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42>

A data dictionary is available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42/blob/main/Appendix/Data%20Dictionary.md>

Docker files are available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42/tree/main/Appendix/Docker%20Files>

Architecture diagrams are available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42/tree/main/Appendix/Architecture%20Diagrams>

Further explanation of the evaluation metrics is available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42/blob/main/Appendix/Evaluation%20Framework%20Explanation.pdf>

Further explanation of the dataset is available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42/blob/main/Appendix/Dataset%20Augmentation%20Explanation.pdf>

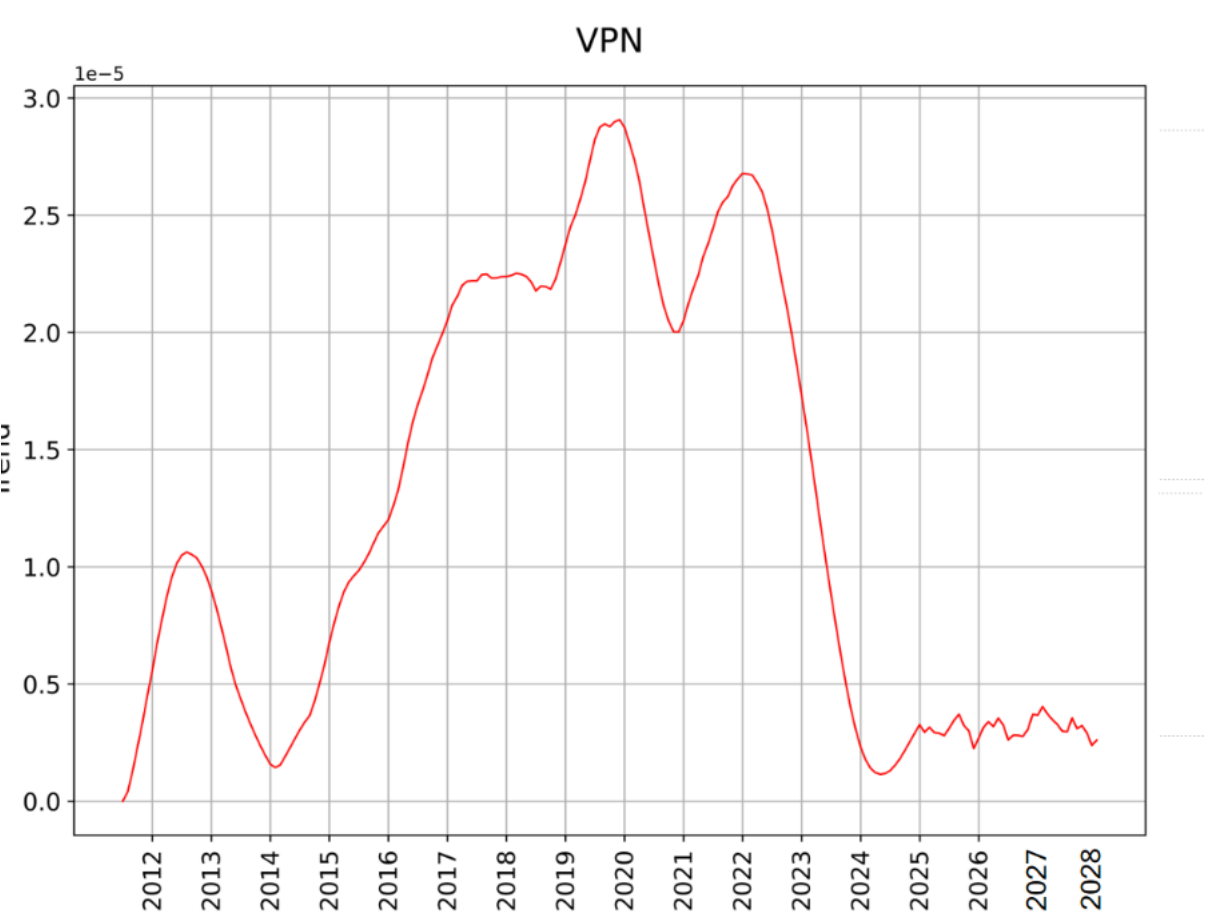
A list of prompts given for the agentic LLM is available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42/blob/main/Appendix/Prompt%20List.txt>

An environment.yml file is available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42/blob/main/Appendix/environment.yml>

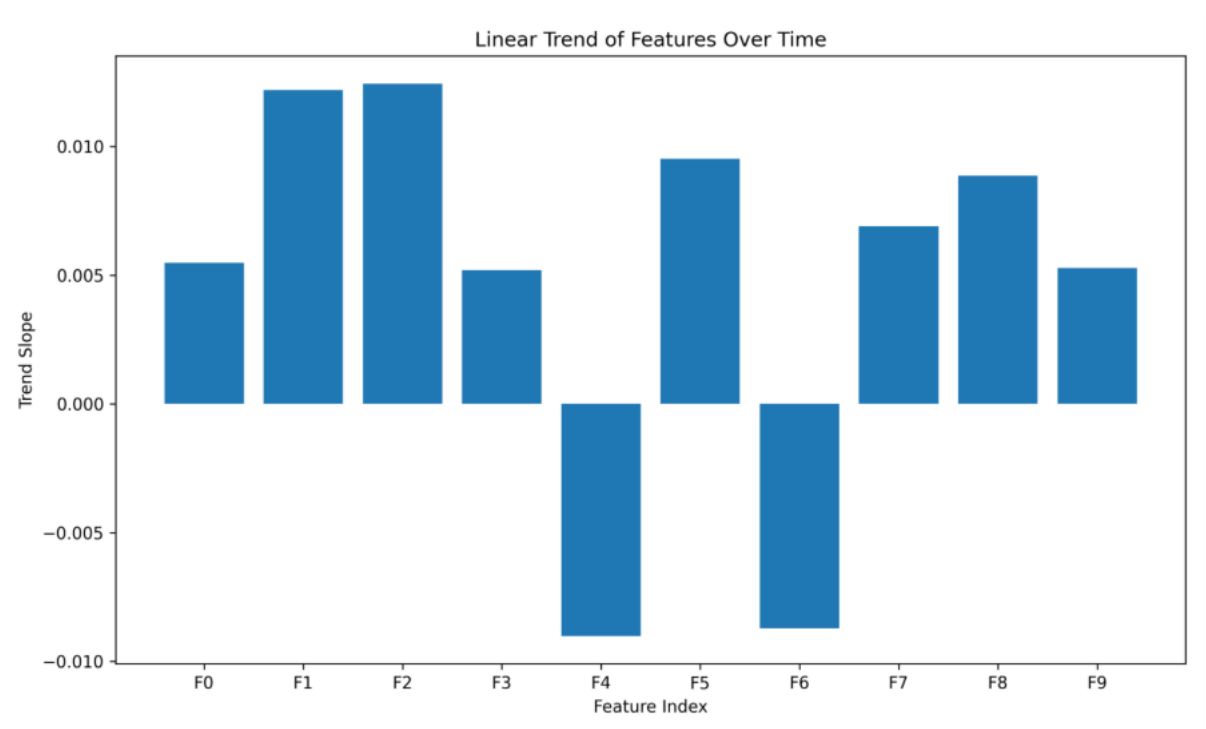
A requirements.txt file is available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42/blob/main/Appendix/requirements.txt>

A data flow diagram is available at: <https://github.com/Birkbeck/msc-project-source-code-files-24-25-bgsbgsbgs42/blob/main/Appendix/data%20flow%20diagram.png>

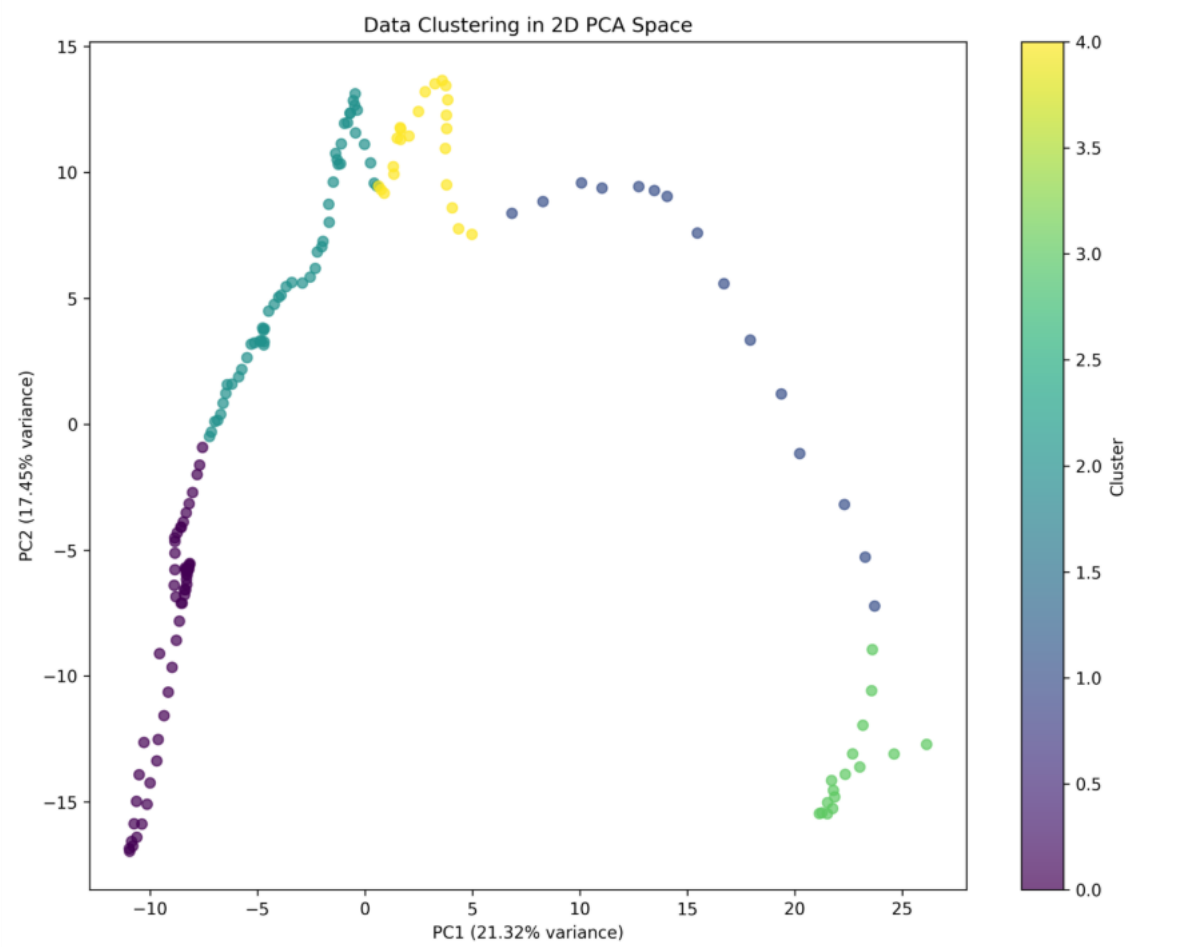
Appendix B



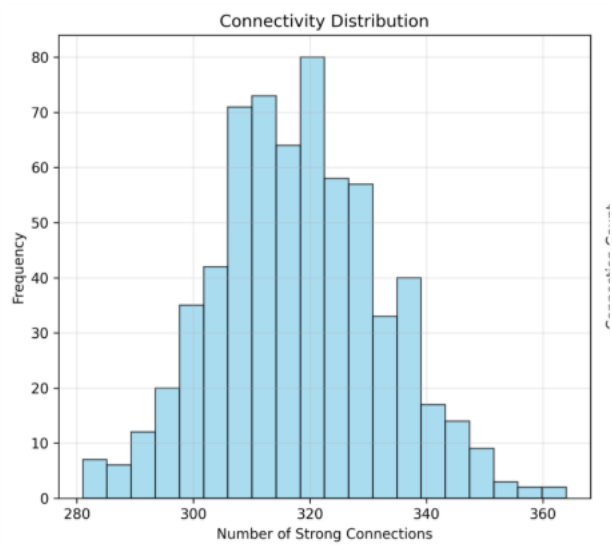
VPN PT Plot (Ensemble)



Linear Trend of Features Over Time Plotted Across Feature Index



PCA Clustering Across Features



BMTGNN Node Connectivity Distribution

Appendix C

Hacking forums scraped:

- Darkode, a hacking forum
- Tor2door, a general marketplace with strong cyber tools section
- Bohemia
- Nemesis, containing exploit listings
- Royal Market, a ransomware-as-a-Service hub
- Vice City, a longstanding marketplace
- ASAP Market, "As Safe As Possible" marketplace
- MGM Grand Market, a casino-themed marketplace
- AlphaBay Market
- Dark Market
- Torrez
- Whitehouse Market
- Abacus
- Dark Matter, an emerging from 2023-2024 platform

RSS feeds scraped:

FEEDS = [

Security Advisories

"http://seclists.org/rss/cert.rss",

"https://www.cisecurity.org/feed/advisories",

"https://www.cyber.gov.au/rss/alerts",

"https://www.cyber.gov.au/rss/advisories",

Security Blogs/News

"https://www.imperialviolet.org/iv-rss.xml",

"https://research.kudelskisecurity.com/feed/",

"http://blog.uncommonsensesecurity.com/feeds/posts/default",

"http://www.darkreading.com/rss/all.xml",

"http://feeds.feedburner.com/schneier/fulltext",

"http://blog.trailofbits.com/feed/",

"https://www.malwaretech.com/feed/",

"https://krebsonsecurity.com/feed/",

"https://www.bleepingcomputer.com/feed/",

"https://threatpost.com/feed/",

"https://www.csoonline.com/index.rss",
"https://www.infosecurity-magazine.com/rss/news/",
"https://www.zdnet.com/topic/security/rss.xml",
"https://www.welivesecurity.com/feed/",
"https://www.tripwire.com/state-of-security/feed/",
"https://www.securityweek.com/rss.xml",
"https://www.helpnetsecurity.com/feed/",
"https://www.scmagazine.com/home/security-news/feed/",
"https://www.itsecurityguru.org/feed/",
"https://www.cnet.com/feeds/security/",
"https://www.techrepublic.com/rssfeeds/topic/security/",
"https://www.thehackernews.com/feeds/posts/default",
"https://www.infosecurity-magazine.com/rss/blogs/",
"https://www.cyberscoop.com/feed/",
"https://www.securityaffairs.co/wordpress/feed",
"https://www.cyberdefensemagazine.com/feed/",
"https://www.infosecurity-magazine.com/rss/reviews/",
"https://www.infosecurity-magazine.com/rss/opinions/",
"https://www.infosecurity-magazine.com/rss/features/",
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"https://www.infosecurity-magazine.com/rss/whitepapers/",
"https://www.infosecurity-magazine.com/rss/infographics/",
"https://www.ncsc.gov.uk/api/1/services/v1/all-rss-feed.xml",
"https://www.us-cert.gov/ncas/alerts.xml",
"https://www.schneier.com/blog/atom.xml",
"https://www.dspace.com/rsssecurity.xml",
"https://www.f-secure.com/en/web/labs_global/rss",

"https://www.grahamcluley.com/feed/",
"https://www.malwarebytes.com/blog/feed/",
"https://seclists.org/rss/basics.rss",
"https://seclists.org/rss/fulldisclosure.rss",
"https://seclists.org/rss/pen-test.rss",
"https://seclists.org/rss/bugtraq.rss",
"https://seclists.org/rss/webappsec.rss",
"https://seclists.org/rss/oss-sec.rss",
"https://seclists.org/rss/crypto.rss",
"https://seclists.org/rss/packetstorm.rss",
"https://seclists.org/rss/zeroday.rss",

Security Advisories & Alerts

"https://www.ncsc.gov.uk/api/1/services/v1/rss-feed.xml",
"https://msrc.microsoft.com/blog/rss/",
"https://www.zerodayinitiative.com/rss/",
"https://cvefeed.io/rssfeed",
"https://www.cisa.gov/cybersecurity-advisories.xml",
"https://www.nist.gov/rss-feeds.xml",

"https://www.bsi.bund.de/SiteGlobals/Forms/RSSFeed/RSSNewsfeed_Formular.xml?nn=784524",

Security Blogs & News

"https://news.sophos.com/en-us/feed/",
"https://aws.amazon.com/blogs/security/feed/",
"https://blogs.cisco.com/security/feed",
"https://feeds.trendmicro.com/TrendMicroSecurityBlog",
"https://blog.qualys.com/feed/",
"https://gbhackers.com/feed/",
"https://www.hackingarticles.in/feed/",

"https://www.sentinelone.com/feed/",
"https://www.troyhunt.com/rss/",
"https://securelist.com/feed/",
"https://cloudsecurityalliance.org/feed/",
"https://www.bankinfosecurity.com/rss-feeds",
"https://www.blackhillsinfosec.com/feed/",
"https://www.govinfosecurity.com/rss-feeds",
"https://www.benthamsgaze.org/feed/",
"https://bishopfox.com/feeds/blog.rss",
"https://www.veracode.com/blog/feed",
"https://marcoramilli.com/feed/",
"https://robert.penz.name/feed/",
"https://feeds.feedburner.com/TheHackerNews",
"https://www.securityweek.com/feed/",
"https://www.tripwire.com/state-of-security/rss",
"https://www.secureworks.com/blog/rss",
"https://securityledger.com/feed/",
"https://blog.skullsecurity.org/feed/",
"https://holisticinfosec.blogspot.com/feeds/posts/default",
"https://taosecurity.blogspot.com/feeds/posts/default",
"https://blog.rootshell.be/feed/",
"https://adamlevin.com/feed/",
"https://johnmasserini.com/feed/",
"https://alstonprivacy.com/feed/",
"https://defensivesecurity.org/feed/",
"https://paloaltonetworks.com/blog/feed/",
"https://security.googleblog.com/feeds/posts/default",
"https://blog.malwarebytes.com/feed/",
"https://www.avast.com/blog/rss.xml",
"https://blog.sucuri.net/feed/",

"https://research.checkpoint.com/feed/",
"https://www.crowdstrike.com/blog/feed/",
"https://blog.f-secure.com/feed/",
"https://blogs.jpccert.or.jp/en/atom.xml",
"https://isc.sans.edu/rssfeed.xml",
"https://www.cert.ssi.gouv.fr/feed/",
"https://www.trendmicro.com/rss/all.xml",
"https://www.darktrace.com/en/rss-feed.xml",
"https://www.elastic.co/security-labs/feed.xml",
"https://www.okta.com/security-blog/rss/",
"https://blog.cloudflare.com/tag/security/rss/",
"https://www.microsoft.com/security/blog/feed/",
"https://www.us-cert.gov/ncas/all.xml",
"https://www.talosintelligence.com/feed",
"https://www.fortinet.com/blog.rss",
"https://www.mcafee.com/blogs/feed/",
"https://community.rsa.com/blogs/feed",
"https://www.eset.com/int/rss/blog/",
"https://www.ibm.com/blogs/security/feed/",
"https://www.tenable.com/blog/feed",
"https://blog.radware.com/feed/",
"https://www.trustwave.com/en-us/rss/blogs/",
"https://www.proofpoint.com/us/rss/blog.xml",
"https://www.carbonblack.com/blog/feed/",
"https://www.splunk.com/en_us/blog/security/_jcr_content.feed",
"https://www.fireeye.com/blog/feed",
"https://www.cybereason.com/blog/rss.xml",

Security Forums & Community

"https://www.reddit.com/r/netsec/.rss",

"https://security.stackexchange.com/feeds/week",
"https://www.reddit.com/r/cybersecurity/.rss",
"https://www.reddit.com/r/AskNetsec/.rss",
"https://www.reddit.com/r/ReverseEngineering/.rss",
"https://www.reddit.com/r/Malware/.rss",

Archival & Mailing List Feeds

"https://catless.ncl.ac.uk/Risks/rss",
"https://web.archive.org/web/20220101000000*/http://www.phrack.org/rss.xml",
"https://marc.info/?rss",
"https://allinfosecnews.com/feed/",
"https://github.com/foorilla/allinfosecnews_sources/blob/main/sources.json",
"https://github.com/thehappydinoa/awesome-threat-intel-rss",
"https://www.cyberseo.net/public-rss-feed-catalogue/Information%20Security/",
"https://archive.cert.org/rss/security-updates.xml",
"https://archives.neohapsis.com/rss/",
"https://lists.openwall.net/rss/all.xml",
"https://www.packetstormsecurity.com/feeds/news/",
"https://exploit-db.com/rss.xml",
"https://seclists.org/rss/oss-security.rss",
"https://vulners.com/rss",
"https://www.exploit-db.com/rss/",
"https://www.securityfocus.com/rss/vulnerabilities.xml",
"https://www.nvd.nist.gov/feeds/xml/cve/misc/nvd-rss.xml",
"https://feeds.feedburner.com/SecurityFocus",
"https://archive.cert.europa.eu/rss/advisories.xml",
"http://www.reddit.com/r/netsec/.rss",
"http://security.stackexchange.com/feeds/week"

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Figure 0.1 Logo of Birkbeck, University of London